Foundations in Accountancy

PRACTICE & REVISION KIT

MA2

MANAGING COSTS AND FINANCES

Welcome to BPP Learning Media's Practice & Revision Kit for MA2. In this Practice & Revision Kit, which has been reviewed by the **ACCA examining team**, we:

- Include **Do you know?** Checklists to test your knowledge and understanding of topics
- Provide you with **two** mock exams including the Specimen Exam
- Provide the ACCA's exam answers to the Specimen Exam as an additional revision aid

FOR EXAMS FROM 1 SEPTEMBER 2020 TO 31 AUGUST 2021

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Helping you with your revision

BPP Learning Media – ACCA Approved Content Provider

As an ACCA **Approved Content Provider**, BPP Learning Media gives you the **opportunity** to use revision materials reviewed by the **ACCA examining team**. By incorporating the ACCA examining team's comments and suggestions regarding the depth and breadth of syllabus coverage, the BPP Learning Media Practice & Revision Kit provides excellent, **ACCA-approved** support for your revision.

These materials are reviewed by the ACCA examining team. The objective of the review is to ensure that the material properly covers the syllabus and study guide outcomes, used by the examining team in setting the exams, in the appropriate breadth and depth. The review does not ensure that every eventuality, combination or application of examinable topics is addressed by the ACCA Approved Content. Nor does the review comprise a detailed technical check of the content as the Approved Content Provider has its own quality assurance processes in place in this respect.

BPP Learning Media does everything possible to ensure the material is accurate and up to date when sending to print. In the event that any errors are found after the print date, they are uploaded to the following website: www.bpp.com/learningmedia/Errata.

Selecting questions

We provide signposts to help you plan your revision.

- A full **question index**

Attempting mock exams

There are two mock exams that provide practice at coping with the pressures of the exam day. We strongly recommend that you attempt them under exam conditions. **Mock exam 1** is the Specimen Exam. **Mock exam 2** reflects the question styles and syllabus coverage of the exam.

Using your BPP Practice & Revision Kit

Aim of this Practice & Revision Kit

To provide the practice to help you succeed in the examination for MA2 *Managing Costs and Finances.*

To pass the examination you need a thorough understanding in all areas covered by the syllabus and teaching guide.

Recommended approach

- Make sure you are able to answer questions on **everything** specified by the syllabus and teaching guide. You cannot make any assumptions about what questions may come up in your exam. The examining team aims to discourage 'question spotting'.
- Learning is an **active** process. Use the **DO YOU KNOW**? Checklists to test your knowledge and understanding of the topics covered in MA2 *Managing Costs and Finances* by filling in the blank spaces. Then check your answers against the **DID YOU KNOW**? Checklists. Do not attempt any questions if you are unable to fill in any of the blanks – go back to your **BPP Interactive Text** and revise first.
- When you are revising a topic, think about the mistakes that you know that you should avoid by writing down **POSSIBLE PITFALLS** at the end of each **DO YOU KNOW**? Checklist.
- You should attempt each bank of questions to ensure you are familiar with their styles and to practise your technique. Ensure you read **Tackling Multiple Choice Questions** on page xi to get advice on how best to approach these types of question.
- Once you have completed all of the questions in the body of this Practice & Revision Kit, you should attempt the **MOCK EXAMS** under examination conditions. Check your answers against our answers to find out how well you did.

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Passing the MA2 exam

MA2 *Managing Costs and Finances* builds on what you learnt in MA1. There is a lot to learn, but none of it is particularly difficult and a good grasp of these topics will help you in higher-level management accounting in FMA (ACCA MA).

To access Foundations in Accountancy syllabi, visit the ACCA website: www.accaglobal.com/students/fia

The exam

All questions in the exam are compulsory. This means you cannot avoid any topic, but also means that you do not need to waste time in the exam deciding which questions to attempt. There are 50 objective test questions (multiple choice, number entry, multiple response and multiple response matching) in the computer-based exam (CBE). This means that the examining team are able to test most of the syllabus at each sitting, and that is what they aim to do. So you need to have revised right across the syllabus for this exam.

Revision

This Practice & Revision Kit has been reviewed by the MA2 examining team and contains questions from the Specimen Exam as Mock Exam 1, so if you just worked through it to the end you would be very well prepared for the exam. It is important to tackle questions under exam conditions. Allow yourself just the number of minutes shown next to the questions in the index and don't look at the answers until you have finished. Then correct your answer and go back to the Interactive Text for any topic you are really having trouble with. Try the same question again a week later – you will be surprised how much better you are getting. Doing the questions like this will really show you what you know, and will make the exam experience less worrying.

Doing the exam

If you have honestly done your revision you can pass this exam. There are certain points which you must bear in mind:

- Read the question properly.
- Don't spend more than the allotted time on each question. If you are having trouble with a question leave it and carry on. You can come back to it at the end.

Approach to examining the syllabus

MA2 is a two-hour exam.

The exam is structured as follows:

	No of marks
50 compulsory objective test questions of 2 marks each	100

The computer-based examination

Computer-based examinations (CBEs) are available for most of the Foundations in Accountancy exams. The CBE exams for the first seven modules can be taken at any time, these are referred to as 'exams on demand'. The Option exams can be sat in June and December of each year, these are referred to as 'exams on sitting'. FTX is moving from paper-based exams (PBE) to CBE format from the June 2020 exam sitting. There will be no parallel running of PBE and CBE exams.

Computer-based examinations must be taken at an ACCA CBE Licensed Centre.

How do CBEs work?

- Questions are displayed on a monitor.
- Candidates enter their answer directly onto the computer.
- Candidates have two hours to complete the examination.
- Candidates sitting exams on demand are provided with a Provisional Result Notification showing their results before leaving the examination room
- The CBE Licensed Centre uploads the results to the ACCA (as proof of the candidate's performance) within 72 hours.
- Candidates sitting the Option exams will receive their results approximately five weeks after the exam sitting once they have been expert marked
- Candidates can check their exam status on the ACCA website by logging into myACCA.

Benefits

- **Flexibility** as the first seven modules, exams on demand, can be sat at any time.
- **Resits** for the first seven modules can also be taken at any time and there is no restriction on the number of times a candidate can sit a CBE.
- **Instant feedback** for the exams on demand as the computer displays the results at the end of the CBE.
- Results are notified to ACCA **within 72 hours**.

For more information on computer-based exams, visit the ACCA website:

www.accaglobal.com/gb/en/student/exam-entry-and-administration/computer-based-exams.html

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Tackling multiple choice questions

MCQs are part of all Foundations in Accountancy exams.

The MCQs in your exam contain four possible answers. You have to **choose the option that best answers the question**. The incorrect options are called distractors. There is a skill in answering MCQs quickly and correctly. By practising MCQs you can develop this skill, giving you a better chance of passing the exam.

You may wish to follow the approach outlined below, or you may prefer to adapt it.

Step 1	Skim read all the MCQs and identify what appear to be the easier questions.
Step 2	Attempt each question – **starting with the easier questions** identified in Step 1. Read the question **thoroughly**. You may prefer to work out the answer before looking at the options, or you may prefer to look at the options at the beginning. Adopt the method that works best for you.
Step 3	Read the options and see if one matches your own answer. Be careful with numerical questions as the distractors are designed to match answers that incorporate common errors. Check that your calculation is correct. Have you followed the requirement exactly? Have you included every stage of the calculation?
Step 4	You may find that none of the options match your answer. • Re-read the question to ensure that you understand it and are answering the requirement • Eliminate any obviously wrong answers • Consider which of the remaining answers is the most likely to be correct and select the option
Step 5	If you are still unsure make a note and continue to the next question.
Step 6	Revisit unanswered questions. When you come back to a question after a break you often find you are able to answer it correctly straight away. If you are still unsure have a guess. You are not penalised for incorrect answers, so **never leave a question unanswered!**

After extensive practice and revision of MCQs, you may find that you recognise a question when you sit the exam. Be aware that the detail and/or requirement may be different. If the question seems familiar read the requirement and options carefully – do not assume that it is identical.

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Using your BPP products

This Practice and Revision Kit gives you the question practice and guidance you need in the exam. Our other products can also help you pass:

- **Interactive Text** introduces and explains the knowledge required for your exam
- **Passcards** provide you with clear topic summaries and exam tips

You can purchase these products by visiting www.bpp.com/learning-media.

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Questions

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Do you know? – Management information and information technology

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- The purpose of management information is to help managers to ……., by …… and …… operations and by allowing ……..
- ……. involves establishing objectives for a company and developing …… in order to achieve those objectives.
- Features of useful management information include the following.

  – …… – ……
  – …… – ……
  – …… – ……
  – …… – ……

- ……. are prepared for individuals external to an organisation…… ..... are prepared for internal managers.

  ……. aims to capture an organisation's costs of operations, departments or products and then classify and analyse the information to produce cost reports. Cost accounting produces information that is used for both financial accounting and management accounting.

  Costs are either …… or ……. depending on whether they change when the volume of production changes, or …… or ……., depending how easily they can be traced to a specific unit of production.

  …… are all methods used to cost end-products.

- The main features of a management information report are as follows.

  – ……
  – ……
  – ……
  – ……
  – ……
  – ……

- The three stages of data input are as follows.

  (1) ……
  (2) ……
  (3) ……

- Common document reading methods include the following.

  – ……
  – ……
  – ……
  – ……
  – ……

- Card reading devices include …… cards and …… cards.
- Data storage methods include the following.

  – ……
  – ……
  – ……

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A management information system is the …….. and …….. used to drive a database system which provides useful information for management. It should have certain essential characteristics which should be defined:

……
……
……

- Possible pitfalls

  Write down the mistakes you know you should avoid.

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Did you know? – Management information and information technology

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- The purpose of management information is to help managers to **manage resources efficiently and effectively**, by **planning** and **controlling** operations and by allowing **informed decision-making**.
- **Planning** involves establishing objectives for a company and developing **strategies** in order to achieve those objectives.
- Features of useful management information include the following.
  - **Relevant**
  - **Complete**
  - **Accurate**
  - **Clear**
  - **Manageable volume**
  - **Timely**
  - **Cost less than value of benefits**
  - **User has confidence in it**
- **Financial accounts** are prepared for individuals external to an organisation. **Management accounts** are prepared for internal managers.

  Cost accounting aims to capture an organisation's costs of operations, departments or products and then classify and analyse the information to produce cost reports. Cost accounting produces information that is used for both financial accounting and management accounting.

  Costs are either **variable** or **fixed**, depending on whether they change when the volume of production changes, or **direct** or **indirect**, depending how easily they can be traced to a specific unit of production. **Job, batch and process costing** are all methods used to cost end-products.
- The main features of a management information report are as follows.
  - **Title**
  - **To – who is the report for?**
  - **Date**
  - **Subject**
  - **Appendix**
  - **From**
- The three stages of data input are as follows.
  (1) **Origination of data**
  (2) **Transcription of data**
  (3) **Data input**
- Common document reading methods include the following.
  - **MICR**
  - **OMR**
  - **Scanners and OCR**
  - **Bar coding and EPOS**
  - **EFTPOS**
- Card reading devices include **magnetic stripe** cards and **smart** cards.
- Data storage methods include the following.
  - **Disks**
  - **CD-ROM**
  - **DVD-ROM**

A management information system is the **hardware** and **software** used to drive a database system which provides useful information for management. It should have certain essential characteristics which should be defined:

The functions of individuals and their responsibilities
Areas of control
Information required for an area of control should flow to the manager who is responsible

- Possible pitfalls
  - **Confusing internal sources and external sources of management information**
  - **Confusing primary and secondary sources of management information**
  - **Not remembering the main features of a management information report**

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1 Management information and information technology 34 mins

1.1 **Which of the following correctly describes a management information system?**

A A system which measures and corrects the performance of activities of subordinates in order to make sure that the objectives of an organisation are being met and the plans devised to attain them are being carried out.

B A system by which managers ensure that information is obtained and used effectively and efficiently in the accomplishment of the organisation's objectives.

C A system which involves selecting appropriate information so that management can prepare a long-term plan to attain the objectives of the organisation.

D A collective term for the hardware and software used to drive a database system with the outputs, both to screen and print, being designed to provide easily assimilated information for management. **(2 marks)**

1.2 **For whom are management accounts prepared?**

A Employees
B Internal managers
C Shareholders
D Providers of finance **(2 marks)**

1.3 When visiting your local supermarket, the items that you have purchased are scanned by a device which acts as a cash register.

What is this device known as?

A MICR
B OCR
C OMR
D EPOS **(2 marks)**

1.4 **Which of the following printers can print a whole page at a time?**

A Bubble jet printers
B Inkjet printers
C Scanners
D Laser printers **(2 marks)**

1.5 **Which of the following is a common feature of cost accounting but not financial accounting?**

A Control accounts
B Cost classification
C Marginal costing
D Periodic stocktaking **(2 marks)**

1.6 **Which TWO of the following are necessary features of useful management information?**

☐ Clear to the user
☐ Detailed and completely accurate
☐ Provided whatever the cost
☐ Relevant for purpose **(2 marks)**

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1.7 **Which TWO of the following are features of graphical user interfaces?**

- ☐ Icons
- ☐ Keyboard
- ☐ Optical mark reading
- ☐ Pull-down menu

(2 marks)

1.8 **What is the purpose of management information?**

A Planning only
B Planning and control only
C Planning, control and decision-making only
D Planning, control, decision-making and research and development **(2 marks)**

1.9 **Which of the following are used for the capture and storage of management accounting data by computer?**

(i) Barcode
(ii) Hard disk
(iii) Printer
(iv) DVD

A (i) and (ii) only
B (i), (ii) and (iv)
C (i), (iii) and (iv)
D (ii), (iii) and (iv) **(2 marks)**

1.10 **Which TWO of the following are characteristics of good management information?**

- ☐ Relevant
- ☐ Detailed
- ☐ Expensive to obtain
- ☐ Timely

(2 marks)

1.11 **Which of the following describes a faceted code?**

A Each digit represents a classification, and each digit further to the right represents a smaller subset than those to the left.

B The code is an abbreviation of the item being coded.

C Each digit of the code gives information about the item.

D A digit, usually the first one, indicates the classification of the item. **(2 marks)**

1.12 **Which of the following correctly describes the card reading method known as MICR?**

A A device that can read text or illustrations printed on paper and translate the information into a form the computer can use

B A device which can sense the marks made by a ballpoint pen or typed line or cross by using an electric current and is able to translate it into machine code

C Groups of marks which, by their spacing and thickness, indicate specific codes or values

D The recognition by a machine of special formatted characters printed in magnetic ink

(2 marks)

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1.13 **In a large business which of the following activities is most likely to be the responsibility of a trainee accountant?**

A Coding invoices
B Determining selling price strategy
C Interpreting cost variances
D Making capital investment decisions **(2 marks)**

1.14 **Which of the following is NOT an internal source of management information?**

A Personnel records
B Production department records
C Financial statements of competitors
D Detailed time records **(2 marks)**

(Total = 28 marks)

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Do you know? – Cost classification and cost behaviour

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- Costs can be divided into the following three elements:
  - .....
  - .....
  - .....
- There are a number of different ways in which costs can be classified.
  - ..... and ..... (or overhead) costs
  - ..... costs (production costs, distribution and selling costs, administration costs and financing costs)
  - Fixed and ..... costs
- A ..... is a unit of product or service which has costs attached to it. ..... are the essential 'building blocks' of a costing system, and they act as a collecting place for certain costs before they are analysed further. A ..... is any activity for which a separate measurement of costs is desired.
- Cost behaviour patterns demonstrate the way in which costs are affected by changes in the level of activity. Costs which are affected by the level of activity are **.....** costs, and those which are not affected by the level of activity are ..... costs or ..... costs.
- Costs which are fixed in nature within certain levels of activity are ..... costs. Some costs are partly fixed and partly variable (and therefore only partly affected by activity level changes), such costs are known as ..... costs (semi-variable/semi-fixed costs).
- The basic principle of cost behaviour is that as the level of activity rises, costs will usually **.....**
- The effect of increasing activity levels on unit costs is as follows. (Tick as appropriate)

	Rises	*Falls*	*Remains constant*
Variable cost per unit	☐	☐	☐
Fixed cost per unit	☐	☐	☐
Total cost per unit	☐	☐	☐

- The fixed and variable elements of semi-variable costs can be determined by the **.....** method.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

Did you know? – Cost classification and cost behaviour

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- Costs can be divided into the following three elements:
  - **Materials**
  - **Labour**
  - **Expenses**
- There are a number of different ways in which costs can be classified.
  - **Direct** and **indirect** (or overhead) costs
  - **Functional** costs (production costs, distribution and selling costs, administration
  - costs and financing costs)
  - Fixed and **variable** costs
- A **cost unit** is a unit of product or service which has costs attached to it. **Cost centres** are the essential 'building blocks' of a costing system, and they act as a collecting place for certain costs before they are analysed further. A **cost object** is any activity for which a separate measurement of costs is desired.
- Cost behaviour patterns demonstrate the way in which costs are affected by changes in the level of activity. Costs which are affected by the level of activity are **variable** costs, and those which are not affected by the level of activity are **fixed** costs or **period** costs.
- Costs which are fixed in nature within certain levels of activity are **step** costs. Some costs are partly fixed and partly variable (and therefore only partly affected by activity level changes), such costs are known as **mixed** costs (semi-variable/semi-fixed costs).
- The basic principle of cost behaviour is that as the level of activity rises, costs will usually **rise**.
- The effect of changing activity levels on unit costs is as follows. (Tick as appropriate)

	Rises	*Falls*	*Remains constant*
Variable cost per unit	☐	☐	✓
Fixed cost per unit	☐	✓	☐
Total cost per unit	☐	✓	☐

- The fixed and variable elements of semi-variable costs can be determined by the **high-low** method.
- Possible pitfalls
  - **Not being able to define key terms encountered in the Interactive Text**
  - **Being unable to distinguish between variable, fixed, step and mixed costs**

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2 Cost classification and cost behaviour 67 mins

2.1 **Which of the following describes the term 'cost unit'?**

A A basis for cost classification
B A production or service department
C A unit of product or service
D The cost of a unit of output **(2 marks)**

2.2 **What is a cost centre?**

A A unit of product or service in relation to which costs are ascertained
B An amount of expenditure attributable to an activity
C A production or service location, function, activity or item of equipment for which costs are accumulated
D A business unit or department where management makes investment decisions **(2 marks)**

2.3 **What is a prime cost?**

A All costs incurred in manufacturing a product
B The total of direct costs
C The material cost of a product
D The cost of operating a department **(2 marks)**

2.4 **Which of the following costs are part of the prime cost for a manufacturing company?**

A The cost of raw materials that go in to the product
B Wages of factory workers engaged in machine maintenance
C Depreciation of lorries used for deliveries to customers
D Cost of indirect production materials **(2 marks)**

2.5 **Which of the following are indirect costs?**

(i) The depreciation of maintenance equipment
(ii) The overtime premium incurred at the specific request of a customer
(iii) The hire of a tool for a specific job

A (i) only
B (i) and (ii) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

2.6 A company has to pay a royalty of $1 per unit to the designer of a product which it manufactures and sells.

How would the royalty charge be classified in the company's accounts?

A A direct expense
B A production overhead
C An administrative overhead
D A selling overhead **(2 marks)**

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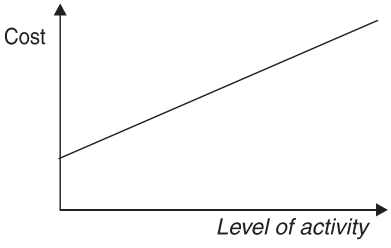
2.7 **Which TWO of the following items might be a suitable cost unit within the credit control department of a company?**

- ☐ Telephone expense
- ☐ Cheque received and processed
- ☐ Customer account
- ☐ Credit control manager's salary **(2 marks)**

2.8 **Which of the following would be classed as indirect labour?**

A Machine operators in a company manufacturing washing machines
B A stores assistant in a factory store
C Plumbers in a construction company
D A team of inspectors ensuring that products meet quality standards **(2 marks)**

2.9 The following is a graph of total cost against level of activity.

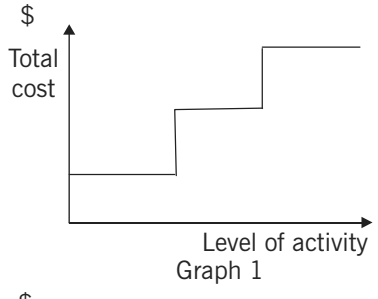


To which of the following costs does the graph correspond?

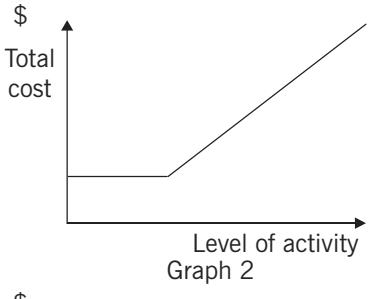
A Electricity bills made up of a standing charge and a variable charge
B Bonus payment to employees when production reaches a certain level
C Sales commissions payable per unit up to a maximum amount of commission
D Bulk discounts on purchases, the discount being given on all units purchased **(2 marks)**

2.10 One supervisor is needed for every five employees.

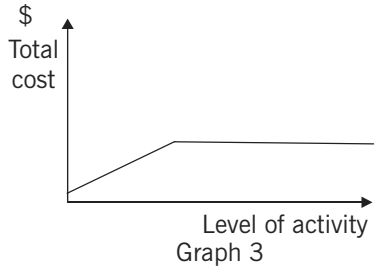
Which of the following graphs depicts the behaviour of the supervisors' salary costs?



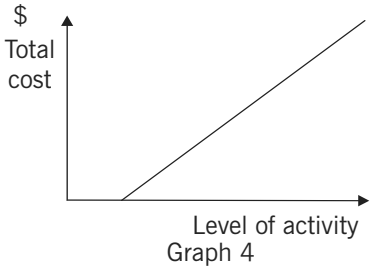
Graph 1



Graph 2



Graph 3



Graph 4

A Graph 1
B Graph 2
C Graph 3
D Graph 4 **(2 marks)**

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2.11 A production worker is paid a salary of $650 per month, plus an extra 5 cents for each unit produced during the month.

Which of the following best describes this labour cost?

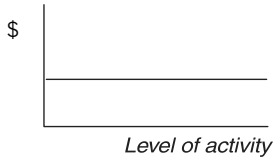
A A variable cost
B A fixed cost
C A step cost
D A semi-variable cost **(2 marks)**

2.12 A hotel has recorded that the laundry costs incurred were $570 when 340 guests stayed for one night. They know that the fixed laundry cost is $400 per night.

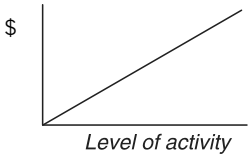
What is the variable laundry cost per guest-night (to two decimal places)?

$ ☐ **(2 marks)**

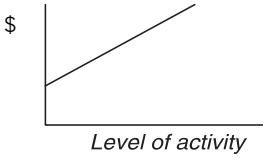
2.13 **Which of these graphs represents a linear variable cost, when the vertical axis represents total cost incurred?**



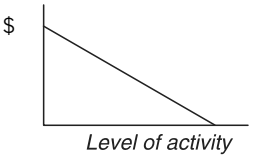
Graph 1



Graph 2



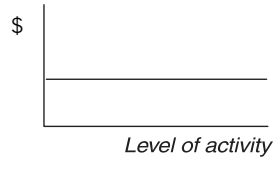
Graph 3



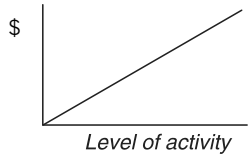
Graph 4

A Graph 1
B Graph 2
C Graph 3
D Graph 4 **(2 marks)**

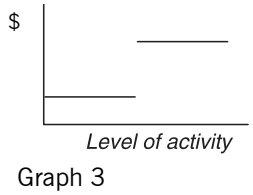
2.14 **Which of these graphs represents a fixed cost, when the vertical axis represents total cost incurred?**



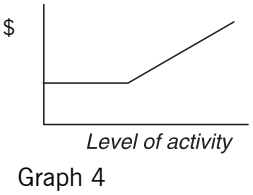
Graph 1



Graph 2



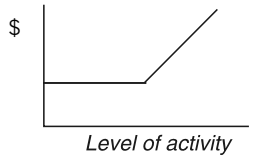
Graph 3



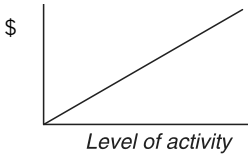
Graph 4

A Graph 1
B Graph 2
C Graph 3
D Graph 4 **(2 marks)**

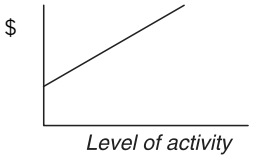
2.15 **Which of these graphs represents a semi-variable or mixed cost, when the vertical axis represents total cost incurred?**



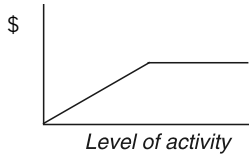
Graph 1



Graph 2



Graph 3

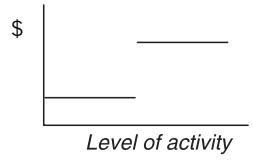


Graph 4

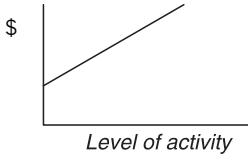
A Graphs 1, 2 and 3
B Graphs 2 and 4
C Graphs 1, 3 and 4
D All of them

(2 marks)

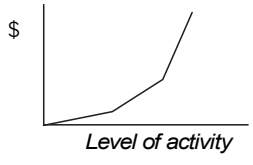
2.16 **Which of these graphs represents a stepped fixed cost, when the vertical axis represents total cost incurred?**



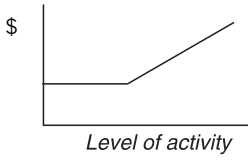
Graph 1



Graph 2



Graph 3



Graph 4

A Graph 1
B Graph 2
C Graph 3
D Graph 4

(2 marks)

2.17 Total production costs and output over three periods have been:

Period	*Production costs*	*Output*
1	$230,485	12,610 units
2	$254,554	14,870 units
3	$248,755	14,350 units

What are the estimated variable production costs per unit if the high-low method is applied (to two decimal places)?

$ ☐

(2 marks)

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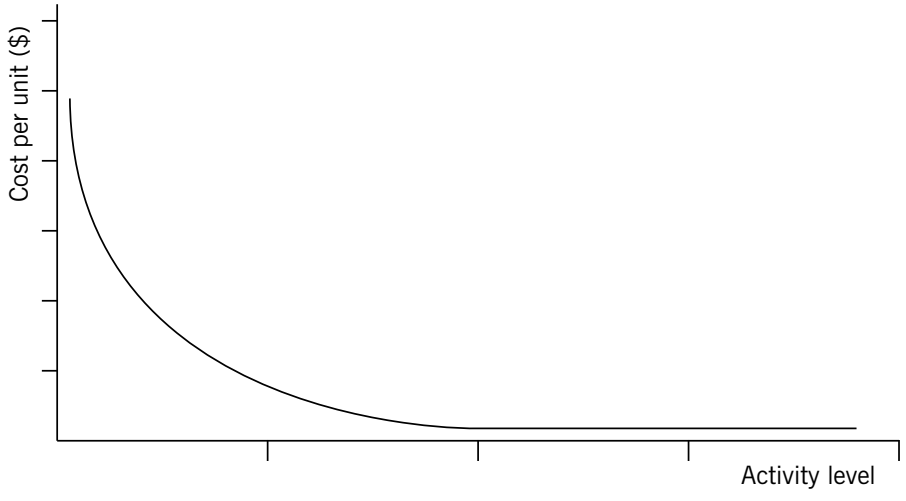
2.18 The following table details the total cost Y, a stepped-fixed cost, for different production levels of Product X.

Units of Product X	*Cost Y*
	$'000
0	100
10	100
20	100
30	150
40	150

What could have been the cause of the increase in the cost?

A Increased storage requirements
B Pay increase for direct labour
C Loss of material discounts
D Increased fuel and maintenance costs for delivery vehicles **(2 marks)**

2.19



Which description best fits the cost curve?

A Direct material cost per unit
B Fixed production cost per unit
C Direct labour cost per unit
D Variable production cost per unit **(2 marks)**

2.20 A particular cost is fixed in total for a period.

What is the effect on cost per unit of a reduction in activity of 50%?

A Cost per unit increases by 50%
B Cost per unit reduces by 50%
C Cost per unit increases by 100%
D Cost per unit is unchanged **(2 marks)**

2.21 **Which TWO of the following classifications could be applied to the cost of raw materials used by a company in the manufacture of its range of products?**

☐ Direct
☐ Fixed
☐ Period
☐ Production **(2 marks)**

2.22 A particular cost is classified as being 'semi-variable'.

If activity increases by 10%, what will happen to the cost per unit?

A Increase
B Reduce but not in proportion to the change in activity
C Reduce in proportion to the change in activity
D Remain constant **(2 marks)**

2.23 **Which of the following would be regarded as a stepped-fixed cost in the operation of a motor vehicle?**

A Hire purchase payments
B Insurance
C Petrol
D Tyre replacement **(2 marks)**

2.24 A small engineering company that makes generators specifically to customers' own designs has had to purchase some special tools for a particular job. The tools will have no further use after the work has been completed and will be scrapped.

How should the cost of these tools be treated?

A As variable production overheads
B As fixed production overheads
C As indirect expenses
D As direct expenses **(2 marks)**

2.25 **Which of the following are examples of responsibility centres within an organisation?**

(i) Profit centre
(ii) Revenue centre
(iii) Investment centre

A (i) and (ii) only
B (i), (ii) and (iii)
C (i) and (iii) only
D (ii) and (iii) only **(2 marks)**

2.26 **Which TWO of the following are cost units?**

☐ A car manufactured in a car factory
☐ A person employed as a hairdresser
☐ An accounts department at a software company
☐ A pizza sold by a pizza restaurant **(2 marks)**

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2.27 **This question has been taken from the July – December 2017 examining team report.**

The following unit costs are incurred in the production of 1,000 units of a product in a period:

	$ per unit
Variable costs	7.80
Semi-variable costs	5.30
Fixed costs	6.50
Total costs	19.60

$3.70 per unit of the semi variable costs are fixed costs.

What total costs would be incurred for production of 1,200 units of the product in a period?

A $23,520
B $21,480
C $21,900
D $21,160 **(2 marks)**

2.28 **This question has been taken from the January – June 2019 examining team report.**

The unit cost structure of a product based on producing and selling 20,000 units in a period is shown below:

	$ per unit
Variable cost	15
Fixed overheads	10
Total unit cost	25

Based on this unit cost structure, what would be the cost per unit if 25,000 units of the product are produced and sold in a period?

$ [ ] per unit **(2 marks)**

(Total = 56 marks)

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Do you know? – Information for comparison

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- Information for comparison may be ……. or ……..
- If the actual level of production is different from the original budget, then …… budgets should be used for comparison.
- Budget comparisons are popular because they show whether budget holders are achieving their ……..
- …… help budget holders to perform their function of control. The reports are especially useful if they separate controllable from non-controllable variances.
- …… highlights variances which might need investigating.
- Variances can be interdependent.
- The three revenue variances are the …… …… ……. variance, the …… variance and the …… …… variance.
- The three cost variances are the ……. ……. ……. variance, the …… variance and the ……. ……./…… ……. ……. variance.
- Revenue variances result from a difference in …… …… or a difference in ……. …….
- Cost variances result from a difference in …… …… or a difference in ……. ……./…… with which resources are used.
- Possible pitfalls.

  Write down examples of mistakes you should avoid.

Did you know? – Information for comparison

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- Information for comparison may be **financial** or **non-financial**.
- If the actual level of production is different from the original budget, then **flexed** budgets should be used for comparison.
- Budget comparisons are popular because they show whether budget holders are achieving their **targets**.
- **Variance reports** help budget holders to perform their function of control. The reports are especially useful if they separate controllable from non-controllable variances.
- **Exception reporting** highlights variances which might need investigating.
- Variances can be interdependent.
- The three revenue variances are the **total sales revenue** variance, the **activity** variance and the **selling price** variance.
- The three cost variances are the **total direct cost** variance, the **activity** variance and the **purchase price/efficiency of usage** variance.
- Revenue variances result in a difference in **quantity sold** or a difference in **selling price**.
- Cost variances result from a difference in **quantity produced** or a difference in **price paid/efficiency** with which resources are used.
- Possible pitfalls include the following *(you may have thought of others)*
  - **Not understanding the concept of flexible budgets**
  - **Not being able to identify whether variances are favourable or adverse**
  - **Not being able to identify possible causes of variances**

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3 Information for comparison 72 mins

3.1 **Which of the following terms defines a flexed budget?**

A A budget that is added to as each month passes
B A budget that has not yet been agreed
C A budget that reflects the actual activity level
D A budget that is in the process of being altered **(2 marks)**

3.2 A business has set its production budget on the basis of production of 15,000 units of its single product. The direct materials budget totals $100,500. In fact production was only 14,600 units during the period.

What is the flexed budget total for direct materials (to the nearest $)?

A $97,820
B $100,100
C $100,500
D $103,253 **(2 marks)**

3.3 A business's budgeted direct labour cost is $84,600 for the month of June 20X3. However production in June was 1,400 more than the budgeted production level of 12,500 units.

What is the flexed budget total for direct labour (to the nearest $)?

A $84,600
B $76,079
C $94,075
D $86,000 **(2 marks)**

3.4 **Which of the following provide meaningful comparisons for revenue analysis for the person receiving the information?**

1 Restaurant takings on Friday night compared to Monday night reported to the owner
2 Monthly sales revenue figures compared to the flexed budget reported to the sales director
3 Quarterly sales compared to those of the same quarter in the previous year reported to the sales director

A 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3 **(2 marks)**

3.5 **What is exception reporting?**

A Reporting of exceptional activities within an organisation
B Reporting only controllable matters to managers
C Reporting only of variances which exceed a certain amount
D Reporting of all variances to the relevant manager **(2 marks)**

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3.6 The following sales data is obtained from a business with three profit centres.

Profit centre	*December 20X6* $	*December 20X5* $	*December 20X4* $
A	185,213	172,586	175,532
B	167,232	178,353	182,268
C	113,415	110,200	105,214
	465,860	461,139	463,014

Which of the following statements is TRUE?

A December sales increased in all profit centres in 20X5 compared to 20X4.
B Profit centres A and B had increased December sales in 20X6 compared to 20X5.
C Profit centres A and B had decreased December sales in 20X5 compared to 20X4.
D Profit centres B and C had increased December sales in 20X6 compared to 20X5.

(2 marks)

3.7 **Are the following statements true or false?**

	True	*False*
Favourable variances are always good for an organisation.	○	○
Variance reporting is the comparison of the actual results with budgeted results.	○	○

(2 marks)

3.8 The budgeted direct materials cost for a product was $12.30 per unit. During the month of June production details were as follows.

Budget	5,000 units
Actual	5,300 units

The total materials cost for the month was $60,000.

What was the favourable direct materials variance comparing actual cost with the flexed budget?

$ ☐ **(2 marks)**

3.9 The budgeted direct labour cost for a product was $6.80 per unit. During the month of September production details were as follows.

Budget	12,000 units
Actual	12,200 units

The total labour cost for the month was $85,200.

What was the direct labour variance as a percentage of the budgeted figure, comparing actual with the flexed budget?

A 2.7% adverse
B 2.7% favourable
C 4.2% adverse
D 4.2% favourable **(2 marks)**

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3.10 A business has a budgeted direct materials cost per unit of $6.00. During the month of October production details were as follows.

Budget	3,400 units
Actual	3,700 units

The actual materials cost for the month was $25,600.

What was the direct materials variance as a percentage of the budgeted figure, comparing actual with the flexed budget?

A 15.3% adverse
B 15.3% favourable
C 25.5% adverse
D 25.5% favourable **(2 marks)**

3.11 A business has a budgeted direct labour cost per unit of $15.50. During the month of December production details were as follows.

Budget 12,600 units

The actual labour cost for the month was $199,400.

What was the total direct labour variance as a percentage of the budgeted figure, comparing actual with the fixed budget (to one decimal place)?

[ ]% adverse **(2 marks)**

3.12 **Which of the following is NOT an example of feedback control?**

A Responding to a price increase by a supplier by sourcing an alternative product
B Responding to an anticipated fall in sales volume by producing fewer units of a product
C Responding to a higher than expected factory rejection rate by correcting faulty factory processes
D Responding to better than expected monthly revenue by flexing the future revenue forecast

(2 marks)

3.13 Given below is a summary of a performance report showing the actual results for a month compared to the flexed budget figures. It is the company's policy that any variance which exceeds 10% of the budgeted figure should be reported and investigated.

	Actual	*Flexed budget*
	$	$
Materials	26,589	24,000
Labour	18,337	16,200
Expenses	2,172	2,100

Which variances should be investigated?

A Materials and labour only
B Materials and expenses only
C Labour and expenses only
D Materials, labour and expenses **(2 marks)**

Data for questions 3.14 to 3.16

A company sells a single product. The sales budget for a period was 8,000 units at a selling price of $2.50 per unit. 8,320 units were actually sold in the period for a total revenue of $19,968.

3.14 **What is the total sales revenue variance comparing the actual revenue against the flexed budget?**

A $32 favourable
B $832 adverse
C $32 adverse
D $832 favourable **(2 marks)**

3.15 **What is the activity variance comparing the fixed budget with the flexed budget?**

A $832 adverse
B $800 favourable
C $800 adverse
D $32 favourable **(2 marks)**

3.16 **What is the selling price variance?**

A $832 adverse
B $602 favourable
C $832 favourable
D $32 adverse **(2 marks)**

Data for questions 3.17 to 3.19

Budgeted and actual production of a product X for a period was:

Budget	60,000 units
Actual	62,400 units

The budgeted direct materials cost of the product was $3.20 per unit and a total of $187,200 was incurred on direct materials in the period.

3.17 **What is the total direct materials cost variance comparing the actual costs against the fixed budget?**

A $4,800 favourable
B $12,480 favourable
C $4,800 adverse
D $12,480 adverse **(2 marks)**

3.18 **What is the activity variance comparing the fixed budget with the flexed budget?**

A $6,000 favourable
B $7,680 adverse
C $12,480 adverse
D $4,800 favourable **(2 marks)**

3.19 **What is the direct materials variance comparing actual with the flexed budget?**

A $4,800 favourable
B $6,000 adverse
C $12,480 favourable
D $12,480 adverse **(2 marks)**

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3.20 M Co sells product L. An extract from its budget for the four-week period ended 28 October 20X1 shows that it planned to sell 500 units at a unit price of $300.

Actual sales were 521 units for the period at an average selling price of $287.

What is the adverse sales price variance (to the nearest $1)?

$ ☐ **(2 marks)**

3.21 **Which of the following would help to explain an adverse direct material price variance?**

(i) The material purchased was of a higher quality than standard.
(ii) A reduction in the level of purchases meant that expected bulk discounts were forgone.
(iii) The budgeted price per unit of direct material was unrealistically high.

A (i), (ii) and (iii)
B (i) and (ii) only
C (ii) and (iii) only
D (i) and (iii) only **(2 marks)**

3.22 **Which of the following would help to explain a favourable direct labour efficiency variance?**

(i) Employees were of a lower skill level than specified in the standard.
(ii) Better quality material was easier to process.
(iii) Suggestions for improved working methods were implemented during the period.

A (i), (ii) and (iii)
B (i) and (ii) only
C (ii) and (iii) only
D (i) and (iii) only **(2 marks)**

3.23 **Which of the following is NOT a reason for an adverse labour efficiency variance?**

A Wage rate increase
B Machine breakdown
C Non-availability of material
D Illness or injury to worker **(2 marks)**

3.24 **Which of the following would help to explain a favourable direct labour rate variance?**

(i) Employees were of a lower grade than standard.
(ii) The budgeted hourly rate of pay was set unrealistically high.
(iii) A pay increase which had been anticipated in the budget was not awarded.

A (i), (ii) and (iii)
B (i) and (ii) only
C (ii) and (iii) only
D (i) and (iii) only **(2 marks)**

3.25 The following variances are extracted from the monthly management accounts of SG Co.

Direct material total variance	$800 (A)
Direct material usage variance	$1,200 (F)

Which of the following statements are consistent with these variances?

	Consistent	*Not consistent*
High quality material was purchased, which led to a lower level of quality control rejects of completed output	○	○
Rapid inflation affected the price of the company's raw material so efforts were made to reduce the quantity of material used per unit of output	○	○

(2 marks)

3.26 **Which of the following is an example of feedforward control?**

A Variance reporting
B Foreseeing a variance and taking corrective action
C Comparing performance measurement ratios with prior periods
D Investigating the reasons behind rejected items **(2 marks)**

3.27 **Which of the following factors would affect a decision as to whether to investigate the variance?**

(i) Controllability of variance
(ii) Cost of investigation
(iii) Personnel involved
(iv) Trend of variance

A (ii) and (iv) only
B (iii) and (iv) only
C (i), (ii) and (iii)
D (i), (ii) and (iv) **(2 marks)**

3.28 The following extract is taken from the distribution cost budget of DC Co:

Volume delivered (units)	8,000	14,000
Distribution cost	$7,200	$10,500

What is the budgeted cost allowance for distribution cost for a delivery volume of 12,000 units?

A $6,600
B $9,000
C $9,400
D $10,800 **(2 marks)**

3.29 Giff Co's budgetary control report for last month is as follows:

	Fixed budget	*Flexed budget*	*Actual results*
	$	$	$
Direct costs	61,100	64,155	67,130
Production overhead	55,000	56,700	54,950
Other overhead	10,000	10,000	11,500
	126,100	130,855	133,580

What was the activity variance for last month?

A $4,755 (A)
B $2,725 (A)
C $4,755 (F)
D $2,725 (F) **(2 marks)**

3.30 Giff Co's budgetary control report for last month is as follows:

	Fixed budget	*Flexed budget*	*Actual results*
	$	$	$
Direct costs	61,100	64,155	67,130
Production overhead	55,000	56,700	54,950
Other overhead	10,000	10,000	11,500
	126,100	130,855	133,580

What was the rate/efficiency variance for last month?

A $7,480 (F)
B $2,725 (F)
C $7,480 (A)
D $2,725 (A) **(2 marks)**

(Total = 60 marks)

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Do you know? – Reporting management information

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- A key consideration in choosing a method of communication is who will be the ..... .
- Regular reports are a regular part of the ....
- ..... reports deal with a one-off issue or problem.
- Types of written communication include the following.
  - .....
  - .....
  - .....
  - .....
- Internal communications will often be sent in the form of a ..... .
- Email is a good method for sending a message efficiently to a number of recipients, but there are issues with the ..... of the information contained in the message.
- Access to ..... information will be restricted. This may be because of the Data Protection Act, or because of company policy. If you have access to restricted information, in whatever form, you are responsible for protecting it.
- Individuals or organisations that control personal data are known as .... Individuals or organisations that have personal data held on them are known as ....
- Types of non-written communication include the following.
  - .....
  - .....
  - .....
- Possible pitfalls.

  Write down examples of mistakes you should avoid.

Did you know? – Reporting management information

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- A key consideration in choosing a method of communication is who will be the **recipient**.
- Regular reports are a regular part of the **management information system**.
- **Ad-hoc** reports deal with a one-off issue or problem.
- Types of written communication include the following.
  - Letters
  - Memos
  - Emails
  - Formal reports
- Internal communications will often be sent in the form of a **memorandum**.
- Email is a good method for sending a message efficiently to a number of recipients, but there are issues with the **security** of the information contained in the message.
- Access to **confidential** information will be restricted. This may be because of the Data Protection Act, or because of company policy. If you have access to restricted information, in whatever form, you are responsible for protecting it.
- Individuals or organisations that control personal data are known as **data users**. Individuals or organisations that have personal data held on them are known as **data subjects**.
- Types of non-written communication include the following.
  - Charts
  - Graphs
  - Telephone calls
- Possible pitfalls include the following *(you may have thought of others)*
  - **Not being able to identify the most appropriate form of communication**
  - **Not understanding the different purposes of documents**
  - **Not identifying when data is confidential**
  - **Not being able to identify who the information is for**

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4 Reporting management information 36 mins

4.1 **When communicating information, which of the following combinations of factors would influence the method of communication used?**

(i) Timeliness
(ii) Confidentiality
(iii) Complexity

A (i), (ii) and (iii)
B (i) and (ii) only
C (i) and (iii) only
D (ii) and (iii) only **(2 marks)**

4.2 **If a letter is started with the greeting 'Dear Mr Smith', how should the letter be signed?**

A Yours truly
B Yours sincerely
C Yours faithfully
D Yours gratefully **(2 marks)**

4.3 **Are the following statements about the use of email as a form of communication, true or false?**

	True	*False*
Email can be used for confidential information.	○	○
Email should not be used where a signature is necessary.	○	○

(2 marks)

4.4 A complaint is to be made to a supplier about quality and reliability of the goods supplied.

Which is the most appropriate form of communication?

A Memo
B Email
C Letter
D Report **(2 marks)**

4.5 **Which of the following is the correct definition of a data subject?**

A An organisation that controls personal data
B An individual who collects and uses personal data
C Information about a living individual
D An individual who is the subject of personal data **(2 marks)**

4.6 **Which of the following is true about a line graph?**

A The independent variable is always shown on the y axis.
B The independent variable is always shown on the x axis.
C There is not necessarily a dependent variable.
D It does not matter on which axis each variable is shown. **(2 marks)**

4.7 Information about the trend in monthly sales and profit for a product for the last two years is to be presented visually.

Which would be the most appropriate method of presentation?

A Pie chart
B Bar chart
C Table
D Line graph **(2 marks)**

4.8 A visual method of presenting the percentage of total sales for the last year made by each of a company's ten divisions is required.

Which of the following would be the most appropriate method?

A Pie chart
B Bar chart
C Line graph
D Spreadsheet **(2 marks)**

4.9 A colleague asks you for your computer password in order to access supplier information which he does not normally deal with. He explains that this is in order to produce a special report for the management accountant.

What should you do?

A Give your password to your colleague
B Access the information yourself and print it out for your colleague
C Give your password to your colleague on this occasion and then change your password
D Refer the query to the management accountant **(2 marks)**

4.10 **Which of the following is NOT considered an advantage of email?**

A Secure method of communication
B Speedy delivery
C Stores messages for a given period after they are received
D Can make a single message available to many persons **(2 marks)**

4.11 **Which of the following methods of communication is most suitable for a message that a managing director wishes to send to all members of staff?**

A Letter
B Memo
C Email
D Telephone **(2 marks)**

4.12 **Which of the following is NOT an element of a formal report?**

A Terms of reference
B Index
C Conclusion
D Recommendations **(2 marks)**

4.13 **Why might an organisation NOT use a house style for its documents?**

A To make each document look completely different
B To present a consistent external image
C To make it easier to read documents
D To make it easier to produce documents **(2 marks)**

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4.14 **Which of the options below does NOT describe the following report?**

REPORT ON PROPOSED UPDATING OF COMPANY POLICY MANUAL

To: Board of Directors, EFG Co
From: A Bigman, Epsilon Management Consultants
Status: Confidential
Date: 3 November 20X2

I INTRODUCTION AND TERMS OF REFERENCE

This report details the results of an investigation commissioned by the Board on 7 October 20X2. We were asked to consider the current applicability of the company's policy manual and to propose changes needed to bring it into line with current practice.

A The report considers the future application of the company's policy manual.
B The report details the results of an investigation commissioned by the Board.
C The report details the results of an investigation conducted by an external body.
D The report considers unnecessary changes that were made to the policy manual.

(2 marks)

4.15 **This question has been taken from the July – December 2017 and the July-December 2018 examining team report**.

Which of the following statements, about presenting information, is/are TRUE?

(i) In a component column chart, the width of each column will be the same.
(ii) On a line graph, the dependent variable is on the x axis.

A Both statements (i) and (ii) are true
B Statement (i) only
C Statement (ii) only
D Neither statement (i) nor statement (ii) **(2 marks)**

(Total = 30 marks)

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Do you know? – Materials, labour and expenses

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- Materials are either a …….. or …….. cost, depending on how easily they can be traced to a specific unit of production.

  There are two approaches to purchasing inventory – …….. and …….. In practice, the most common approach is to use ……..

  Materials issued from inventory can be valued using either ……., ……. or …….. methods.

- FIFO prices materials issues at the prices of the newest/oldest items in inventory, and values closing inventory at the value of the most recent/oldest items in inventory. (Delete as appropriate.)

- LIFO prices materials issues at the prices of the newest/oldest items in inventory and values closing inventory at the value of the most recent/oldest items. (Delete as appropriate.)

- …….. is usually carried out annually, when all items of inventory are counted on a specific date. …….. involves counting and checking a number of inventory items on a regular basis so that each item is checked at least once a year.

- Inventory control levels are calculated in order to maintain inventory at the optimum level. The four critical control levels are as follows.

  – …….. (maximum usage × maximum lead time)
  – …….. (quantity of inventory to be reordered when inventory reaches reorder level)
  – …….. (reorder level – (average usage × average lead time))
  – …….. (reorder level + reorder quantity – (min usage × min lead time))

- The …….. is the ordering quantity which minimises inventory costs (holding costs and ordering costs), and is calculated as follows.

  $Q = \sqrt{\frac{2cd}{h}}$ Where h = ……..
  c = ……..
  d = ……..
  Q = ……..

- Labour attendance time is recorded on an …….. or on a …….. Job time is recorded on the following documents:

  – ……..
  – ……..
  – ……..
  – ……..

- The labour cost of work done by pieceworkers is recorded on a …….. (operation card). Piecework, time-saved bonus, discretionary bonus, group bonus scheme and profit-sharing are all different types of …….. scheme.

  …….. is a measure of the number of employees leaving/being recruited in a period of time, expressed as a percentage of the total labour force.

- Labour performance can be measured using the following ratios:
  ……
  ……
  ……
- Expenditure which results in the acquisition of non-current assets is known as **……** expenditure, and is charged to the statement of profit or loss (income statement) via a **……** over a period of time. Expenditure which is incurred for the purpose of the trade of the business or in order to maintain the existing earning capacity of non-current assets is known as **……** expenditure.

  There are two methods of calculation depreciation: …… and ……..
- In a job costing system**……** expenses are recorded by coding them to the appropriate job or client. **……** expenses are initially allocated to appropriate cost centres and then spread out or **……** to the cost centres that have benefited from the expense.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

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Did you know? – Materials, labour and expenses

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- Materials are either a **direct** or **indirect** cost, depending on how easily they can be traced to a specific unit of production.

  There are two approaches to purchasing inventory – **Just-in-time and buffer inventory**. In practice, the most common approach is to use **buffer inventory**.

  Materials issued from inventory can be valued using either **FIFO, LIFO** or **weighted average** methods.

- FIFO prices materials issues at the prices of the ~~newest~~/**oldest** items in inventory, and values closing inventory at the value of the most **recent**/~~oldest~~ items in inventory.

- LIFO prices materials issues at the prices of the **newest**/~~oldest~~ items in inventory and values closing inventory at the value of the most ~~recent~~/**oldest** items.

- **Periodic inventory taking** is usually carried out annually, when all items of inventory are counted on a specific date. **Continuous inventory taking** involves counting and checking a number of inventory items on a regular basis so that each item is checked at least once a year.

- Inventory control levels are calculated in order to maintain inventory at the optimum level. The four critical control levels are as follows.

  – **Reorder level** (maximum usage × maximum lead time)
  – **Reorder quantity** (quantity of inventory to be reordered when inventory reaches reorder level)
  – **Minimum inventory level** (reorder level – (average usage × average lead time))
  – **Maximum inventory level** (reorder level + reorder quantity – (min usage × min lead time))

- The **economic order quantity** is the ordering quantity which minimises inventory costs (holding costs and ordering costs), and is calculated as follows.

  $Q = \sqrt{\frac{2cd}{h}}$ Where h = **holding costs of one unit of inventory for one year**
  c = **cost of ordering a consignment**
  d = **annual demand**
  Q = **economic order quantity**

- Labour attendance time is recorded on an **attendance card** or on a **clock card.** Job time is recorded on the following documents.

  – **Daily time sheets**
  – **Weekly time sheets**
  – **Job cards**
  – **Route cards**

- The labour cost of work done by pieceworkers is recorded on a **piecework ticket** (operation card). Piecework, time-saved bonus, discretionary bonus, group bonus scheme and profit-sharing are all different types of **incentive** scheme.

  Labour turnover is a measure of the number of employees leaving/being recruited in a period of time, expressed as a percentage of the total labour force.

  Labour performance can be measured using the following ratios:

  Efficiency ratio

  Capacity utilisation ratio

  Production volume ratio

- Expenditure which results in the acquisition of non-current assets is known as **capital** expenditure, and is charged to the statement of profit or loss (income statement) via a **depreciation charge** over a period of time. Expenditure which is incurred for the purpose of the trade of the business or in order to maintain the existing earning capacity of non-current assets is known as **revenue** expenditure.

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There are two methods of calculation depreciation: **straight line** and **reducing balance**.

- In a job costing system **direct** expenses are recorded by coding them to the appropriate job or client. **Indirect** expenses are initially allocated to appropriate cost centres and then spread out or **apportioned** to the cost centres that have benefited from the expense.

- Possible pitfalls
  - **Confusing FIFO with LIFO**
  - **Not being able to reproduce the inventory control formulae**
  - **Confusing the meaning of 'c', 'd', and 'h' in the economic order quantity equation**

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5 Materials 70 mins

5.1 **In the context of calculating inventory control levels, what is meant by the term 'lead time'?**

A The time between raising a purchase requisition and a corresponding purchase order being completed by the purchasing department

B The time between materials inventory reaching the minimum level and a stock-out occurring

C The time between placing an order for materials and the relevant materials being received into inventory

D The time between the current date and the date at which a stock-out will occur at current levels of usage **(2 marks)**

5.2 **In the context of inventory control, what is the reorder level?**

A The amount of inventory to be ordered on each occasion that an order is placed with a supplier
B An inventory level which inventory held should not exceed
C An inventory level below which inventory held should not fall
D An inventory level at which a replenishment order should be placed **(2 marks)**

5.3 A component has a buffer inventory of 280, a reorder quantity of 1,200 and a rate of demand which varies between 100 and 400 per day.

What is the average inventory of the component?

[ ] **(2 marks)**

5.4 Fall Co wishes to minimise its inventory costs. Order costs are $10 per order and holding costs are $0.10 per unit per month. Fall Co estimates annual demand to be 5,400 units.

What is the economic order quantity?

[ ] units **(2 marks)**

Data for questions 5.5 and 5.6

The re-order level of Material M is 1,600 kg and the order quantity is 1,400 kg. Lead times and usage are as follows:

Lead time:	minimum	1 week
	average	1.5 weeks
	maximum	2 weeks
Usage:	minimum	600 kg per week
	average	700 kg per week
	maximum	800 kg per week

5.5 **What is the maximum inventory control level of Material M?**

[ ] kg **(2 marks)**

5.6 **What is the minimum inventory control level of Material M?**

A Nil
B 350 kg
C 550 kg
D 1,000 kg **(2 marks)**

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5.7 **Which of the following, relating to the management of raw materials, would result from the introduction of buffer (safety) inventory?**

(i) Holding costs per unit of inventory would increase.
(ii) The economic order quantity would decrease.
(iii) Average inventory levels would increase.
(iv) Total ordering costs would decrease.

A (iii) only
B (ii) and (iii) only
C (ii) and (iv) only
D (i), (ii), (iii) and (iv) **(2 marks)**

Data for questions 5.8 and 5.9

2,400 units of component C, valued at a price of $6 each, were in inventory on 1 March. The following receipts and issues were recorded during March.

3 March	Received	4,000 units @ $6.20 per unit
12 March	Received	2,000 units @ $6.86 per unit
23 March	Issued	5,100 units

5.8 **Using the weighted average price method of inventory valuation, what is the total value of the components remaining in inventory on 23 March?**

A $19,800
B $20,460
C $20,790
D $22,638 **(2 marks)**

5.9 **Using the FIFO method of inventory valuation, what is the total value of the components issued on 23 March?**

A $30,600
B $31,140
C $31,620
D $34,986 **(2 marks)**

5.10 **In a period of rising prices, which of the following will be true with a first in first out (FIFO) system of pricing inventory issues?**

A Product costs are overstated and profits understated.
B Product costs are overstated and profits overstated.
C Product costs are understated and profits understated.
D Product costs are understated and profits overstated. **(2 marks)**

5.11 The following information is available for part LP42.

Minimum usage per day	300 units
Average usage per day	400 units
Maximum usage per day	600 units
Lead time for replenishment	3–4 days
Reorder quantity	1,900 units

What is the maximum level of inventory?

☐ units **(2 marks)**

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5.12 A company uses the first-in, first-out (FIFO) method to price issues of raw material to production and to value its closing inventory.

Which of the following statements best describes the first-in, first-out method?

A The last materials received will be the first issued to production.
B The first materials issued will be priced at the cost of the most recently received materials.
C The last materials issued will be those that were most recently received.
D The first materials issued will be priced at the cost of the earliest goods still in inventory.

(2 marks)

5.13 A company uses two very similar types of fixing bracket, Z99 and Z100. The brackets are purchased from an outside supplier. When the company undertakes an inventory check it finds some differences as shown below.

Product	*Inventory record*	*Inventory count*
Z99	100	79
Z100	80	101

What is the most likely reason for the differences between the inventory record and the inventory count for each bracket?

A Production was higher than expected.
B Some brackets were damaged during production.
C A customer asked the company to supply some extra brackets of both types.
D Some brackets were put in the incorrect storage racks. **(2 marks)**

5.14 **If FIFO rather than LIFO was used when material prices are falling, which of the following combinations would be correct?**

	Production costs	*Profits*
A	Will be lower	Will be higher
B	Will be higher	Will be lower
C	Will be lower	Will be lower
D	Will be higher	Will be higher

(2 marks)

5.15 **Which of the following costs would be needed in order to calculate the economic order quantity?**

(i) The cost of storing materials
(ii) The cost of interest incurred in financing materials
(iii) The cost of ordering materials
(iv) The cost of insuring materials

A Items (i) and (ii) only
B Items (iii) and (iv) only
C Items (i), (iii) and (iv) only
D Items (i), (ii), (iii) and (iv) **(2 marks)**

5.16 The following are statements relating to raw material pricing in a situation where raw material prices are rising consistently.

1 Production costs will be lower using weighted average pricing rather than LIFO
2 Profit will be higher using LIFO pricing rather than FIFO
3 Inventory values will be lower using FIFO pricing rather than weighted average

Are the statements true or false?

A Statement 1 is true but Statements 2 and 3 are false
B Statements 1 and 2 are true but Statement 3 is false
C Statements 1 and 3 are true but Statement 2 is false
D Statements 2 and 3 are true but Statement 1 is false **(2 marks)**

5.17 A company orders a particular raw material in order quantities of 250 units. No safety inventory is held, the inventory holding cost is $3 per unit per annum and the annual demand is 2,500 units.

What is the total annual inventory holding cost of the material?

$ [ ] **(2 marks)**

5.18 **Which of the following is NOT relevant to the calculation of the economic order quantity of a raw material?**

A Ordering cost
B Purchase price
C Inventory holding cost
D Usage **(2 marks)**

5.19 The order quantity of a raw material is 2,000 kg. Buffer inventory of 1,200 kg is held. The inventory holding cost of the raw material is $1.20 per kg per annum.

What is the total annual inventory holding cost of the raw material?

A $1,200
B $1,920
C $2,640
D $3,840 **(2 marks)**

5.20 The following data refers to a particular inventory item:

	Receipts (units)	*Issues (units)*	*Total cost*
			$
Day 1	200		1,100
Day 3		140	
Day 6	150		840
Day 7		140	

The weighted average method is used to value inventory issues. A revised weighted average price is calculated after each purchase.

At what cost price per unit (to two decimal places of $) will the issue on Day 7 be made?

A $5.54
B $5.56
C $5.57
D $5.60 **(2 marks)**

5.21 The following statements relate to the use of different raw material pricing methods in a period of consistently rising prices.

Are the statements true or false?

	True	*False*
Production costs will be higher using Last-in First-out (LIFO) rather than First-in First-out (FIFO).	○	○
Raw material inventory values will be lower using Last-in First-out (LIFO) rather than weighted average.	○	○

(2 marks)

5.22 **Which of the following is a calculation of the minimum inventory control level (buffer inventory)?**

A Re-order level minus average usage in average lead time
B Re-order level minus maximum usage in maximum lead time
C Re-order quantity minus maximum usage in maximum lead time
D Re-order quantity plus re-order level minus minimum usage in minimum lead time

(2 marks)

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5.23 Holding costs are included in the Economic Order Quantity formula.

Which of the following are examples of holding costs?

1 Warehouse rent
2 Interest on inventory investment
3 Carriage inwards
4 Inventory theft

A 1 and 2 only
B 1 and 3
C 3 and 4
D 1, 2 and 4 **(2 marks)**

5.24 **Which of the following documents will be completed by goods inwards staff on the basis of a physical check, which involves counting the items received and seeing that they are not damaged?**

A Despatch note
B Goods received note
C Purchase requisition form
D Purchase order **(2 marks)**

5.25 **Which of the following is a typical measure of output wastage?**

A Rework costs as a percentage of production costs
B Quantity of materials wasted as a percentage of the quantity of material used
C Number of quality rejects as a percentage of total output
D Price of raw materials **(2 marks)**

5.26 **Which of the following correctly describes a perpetual inventory system?**

A Designed to ensure that inventory levels never reach zero
B Automatically orders replenishment inventory whenever an issue is made
C Records every receipt and issue of inventory as they occur
D Designed to maintain inventory holdings at a constant level **(2 marks)**

5.27 **This question has been taken from the January – June 2014 examining team report**.

A company's single product requires 1.6 litres of material M in each finished unit. 10% of material is lost in processing. Sales in Period 2 are expected to be 5,700 units and product inventory is planned to be reduced by 200 units over the period.

What is the expected usage of material M in Period 2?

A 9,778 litres
B 10,489 litres
C 10,133 litres
D 9,680 litres **(2 marks)**

5.28 **This question has been taken from the January – June 2015 examining team report**.

Which of the following statements, concerning inventory management, is TRUE?

A Setting a minimum control level seeks to ensure that a stockout does not occur

B The reorder level is the amount purchased to replenish inventory

C With a continuous stocktaking system there is a continuous record of the actual quantity of each item of inventory

D Periodic stocktaking involves frequent counts and checks on a proportion of the inventory items **(2 marks)**

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5.29 **This question has been taken from the January – June 2019 examining team report.**

Are the following variables relevant in the calculation of economic order quantity (EOQ)?

	Yes	*No*
Order costs	○	○
Finance charge on capital	○	○

(2 marks)

(Total = 58 marks)

6 Labour 41 mins

6.1 Gross wages incurred in department 1 in June were $54,000. The wages analysis shows the following summary breakdown of the gross pay.

	Paid to direct labour	*Paid to indirect labour*
	$	$
Ordinary time	25,185	11,900
Overtime: basic pay	5,440	3,500
Premium	1,360	875
Bonus payments	2,700	1,360
Sick pay	1,380	300
	36,065	17,935

What is the direct wages cost for department 1 in June?

A $25,185
B $30,625
C $34,685
D $36,065

(2 marks)

6.2 **Which of the following is NOT a cost of labour turnover?**

A The cost of recruiting new employees to replace those leaving
B The cost of increased wastage due to lack of expertise among new staff
C The contribution forgone on the output lost due to slower working
D The salary paid to the personnel manager **(2 marks)**

6.3 **Which of the following statements is/are true about group bonus schemes?**

(i) Group bonus schemes are appropriate when increased output depends on a number of people all making extra effort.

(ii) With a group bonus scheme, it is easier to reward each individual's performance.

(iii) Non-production employees can be rewarded as part of a group incentive scheme.

A (i) only
B (ii) only
C (i) and (iii) only
D (ii) and (iii) only **(2 marks)**

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6.4 Guilt Trips Co budgets to make 50,000 units of output (in eight hours each) during a budget period of 400,000 hours.

Actual output during the period was 54,000 units which took 480,000 hours to make.
Which of the following options shows the correct efficiency and capacity utilisation ratios?

	Efficiency ratio	*Capacity utilisation ratio*
A	90%	83%
B	90%	120%
C	111%	83%
D	111%	120%

(2 marks)

6.5 A manufacturing firm has temporary production problems and overtime is being worked.

The amount of overtime premium contained in the wages of direct workers would normally be classed as which of the following?

A Direct expenses
B Production overheads
C Direct labour costs
D Administrative overheads **(2 marks)**

6.6 An employee is paid $8 per piecework hour produced. In a 40 hour week he produces the following output.

	Piecework time allowed per unit
6 units of Product X	2 hours
10 units of Product Z	4 hours

What is the employee's pay for the week?

$ ☐ **(2 marks)**

6.7 A company operates a piecework scheme to pay its staff. The staff receive $0.20 for each unit produced. However, the company guarantees that every member of staff will receive at least $15 per day.

Shown below is the number of units produced by Operator A during a recent week.

Day	*Monday*	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
Units produced	90	70	75	60	90

What are Operator A's earnings for the week?

A $75
B $77
C $81
D $152 **(2 marks)**

6.8 **Which of the following statements is correct?**

A Idle time cannot be controlled because it is always due to external factors.
B Idle time is always controllable because it is due to internal factors.
C Idle time is always due to inefficient production staff.
D Idle time is not always the fault of production staff. **(2 marks)**

6.9 A company has calculated that its production volume ratio is 103.5% and that its efficiency ratio is 90%.

What is the capacity utilisation ratio?

A 86.96%
B 93.15%
C 115.00 %
D 193.50% **(2 marks)**

6.10 A company employs 20 direct production operatives and 10 indirect staff in its manufacturing department. The normal operating hours for all employees is 38 hours per week and all staff are paid $5 per hour. Overtime working is to satisfy general production requirements and hours are paid at the basic rate plus 50%.

During a particular week all employees worked for 44 hours.

What amount would be charged to production overhead?

A $2,650
B $2,350
C $450
D $300 **(2 marks)**

6.11 A company planned to produce 5,000 units of Product X (in 0.65 hours each) and set a time budget of 3,250 hours. The actual output for the month was 5,316 units, which was actually achieved in 3,570 hours.

What was the production volume ratio (to two decimal places)?

[ ] % **(2 marks)**

6.12 Employee A works a normal working week of 36 hours at a basic rate of $3.60 per hour. A premium of 50% of the basic hourly rate is paid for all hours worked in excess of 36 hours per week. Employee A worked for a total of 42 hours last week. The reasons for the overtime were:

Machine breakdown	4 hours
Completion of a special job at the request of a customer	2 hours

How much of Employee A's earnings for the last week should be treated as direct wages?

A $129.60
B $140.40
C $151.20
D $162.00 **(2 marks)**

6.13 A company pays direct operatives a basic wage of $8.50 per hour plus a productivity bonus. The bonus is calculated as:

[(time allowed – time taken) × (basic rate per hour ÷ 3)]

The time allowed is 2.4 minutes per unit of output. An operative produced 1,065 units in a 37.5 hour week.

What were the total earnings of the operative in the week?

A $318.75
B $333.20
C $340.40
D $362.10 **(2 marks)**

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6.14 **Which TWO of the following items will usually be treated as production overhead costs?**

- ☐ Training of direct operatives
- ☐ Wages of distribution staff
- ☐ Normal idle time in the factory
- ☐ Productive time of direct operatives
- ☐ Sales personnel salaries

(2 marks)

6.15 A differential piecework scheme has a basic rate of $0.50 per unit. Output in addition to 500 units is paid at higher rates. The premiums over and above the basic rate, which apply only to additional units over the previous threshold, are:

Output (units)	*Premium (per unit)*
501–600	$0.05
above 600	$0.10

What is the total amount paid if output is 620 units?

A $317
B $318
C $322
D $372

(2 marks)

6.16 Labour costs may include:

(i) Overtime hours of direct operatives at basic rate
(ii) Overtime premiums of factory employees
(iii) Productive time of direct operatives
(iv) Training of direct operatives

Which of the above items will usually be treated as a direct cost?

A (i) and (ii) only
B (i) and (iii) only
C (ii) and (iv) only
D (i), (iii) and (iv)

(2 marks)

6.17 **This question has been taken from the January – June 2016 examining team report**.

Labour costs for a period in production cost centre P1 include:

Gross wages of direct operatives:	
Normal hours	$76,200
Overtime hours (at a premium of 50% over the normal rate)	$9,600
Gross wages of indirect operatives:	
Normal hours	$14,300
Overtime hours (at a premium of 50% over the normal rate	$1,650

Overtime is worked, as necessary to satisfy seasonal sales demand.

What amount of the above labour costs would be charged to production overhead?

A $19,150
B $15,950
C $25,550
D $15,400

(2 marks)

(Total = 34 marks)

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7 Expenses 24 mins

7.1 **Which TWO of the following are examples of capital expenditure?**

- ☐ Purchase of a building
- ☐ Extension to a building
- ☐ Fixing broken windows
- ☐ Replacing missing roof tiles

(2 marks)

7.2 During 20X0, Joe Co bought new machinery for $40,000 and built an extension on its head office at a cost of $20,000. Machinery was maintained at a cost of $4,000 during the year and the head office was repainted at a cost of $5,000.

What was Joe Co's capital expenditure in 20X0?

A $40,000
B $60,000
C $64,000
D $69,000

(2 marks)

7.3 New England plc purchases an asset for $25,000 which is depreciated over four years using the straight line method. Assume a residual value of $5,000 after four years.

What is the net book value of the asset after three years?

A $5,000
B $10,000
C $15,000
D $20,000

(2 marks)

7.4 **Which of the following statements about capital expenditure is correct?**

(i) Capital expenditure is charged to the statement of profit or loss (income statement) at the end of an accounting period.

(ii) Capital expenditure is shown as an asset on the statement of financial position.

(iii) Non-current assets are assets acquired to provide benefit in more than one accounting period.

(iv) The annual depreciation charge is shown as a liability in the statement of financial position.

A (i) and (ii)
B (i) and (iii)
C (ii) and (iii)
D (i) and (iv)

(2 marks)

7.5 **Which TWO of the following are items of revenue expenditure?**

- ☐ Administration expenses
- ☐ Plant maintenance costs
- ☐ Purchase of a new factory
- ☐ Purchase of managing director's second-hand company car

(2 marks)

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7.6 A machine costs $200,000 and it is estimated that it will be sold as scrap for $10,000 at the end of its useful life. Such machines have been seen to run for approximately 40,000 hours before they wear out. The machine hour method of depreciation is used.

If the machine was used for 3,000 hours in year 1, what is the depreciation charge for the year (to the nearest $)?

A $14,250
B $15,000
C $63,333
D $66,667 **(2 marks)**

7.7 A new machine has an estimated five year life and a nil disposal value at the end of its life. Depreciation methods being considered are:

(i) Reducing balance at 25% per annum
(ii) Straight line

Which of the following statements is correct?

A Depreciation in each year would be greater using the reducing balance method.

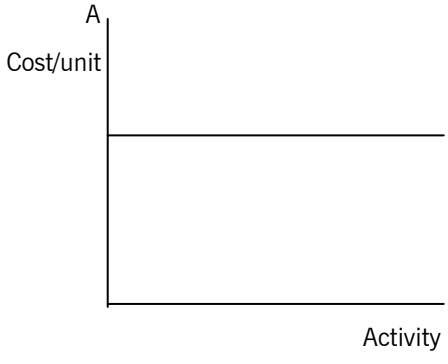
B Depreciation in each year would be greater using the straight-line method.

C Depreciation would be greater in year 1 but less in year 5 if the reducing balance method, rather than the straight line method, was used.

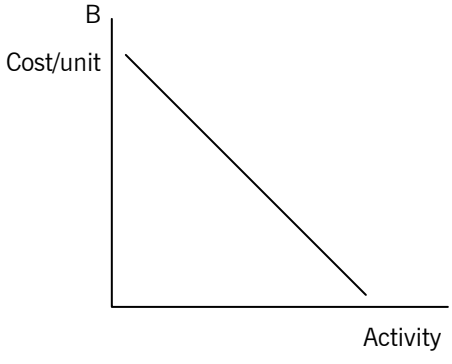
D Depreciation would be greater in year 1 but less in year 5 if the straight line method, rather than the reducing balance method, was used. **(2 marks)**

7.8 **Which chart shows the unit cost behaviour of straight-line depreciation costs?**

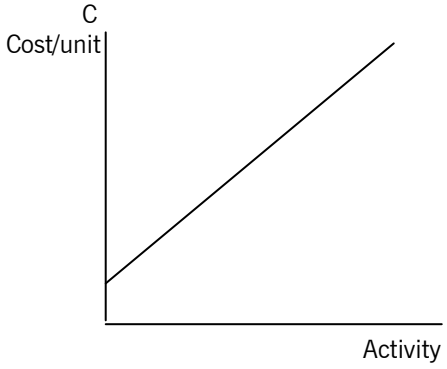
A



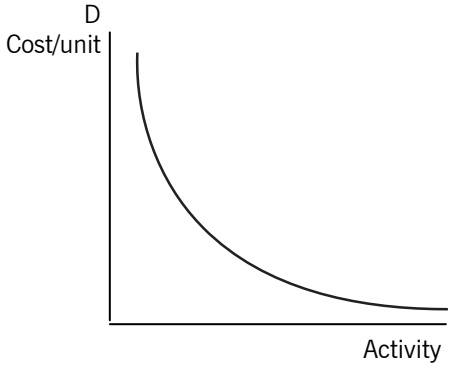
B



C



D



A Chart A
B Chart B
C Chart C
D Chart D **(2 marks)**

7.9 A non-current asset has an expected life of ten years with a nil residual value. The asset is due to be depreciated using the straight-line method.

Which of the following statements is correct regarding the use of the reducing balance method instead?

A Depreciation will be higher in each year using 20% reducing balance.
B Depreciation will be higher in each of years 1, 2 and 3 using 25% reducing balance.
C Depreciation will be lower in each year using 15% reducing balance.
D Depreciation will be lower in year 2 using 18% reducing balance. **(2 marks)**

7.10 The following statements relate to depreciation:

1 Using the reducing balance method, product unit costs decline from year to year if output stays the same.

2 Using the straight-line method, product unit costs decline as output increases.

Are the statements TRUE or FALSE?

	Statement 1	*Statement 2*	
A	True	True	
B	False	False	
C	True	False	
D	False	True	**(2 marks)**

(Total = 20 marks)

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Do you know? – Absorption costing and marginal costing

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- Costs incurred during production or while providing a service that cannot be traced directly and in full to the product or service are known as ……., and the four main types of ……. are production, administration, ……. and distribution.
- The three stages of calculating the costs of overheads to be charged to manufactured output are as follows: …… ; …… ; and ……
- The procedure whereby indirect costs (overheads) are spread fairly between cost centres is known as …… Service cost centres may be apportioned to production cost centres by the …… method or by the ……. method of reapportionment.
- Overheads are absorbed into cost unites using a ……

  ……
  (calculated by dividing budgeted overhead by budgeted level of activity).
- Under and over absorption of overhead occurs when actual overhead incurred is different to absorbed overhead. ……-absorbed overhead occurs when actual overhead is less than absorbed overhead, and therefore too ……. overhead has been charged to production. ……-absorbed overhead occurs when actual overhead is greater than absorbed overhead, and therefore too ……. overhead has been charged to production. Under or over absorption of overheads occurs because the predetermined overhead absorption rates are based on forecasts (estimates).
- Marginal cost is the ……. cost of one unit of product or service. ……. is the difference between the sales value and the marginal cost of one unit of product or service.
- In marginal costing, fixed production costs are treated as …… costs and are written off as they are incurred. In absorption costing fixed production costs are …….. the cost of units and are carried forward in inventory to be charged against the sales revenue for the next period. Inventory values using absorption costing are therefore …… than those calculated using marginal costing.
- Marginal costing and absorption costing will report different profit figures if there is any change in the volume of inventory during the period. If closing inventory is greater than opening inventory, absorption costing will report a …… profit than marginal costing. If opening inventory is greater than closing inventory (ie inventory levels ……), then absorption costing will report a ……. profit than marginal costing.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

Did you know? – Absorption costing and marginal costing

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- Costs incurred during production or while providing a service that cannot be traced directly and in full to the product or service are known as **overheads**, and the four main types of **overhead** are production, administration, **selling** and distribution.
- The three stages of calculating the costs of overheads to be charged to manufactured output are as follows: **allocation**; **apportionment**; and **absorption**.
- The procedure whereby indirect costs (overheads) are spread fairly between cost centres is known as **apportionment**. Service cost centres may be apportioned to production cost centres by the **direct** method or by the **step down** method of reapportionment.
- Overheads are absorbed into cost units using a **predetermined overhead absorption rate** (calculated by dividing budgeted overhead by budgeted level of activity).
- Under and over absorption of overhead occurs when actual overhead incurred is different to absorbed overhead. **Over**-absorbed overhead occurs when actual overhead is less than absorbed overhead, and therefore too **much** overhead has been charged to production. **Under**-absorbed overhead occurs when actual overhead is greater than absorbed overhead, and therefore too **little** overhead has been charged to production. Under or overabsorption of overheads occurs because the predetermined overhead absorption rates are based on forecasts (estimates).
- Marginal cost is the **variable** cost of one unit of product or service. **Contribution** is the difference between the sales value and the marginal cost of one unit of product or service.
- In marginal costing, fixed production costs are treated as **period** costs and are written off as they are incurred. In absorption costing fixed production costs are **absorbed into** the cost of units and are carried forward in inventory to be charged against the sales revenue for the next period. Inventory values using absorption costing are therefore **greater** than those calculated using marginal costing.
- Marginal costing and absorption costing will report different profit figures if there is any change in the volume of inventory during the period. If closing inventory is greater than opening inventory, absorption costing will report a **higher** profit than marginal costing. If opening inventory is greater than closing inventory (ie inventory levels **decrease**), then absorption costing will report a **lower** profit than marginal costing.
- Possible pitfalls
  - **Including an element of fixed overheads in the inventory valuation in marginal costing statements**
  - **Selecting inappropriate bases when calculating overhead absorption rates**
  - **Confusing under recovery and over recovery of overheads**

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8 Overheads and absorption costing 53 mins

8.1 A method of sharing overhead costs involves spreading common costs over cost centres on the basis of benefit received.

What is this known as?

A Overhead absorption
B Overhead apportionment
C Overhead allocation
D Overhead analysis **(2 marks)**

8.2 The following extract of information is available concerning the four cost centres of EG Co.

	Production cost centres			*Service cost centre*
	Machinery	*Finishing*	*Packing*	*Canteen*
Number of direct employees	7	6	2	–
Number of indirect employees	3	2	1	4
Overhead allocated and apportioned	$28,500	$18,300	$8,960	$8,400

The overhead cost of the canteen is to be re-apportioned to the production cost centres on the basis of the number of employees in each production cost centre.

After the re-apportionment, what is the total overhead cost of the packing department, to the nearest $?

A $1,200
B $9,968
C $10,080
D $10,160 **(2 marks)**

8.3 **Which of the following bases of apportionment would be MOST appropriate for apportioning heating costs to production cost centres?**

A Floor space occupied (square metres)
B Volume of space occupied (cubic metres)
C Number of employees
D Labour hours worked **(2 marks)**

8.4 **When do over-absorbed overheads occur?**

A When absorbed overheads exceed actual overheads
B When absorbed overheads exceed budgeted overheads
C When actual overheads exceed budgeted overheads
D When budgeted overheads exceed absorbed overheads **(2 marks)**

8.5 The production overhead of department D is absorbed using a machine hour rate. Budgeted production overheads for the department were $280,000 and the actual machine hours were 70,000. Production overheads were under absorbed by $9,400.

If actual production overheads were $295,000 what was the overhead absorption rate per machine hour (in $s to the nearest cent)?

$ ☐ **(2 marks)**

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8.6 **What is overhead apportionment used for?**

A To charge whole items of costs to cost centres
B To charge cost units with an appropriate share of overheads
C To charge whole items of costs to cost units
D To spread common costs over cost centres **(2 marks)**

8.7 A vehicle repair company recovers overheads on the basis of labour hours. Budgeted overheads were $615,000 and actual labour hours were 48,225. Overheads were over absorbed by $35,000.

If actual overheads were $640,150, what was the budgeted overhead absorption rate per hour (to two decimal places)?

$ ☐ **(2 marks)**

8.8

Actual overheads	$496,980
Actual machine hours	16,566
Budgeted overheads	$475,200

Based on the data above, and assuming that the budgeted overhead absorption rate was $32 per hour, what were the number of machine hours (to the nearest hour) budgeted to be worked?

A 14,850
B 15,531
C 16,566
D 30,381 **(2 marks)**

8.9 **What is an overhead absorption rate used for?**

A To share out common costs over benefiting cost centres
B To find the total overheads for a cost centre
C To charge overheads to products
D To control overheads **(2 marks)**

8.10 A company has two production departments, Cutting and Finishing. The budgeted overheads and operating hours for the two departments for next year are:

Cutting	$210,000	60,000 machine hours	4,000 labour hours
Finishing	$200,000	5,000 machine hours	14,000 labour hours

From the information given, what should the pre-determined overhead absorption rates for the departments be based on?

A Both be based on machine hours

B Both be based on labour hours

C Be based on machine hours for the cutting department and labour hours for the finishing department

D Be based on labour hours for the cutting department and machine hours for the finishing department **(2 marks)**

8.11 A company absorbs production overheads using a machine hour basis.

In order to calculate any over or under absorbed overheads which of the following would be needed, in addition to the pre-determined machine hour rate?

A Budgeted overheads and actual overheads incurred
B Budgeted overheads and actual hours worked
C Actual overheads incurred and budgeted hours
D Actual overheads incurred and actual hours worked **(2 marks)**

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8.12 A company had the following budgeted and actual production overhead costs in its two production cost centres, Machining and Assembly.

	Budget	*Actual*
Machining	$210,000	$212,000
Assembly	$136,000	$134,000

Which of the following statements is true?

A From the data available it is not possible to determine over/under absorption.
B Machining overheads were over-absorbed: Assembly overheads were under-absorbed.
C Machining overheads were over-absorbed: Assembly overheads were over-absorbed.
D Machining overheads were under-absorbed: Assembly overheads were over-absorbed.

(2 marks)

8.13 The following production overhead costs relate to a production cost centre:

Budget	$124,000
Actual	$126,740
Absorbed	$125,200

Which of the following statements is true?

A Overheads were over-absorbed by $1,200.
B Overheads were over-absorbed by $1,540.
C Overheads were under-absorbed by $1,200.
D Overheads were under-absorbed by $1,540. **(2 marks)**

8.14 There are two production cost centres and two service cost centres in a factory. Production overheads have been allocated and apportioned to cost centres and now require re-apportionment from service cost centres to production cost centres. Relevant details are:

	Service cost Centre A	*Service cost Centre B*
Total overhead	$42,000	$57,600
% to production cost centre X	40	55
% to production cost centre Y	60	45

What is the total re-apportionment to production cost centre Y?

A $42,720
B $48,480
C $51,120
D $56,880 **(2 marks)**

8.15 Overheads are absorbed at a pre-determined rate based on direct labour hours. The following additional information is available for a period:

Budget	$164,000 overhead expenditure	10,000 direct labour hours
Actual	$158,000 overhead expenditure	9,800 direct labour hours

What was the overhead over-/under-absorption in the period?

A $2,720 over-absorbed
B $3,224 over-absorbed
C $3,280 under-absorbed
D $6,000 under-absorbed **(2 marks)**

8.16 A company uses absorption costing. In a period, 34,000 units of the company's single product were manufactured and 33,000 units were sold.

Consider the following two statements:

1 Fixed production overheads would be over-absorbed.
2 Profit would be higher than in the previous period.

Are the statements true in relation to the situation described or is it not possible to determine whether or not they are true?

	Statement 1	*Statement 2*
A	Cannot determine	Cannot determine
B	Cannot determine	True
C	True	Cannot determine
D	True	True

(2 marks)

8.17 **Which of the following would be the most appropriate basis for reapportioning the cost of personnel services in a factory?**

A Floor space occupied
B Hours worked by direct operatives
C Number of direct operatives
D Number of employees **(2 marks)**

8.18 **Which TWO of the following are practical reasons for using absorption costing?**

☐ To establish the profitability of different products
☐ To aid decision making by focusing on product contribution
☐ To encourage management to produce more goods in order to absorb all allocated overheads
☐ To value closing inventory in accordance with accounting principles **(2 marks)**

8.19 A product has direct material costs of $7.15 per unit and direct labour costs of $8.25 per unit. Each unit spends 3 machine hours in the assembly cost centre and 1.5 labour hours in the finishing cost centre. Production overhead absorption rates are as follows.

Assembly $1.76 per machine hour
Finishing $3.28 per labour hour

What is the full production cost per unit (to two decimal places)?

$ ☐ **(2 marks)**

8.20 **This question has been taken from the January – June 2014 examining team report.**

What will result in over-absorption of fixed production overhead using an absorption rate based on direct labour hours?

A Actual expenditure below budget and actual direct labour hours above budget

B Actual expenditure above budget and actual direct labour hours below budget

C Absorption based on actual expenditure and actual direct labour hours both of which are above budget

D Absorption based on actual expenditure and actual direct labour hours both of which are below budget **(2 marks)**

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8.21 **This question has been taken from the January – June 2015 examining team report**.

Direct labour hours are used as the basis for overhead absorption in a production cost centre. The following data are available for a period:

Actual direct labour hours worked	9,760
Actual overheads incurred	$86,920
Overhead under-absorption	$2,496
Budgeted direct labour hours	10,000

What was the overhead absorption rate in the period?

A $8.65
B $9.16
C $8.44
D $8.94 **(2 marks)**

8.22 **This question has been taken from the July – December 2017 examining team report**.

Direct labour hours are used to absorb overhead in production cost centre PCC6. Data for a period, relating to the cost centre, include:

Budgeted overhead expenditure	$73,820
Actual overhead expenditure	$74,960
Actual direct labour	3,670 hours
Overhead under-absorbed	$3,762

What were the budgeted direct labour hours for the period?

A 3,430
B 3,614
C 3,805
D 3,441 **(2 marks)**

(Total = 44 marks)

9 Marginal costing and absorption costing 29 mins

9.1 The fixed production overhead absorption rate for product Y is $2.50 per direct labour hour. Each unit of Y requires 3 direct labour hours. Inventory of product Y at the beginning of the month was 200 units and at the end of the month was 250 units.

What is the difference in the profits reported for the month using absorption costing compared with marginal costing?

A The absorption costing profit would be $375 less.
B The absorption costing profit would be $125 greater.
C The absorption costing profit would be $375 greater.
D The absorption costing profit would be $1,875 greater. **(2 marks)**

9.2 A company produces a single product for which cost and selling price details are as follows.

	$ per unit	*$ per unit*
Selling price		28
Direct material	10	
Direct labour	4	
Variable production overhead	2	
Fixed production overhead	5	
		21
Profit per unit		7

Last period, 8,000 units were produced and 8,500 units were sold. The opening inventory was 3,000 units and profits reported using marginal costing were $60,000.

What profits would be reported using an absorption costing system?

A $47,500
B $57,500
C $59,500
D $62,500 **(2 marks)**

9.3 A company had opening inventory of 48,500 units and closing inventory of 45,500 units. Profits based on marginal costing were $315,250 and on absorption costing were $288,250.

What is the fixed production overhead absorption rate per unit (to two decimal places)?

$ ☐ **(2 marks)**

9.4 **Which of the following are arguments in favour of the use of absorption costing?**

(i) Closing inventory is valued in accordance with accounting standards.

(ii) There is no under or over absorption of overheads.

(iii) When sales fluctuate but production is constant, absorption costing smoothes out profit fluctuations.

A (i) only
B (i) and (ii)
C (i) and (iii)
D (ii) and (iii) **(2 marks)**

9.5 A company currently uses absorption costing. The following information relates to Product X for Month 1:

Opening inventory	Nil
Production	900 units
Sales	800 units

If the company had used marginal costing, which of the following combinations would be true?

	Profit	*Inventory valuation*
A	Would be higher	Would be higher
B	Would be higher	Would be lower
C	Would be lower	Would be higher
D	Would be lower	Would be lower **(2 marks)**

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9.6 When opening inventory was 8,500 litres and closing inventory 6,750 litres, a firm had a profit of $27,400 using marginal costing.

Assuming that the fixed production overhead absorption rate was $2 per litre, what profit would be reported using absorption costing?

A $30,900
B $30,500
C $23,900
D $27,400 **(2 marks)**

9.7 **Are the following statements about marginal costing true or false?**

	True	*False*
Inventory value will always be lower than when using absorption costing.	○	○
Profit will always be higher than when using absorption costing.	○	○

(2 marks)

9.8 A company manufactures a single product. Production and sales quantities for a period were:

	Production	*Sales*
Budget	100,000 units	102,000 units
Actual	97,000 units	96,000 units

The fixed production overhead absorption rate is $1.40 per unit.

If marginal costing had been used instead of absorption costing how would the profit for the period have differed?

A $1,400 less using marginal costing
B $1,400 more using marginal costing
C $4,200 less using marginal costing
D $4,200 more using marginal costing **(2 marks)**

9.9 A company sold 82,000 units of its single product in a period in which 84,000 units were manufactured.

Consider the following statements:

1 Inventory value at the end of the period would be higher than at the beginning of the period.

2 Inventory values both at the beginning and at the end of the period would be higher using absorption rather than marginal costing.

Are the statements true or false in relation to the situation described?

	Statement 1	*Statement 2*
A	False	False
B	False	True
C	True	False
D	True	True

(2 marks)

9.10 **What distinguishes absorption costing from marginal costing?**

A Product costs include both prime cost and variable production overhead.
B Product costs include both production and non-production costs.
C Inventory valuation includes a share of all production costs.
D Inventory valuation includes a share of all costs. **(2 marks)**

9.11 A company uses a marginal costing system. 10,000 units of its single product were manufactured in a period during which 9,760 units were sold.

If absorption costing is applied instead what would be the effect on profit?

A Higher by (240 units × fixed production overhead cost per unit)

B Lower by (240 units × fixed production overhead cost per unit)

C Higher by [240 units × (fixed production overhead cost per unit + fixed non-production overhead cost per unit)]

D Lower by [240 units × (fixed production overhead cost per unit + fixed non-production overhead cost per unit)] **(2 marks)**

9.12 **Which of the following correctly describes the concept of contribution?**

A It is the cost of a unit of product or service which would be avoided if that unit were not produced or provided.

B It is the difference between sales value and the marginal cost of sales.

C It is the difference in units between the expected sales volume and the breakeven sales volume.

D It is the difference between total sales revenue and total fixed costs. **(2 marks)**

(Total = 24 marks)

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Do you know? – Cost bookkeeping

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- ..... are a detailed breakdown of the information contained in the purchases account, wages and salaries account and all the expense accounts in the nominal ledger.
- An account which controls total cost is called a ..... account. For example, the wages ..... account acts as a sort of ..... for net wages paid and deductions made from gross pay. The gross pay is then usually analysed between ..... wages (which will be debited to the..... account) and ..... wages (which will be debited to the ..... account).
- There are two types of cost bookkeeping system, the interlocking and the integrated.
  - The ..... accounts system combines the financial and cost accounts in one set of self-balancing ledger accounts.
  - An interlocking system features two ledgers: the ..... ledger; and the ..... ledger.
- The accounting entries for the various stages of the production process are as follows.
  - Resources (direct materials, labour and expenses) are allocated to work in progress by ..... the work in progress control account and ..... the resource accounts.
  - Indirect materials, labour and expenses (overheads incurred) are allocated to production by ..... the raw materials and wages and salaries accounts and ..... the production overhead account.
  - The amount of overhead absorbed is ..... to the work in progress account, and ..... to the production overhead account. Overheads incurred have therefore been ..... to the production overhead account, and those absorbed have been ..... to the production overhead account. If overhead absorbed differs from overhead incurred, the difference is written off to an ..... account.
  - The production of finished goods is recorded by debiting the ..... account and crediting the .....account.
  - The cost of goods sold is established by transferring the balance on the ..... account to the cost of sales account.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

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Did you know? – Cost bookkeeping

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- **Cost records** are a detailed breakdown of the information contained in the purchases account, wages and salaries account and all the expense accounts in the nominal ledger.
- An account which controls total cost is called a **control** account. For example, the wages **control** account acts as a sort of **collecting place** for net wages paid and deductions made from gross pay. The gross pay is then usually analysed between **direct** wages (which will be debited to the **work in progress** account) and **indirect** wages (which will be debited to the **production overhead** account).
- There are two types of cost bookkeeping system, the interlocking and the integrated.
  - The **integrated** accounts system combines the financial and cost accounts in one set of self-balancing ledger accounts.
  - An interlocking system features two ledgers: the **financial** ledger; and the **cost** ledger**.**
- The accounting entries for the various stages of the production process are as follows.
  - Resources (direct materials, labour and expenses) are allocated to work in progress by **debiting** the work in progress control account and **crediting** the resource accounts.
  - Indirect materials, labour and expenses (overheads incurred) are allocated to production by **crediting** the raw materials and wages and salaries accounts and **debiting** the production overhead account.
  - The amount of overhead absorbed is **debited** to the work in progress account, and **credited** to the production overhead account. Overheads incurred have therefore been **debited** to the production overhead account, and those absorbed have been **credited** to the production overhead account. If overhead absorbed differs from overhead incurred, the difference is written off to an **under-/over-absorbed overhead** account.
  - The production of finished goods is recorded by debiting the **finished goods** account and crediting the **work in progress** account.
  - The cost of goods sold is established by transferring the balance on the **finished goods** account to the cost of sales account.
- Possible pitfalls
  - **Posting debit entries as credits and credit entries as debits when completing interlocking and integrated accounts**
  - **Being unable to identify which costs are indirect costs when preparing the production overhead account**

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10 Cost bookkeeping 29 mins

10.1 A company's accounting system operates so that the cost accounts are independent of the financial accounts. The two sets of accounts are reconciled on a regular basis to keep them continuously in agreement.

Which of the following terms correctly describes this accounting system?

A Independent accounts
B Interlocking accounts
C Reconciled accounts
D Integrated accounts **(2 marks)**

10.2 **Are the following statements about integrated accounts true or false?**

	True	*False*
Integrated systems save time and administrative effort.	○	○
Integrated systems avoid the need for periodic profit reconciliations.	○	○

(2 marks)

10.3 The wages control account for A Co for February is shown below.

WAGES CONTROL ACCOUNT

	$		$
Bank	128,400	Work in progress control	79,400
Balance c/d	12,000	Production overhead control	61,000
	140,400		140,400
		Balance b/d	12,000

Which of the following statements about wages for February is NOT correct?

A Wages paid during February amounted to $128,400.
B Wages for February were prepaid by $12,000.
C Direct wages cost incurred during February amounted to $79,400.
D Indirect wages cost incurred during February amounted to $61,000. **(2 marks)**

10.4 The material stores control account for J Co for March looks like this:

MATERIAL STORES CONTROL ACCOUNT

	$		$
Balance b/d	12,000	Work in progress	40,000
Payables	49,000	Overhead control	12,000
Work in progress	18,000	Balance c/d	27,000
	79,000		79,000
Balance b/d	27,000		

Which of the following statements are correct?

(i) Issues of direct materials during March were $18,000.
(ii) Issues of direct materials during March were $40,000.
(iii) Issues of indirect materials during March were $12,000.
(iv) Purchases of materials during March were $49,000.

A (i) only
B (ii) and (iv) only
C (ii), (iii) and (iv) only
D (i), (ii), (iii) and (iv) **(2 marks)**

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10.5 **What is a record of total actual expenditure incurred on indirect costs and the amount absorbed into individual units, jobs or processes known as?**

A Stores control account
B Wages control account
C Work in progress control account
D Production overhead control account **(2 marks)**

10.6 **In an integrated cost and financial accounting system, what would the accounting entries at the end of the period for production overhead over-absorbed be?**

	Debit	*Credit*
A	Overhead control account	Work in progress account
B	Overhead control account	Statement of profit or loss (income statement)
C	Work in progress account	Overhead control account
D	Statement of profit or loss (income statement)	Overhead control account

(2 marks)

10.7 **In a typical cost ledger, what is the correct double entry for indirect labour cost incurred?**

	Debit	*Credit*
A	Wages control	Overhead control
B	WIP control	Wages control
C	Overhead control	Wages control
D	Wages control	WIP control

(2 marks)

10.8 **Which of the following may be included in the cost accounts but excluded from the financial accounts?**

A Depreciation of equipment
B Distribution expenses
C Factory manager's salary
D Notional rent **(2 marks)**

10.9 **In an interlocking system, what would be the entry for the issue of indirect material from inventory?**

	Account debited	*Account credited*
A	Material inventory	Production overhead
B	Material inventory	Work-in-progress
C	Production overhead	Material inventory
D	Work-in-progress	Material inventory

(2 marks)

10.10 **In a cost bookkeeping system what would be the entry for the absorption of production overhead?**

	Debit	*Credit*
A	Cost Ledger Control Account	Production Overhead Account
B	Production Overhead Account	Work-in-Progress Account
C	Work-in-Progress Account	Cost Ledger Control Account
D	Work-in-Progress Account	Production Overhead Account

(2 marks)

10.11 **In an integrated cost and financial accounting system what would be the entry to record direct labour costs being charged to production?**

	Debit	*Credit*
A	Financial ledger control	Work-in-progress
B	Production overhead	Wages control
C	Finished goods	Work-in-progress
D	Work-in-progress	Wages control

(2 marks)

10.12 **In an interlocking accounting system what is the double-entry in the cost accounts to record the purchase of raw materials on credit?**

	Debit	*Credit*
A	Cost ledger control	Materials control
B	Materials control	Cost ledger control
C	Materials control	Payables
D	Payables	Materials control

(2 marks)

(Total = 24 marks)

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Do you know? – Job, batch and service costing

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- Job costing is the costing method used where each cost unit is separately identifiable. Costs for each job are collected on a …….. or ……. Overhead is absorbed into the cost of jobs using the …….. rate.
- The usual method of fixing prices within a jobbing concern is ……..
- Batch costing is similar to job costing in that each batch of similar articles is separately identifiable. The cost per unit manufactured in a batch is calculated by dividing the …….. by the …….. in the batch.
- Service costing is used by companies operating in a service industry or by companies wishing to establish the cost of services carried out by some of their departments.
- Characteristics of services
  - ……
  - ……
  - ……
  - ……
- If a service is a function of two activity variables, a ……. cost unit might be appropriate.
- A difficulty with service costing is the selection of an appropriate cost unit. The cost per unit is calculated by dividing the …….. for the period by the …….. in the period.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

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Did you know? – Job, batch and service costing

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- Job costing is the costing method used where each cost unit is separately identifiable. Costs for each job are collected on a **job cost sheet** or **job card**. Overhead is absorbed into the cost of jobs using the **predetermined overhead absorption** rate.
- The usual method of fixing prices within a jobbing concern is **cost plus pricing**.
- Batch costing is similar to job costing in that each batch of similar articles is separately identifiable. The cost per unit manufactured in a batch is calculated by dividing the **total batch cost** by the **number of units** in the batch.
- Service costing is used by companies operating in a service industry or by companies wishing to establish the cost of services carried out by some of their departments.
- Characteristics of services
  - **Intangibility**
  - **Simultaneity**
  - **Perishability**
  - **Heterogeneity**
- If a service is a function of two activity variables, a **composite** cost unit might be appropriate.
- A difficulty with service costing is the selection of an appropriate cost unit. The cost per unit is calculated by dividing the **total costs** for the period by the **number of service units** in the period.
- Possible pitfalls
  - **Posting amounts brought forward (plant on site, materials on site, work in progress) as credits instead of debits in the contract account**
  - **Forgetting to include accruals brought forward and carried forward in the contract account**

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11 Job, batch and service costing 31 mins

11.1 **Which of the following is a feature of job costing?**

A Production is carried out in accordance with the wishes of the customer
B Associated with continuous production of large volumes of low-cost items
C Establishes the cost of services rendered
D Costs are charged over the units produced in the period **(2 marks)**

11.2 A rug-making business manufactures quality rugs to customers' orders. It has two production departments (Department X and Department Y) which have overhead absorption rates (per direct labour hour) of $12.86 and $12.40 respectively.

Direct costs for job LJ1 are as follows.

	Job LJ1
Direct Material	$154.00
Direct Labour (Department X)	20 hours
Direct Labour (Department Y)	12 hours

Labour rates per hour are $3.80 for Department X and $3.50 for Department Y.

The company quotes prices to customers that reflect a required gross profit margin of 25% on selling price.

What is the selling price of job LJ1 (to two decimal places)?

$ ☐ **(2 marks)**

11.3 P Co manufactures ring binders which are embossed with the customer's own logo. A customer has ordered a batch of 300 binders. The following data illustrate the cost for a typical batch of 100 binders.

	$
Direct materials	30
Direct wages	10
Machine set up	3
Design and artwork	15
	58

Direct employees are paid on a piecework basis.

P Co requires a gross profit margin of 25 per cent of sales value.

What is the selling price for a batch of 300 binders?

A $172.50
B $138.00
C $184.00
D $232.00 **(2 marks)**

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11.4 JC operates a job costing system. The company's standard gross profit margin is 20% of sales value.

The estimated costs for job B124 are as follows.

Direct materials	3 kg @ $5 per kg
Direct labour	4 hours @ $9 per hour

Production overheads are budgeted to be $240,000 for the period, to be recovered on the basis of a total of 30,000 labour hours.

What should the price for job B124 be quoted at (to two decimal places)?

$ ☐ **(2 marks)**

11.5 **Which of the following are characteristics of service costing?**

(i) High levels of indirect costs as a proportion of total costs
(ii) Use of composite cost units
(iii) Use of equivalent units

A (i) only
B (i) and (ii) only
C (ii) only
D (ii) and (iii) only **(2 marks)**

11.6 **Which of the following costs would be appropriate for cost control purposes in a transport business?**

(i) Cost per tonne-kilometre
(ii) Fixed cost per kilometre
(iii) Maintenance cost of each vehicle per kilometre

A (i) only
B (ii) only
C (i) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

11.7 **For which of the following organisations would service costing NOT be appropriate?**

A Hotel
B Hospital
C Maintenance division of a manufacturing company
D A light engineering company **(2 marks)**

11.8 The following information relates to the distribution division of a multinational company.

What is the most appropriate unit cost for the distribution division?

Kilometres travelled	636,500
Tonnes carried	2,479
Number of drivers	20
Hours worked by drivers	35,520
Tonne/kilometres carried	375,200
Costs incurred	$562,800

A $0.88 per kilometre travelled
B $1.50 per tonne-kilometre
C $15.84 per hour worked by drivers
D $28,140 per driver employed **(2 marks)**

11.9 **Which of the following would be a typical cost unit for a hotel?**

A Meal served
B Guest stay
C Bed available/night
D Bed occupied/night **(2 marks)**

11.10 The following items are recorded in a costing system:

(1) Actual manufacturing overheads
(2) Absorbed manufacturing overheads

Which of the items are contained in a typical job cost?

A (1) and (2)
B (1) only
C (2) only
D Neither (1) nor (2) **(2 marks)**

11.11 **This question has been taken from the January – June 2014 and the July – December 2018 examining team report**.

The direct costs of job J76 comprise the following:

	$
Direct labour (45 hours)	520
Direct material M1 (72 kg)	430
Direct material M2 (25 litres)	120

Production overheads are absorbed at a rate of $8.60 per direct labour hour. Non-production overheads are absorbed at 10% of sales revenue. Selling prices are determined so as to achieve a net profit margin of 12% of sales revenue.

What is the selling price of job J76 (to the nearest $)?

A $1,868
B $1,778
C $1,821
D $1,813 **(2 marks)**

11.12 **This question has been taken from the July – December 2018 examining team report**.

Average bed occupancy, in a hospital ward with 20 beds, was 90% over a 30-day period. Costs in the period totalled $194,400.

What was the cost per patient per day?

A $6,480
B $324
C $5,832
D $360 **(2 marks)**

11.13 **This question has been taken from the January – June 2019 examining team report**.

A tour company operates a bus between two towns which are 25 kilometres apart. The bus has seating for 40 passengers. The average occupancy is 80%. The bus makes three return trips every day.

What are the total passenger-kilometres per day?

☐ passenger-kilometres **(2 marks)**

(Total = 26 marks)

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Do you know? – Process costing

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- Process costing is a costing method used where it is not possible to identify separate units of production usually because of the ..... of the production processes involved.
- Three reasons why losses occur include the following:

  (1) .....

  (2) .....

  (3) .....
- ..... loss is the loss expected during a process and it is not given a cost. If it has a scrap value then it is valued at this amount.
- ..... loss is the extra loss resulting when actual loss is greater than the loss anticipated. It is given a cost.
- Loss may have a scrap value. Revenue from normal scrap is treated as a ..... in costs.
- When dealing with process costing questions, the following four-step approach should be used.

  Step 1 .....

  Step 2 .....

  Step 3 .....

  Step 4 .....
- The costs of labour and overhead are sometimes referred to as ..... costs.
- ..... products are two or more products separated in a process, each of which has a significant value compared to the other.
- A ..... is an incidental product from a process which has an insignificant value compared to the main product.
- The point at which joint and by-products become separately identifiable is known as the ..... or the ..... point.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

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Did you know? – Process costing

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- Process costing is a costing method used where it is not possible to identify separate units of production usually because of the **continuous nature** of the production processes involved.
- Three reasons why losses occur include the following:

  (1) **Wastage**
  (2) **Spoilage**
  (3) **Evaporation**

- **Normal** loss is the loss expected during a process and it is not given a cost. If it has a scrap value then it is valued at this amount.
- **Abnormal** loss is the extra loss resulting when actual loss is greater than the loss anticipated. It is given a cost.
- Loss may have a scrap value. Revenue from normal scrap is treated as a **reduction** in costs.
- When dealing with process costing questions, the following four-step approach should be used.

Step 1	**Determine output and losses**
Step 2	**Calculate cost per unit of output, and losses**
Step 3	**Calculate total cost of output, and losses**
Step 4	**Complete accounts**

- The costs of labour and overhead are sometimes referred to as **conversion** costs.
- **Joint** products are two or more products separated in a process, each of which has a significant value compared to the other.
- A **by-product** is an incidental product from a process which has an insignificant value compared to the main product.
- The point at which joint and by-products become separately identifiable is known as the **point of separation** or the **split-off** point.
- Possible pitfalls

  – **Forgetting how to calculate abnormal gains and losses**
  – **Not using the suggested four-step approach when answering process costing questions**

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12 Process costing 41 mins

12.1 **In process costing, what is a joint product?**

A A product which is later divided into many parts

B A product which is produced simultaneously with other products and is of significant value

C A product which is produced simultaneously with other products but which is of a greater value than any of the other products

D A product produced jointly with another organisation **(2 marks)**

12.2 **What is a by-product?**

A A product produced at the same time as other products which has no value

B A product produced at the same time as other products which requires further processing to put it in a saleable state

C A product produced at the same time as other products which has a relatively low volume compared with the other products

D A product produced at the same time as other products which has a relatively low value compared with the other products **(2 marks)**

12.3 **In process costing, if an abnormal loss arises, what action needs to be taken in the process account?**

A Debited with the scrap value of the abnormal loss units

B Debited with the full production cost of the abnormal loss units

C Credited with the scrap value of the abnormal loss units

D Credited with the full production cost of the abnormal loss units **(2 marks)**

12.4 A company makes a product, which passes through a single process.

Details of the process for the last period are as follows.

Materials	5,000 kg at 50c per kg
Labour	$700
Production overheads	200% of labour

Normal losses are 10% of input in the process, and without further processing any losses can be sold as scrap for 20c per kg.

The output for the period was 4,200 kg from the process.

There was no work in progress at the beginning or end of the period.

What will the value credited to the process account for the scrap value of the normal loss for the period be?

$ ☐ **(2 marks)**

12.5 **In process costing the 'point of separation' is relevant to which of the following?**

A Abnormal losses

B Normal losses

C Joint products

D Abnormal gains **(2 marks)**

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12.6 A company discovers, at the end of a process, that abnormal losses had occurred.

At what value would a unit of abnormal loss be recorded in the process account?

A The total cost per unit of normal output
B Scrap value
C The direct cost per unit of normal output
D Nil value **(2 marks)**

12.7 **What are conversion costs?**

A Rework costs
B Direct costs only
C Indirect costs only
D Production costs excluding direct materials **(2 marks)**

12.8 340 litres of Chemical X were produced in a period. There is a normal loss of 10% of the material input into the process. There was an abnormal loss in the period of 5% of the material input.

How many litres of material were input into the process during the period?

A 357 litres
B 374 litres
C 391 litres
D 400 litres **(2 marks)**

12.9 **How are abnormal GAINS recorded in a process account?**

A Credited at a cost per unit based on total production cost divided by actual output
B Credited at a cost per unit based on total production cost divided by normal output
C Debited at a cost per unit based on total production cost divided by actual output
D Debited at a cost per unit based on total production cost divided by normal output

(2 marks)

12.10 Products A and B are manufactured in a joint process. The following data is available for a period:

Joint process costs		$30,000
Output:	Product A	2,000 kg
	Product B	4,000 kg
Selling price	Product A	$12 per kg
	Product B	$18 per kg

What is Product B's share of the joint process costs if the sales value method of cost apportionment is used?

A $7,500
B $18,000
C $20,000
D $22,500 **(2 marks)**

12.11 **Are the following statements relating to process costing true or false?**

	True	*False*
The higher the net realisable value of losses the lower will be the cost per unit of normal output.	○	○
The higher the abnormal losses the higher will be the cost per unit of normal output.	○	○

(2 marks)

12.12 **Which of the following signify an abnormal gain?**

A Actual loss less than normal loss
B Actual loss more than normal loss
C Actual loss equal to normal loss
D None of the above **(2 marks)**

12.13 800 litres of a chemical were input in a period. There is a normal loss of 25% of material input into the process. Output in the period was 500 litres.

What was the abnormal loss in the period?

[ ] litres **(2 marks)**

12.14 Two products are produced from a common process:

Product X 15,000 litres

Product Y 7,500 kg

The joint costs to the point of separation are $180,000.

Product Y can be sold immediately after separation for $16.50 per kg. Product X requires further processing at a cost of $8 per litre before it can be sold for $23 per litre.

What are the common costs apportioned to Product X using net realisable values?

A $63,871
B $180,000
C $132,480
D $116,129 **(2 marks)**

12.15 **This question has been taken from the July – December 2014 examining team report**.

Two joint products (A and B) are manufactured from a common process. Data for a period includes:

	Product A	*Product B*	*Total*
Weight of output	3,000 kg	5,000 kg	8,000 kg
Selling price per kg	$20	$12	
Final sales value	$60,000	$60,000	$120,000
Joint process costs			$60,000
Further processing costs	$15,000	$5,000	$20,000

Using the net realisable value method of apportionment, what percentage of the joint process costs would be apportioned to Product A?

A 50.0%
B 37.5%
C 62.5%
D 45.0% **(2 marks)**

12.16 **This question has been taken from the July – Dec 2015 examining team report**.

Two products (X and Y) are produced jointly in a manufacturing process. Common process costs are apportioned on the basis of sales value. The following data relate to a period in which the common costs were $38,600:

	Product X	*Product Y*
Selling price at split-off point	$8.00 per unit	$15.00 per unit
Selling price after further processing	$10.00 per unit	$18.00 per unit
Output	2,500 units	2,300 units

What amount of the common process costs in the period will be apportioned to Product X?

A $14,165
B $14,533
C $20,104
D $13,426 **(2 marks)**

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12.17 **This question has been taken from the Sep – Jun 2018 examining team report**.

A process with costs of $15,600 has resulted in 1,000 kilogrammes (kg) of product A and 4,000 kg of product B. Product A can be sold for $50 per kg, but only after further processing at a cost of $10 per kg. Product B can be sold for $20 per kg with no further processing.

How much of the process costs will be apportioned to product A, using the net realisable value method?

A $5,200
B $6,000
C $3,120
D $10,400 **(2 marks)**

(Total = 34 marks)

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Do you know? – Cost-volume-profit (CVP) analysis and decision making

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- The number of units of sale required to break even is known as the ..... .
- The ratio which measures how much contribution is earned from each $1 of sales is known as the ..... or ..... ratio.
- The difference between the budgeted sales volume (or revenue) and the breakeven sales volume (or revenue) is known as the ..... .
- The total contribution required for a target profit = required profit + ...
- Breakeven analysis is a useful technique for managers as it can provide simple and quick estimates. A graphical representation of breakeven arithmetic can be provided by a ..... .
- Future costs, cash flows, incremental costs, differential costs and opportunity costs are all known as ..... costs.
- A past cost which is not relevant in decision making is known as a ..... .
- In general, when deciding which costs are relevant to a decision, variable costs will be ..... costs and fixed costs will be ..... to the decision.
- A factor which limits an organisation's activities is known as a ..... .

  In a make/buy-in problem with no limiting factors, the relevant costs for the decision are the ..... between the two options.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

Did you know? – Cost-volume-profit (CVP) analysis and decision making

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- The number of units of sale required to break even is known as the **breakeven point**.
- The ratio which measures how much contribution is earned from each $1 of sales is known as the **contribution/sales** or **profit/volume** ratio.
- The difference between the budgeted sales volume (or revenue) and the breakeven sales volume (or revenue) is known as the **margin of safety**.
- The total contribution required for a target profit = required profit + **fixed costs**.
- Breakeven analysis is a useful technique for managers as it can provide simple and quick estimates. A graphical representation of breakeven arithmetic can be provided by a **breakeven chart.**
- Future costs, cash flows, incremental costs, differential costs and opportunity costs are all known as **relevant** costs.
- A past cost which is not relevant in decision making is known as a **sunk cost**.
- In general, when deciding which costs are relevant to a decision, variable costs will be **relevant** costs and fixed costs will be **irrelevant** to the decision.
- A factor which limits an organisation's activities is known as a **limiting factor**.

  In a make/buy-in problem with no limiting factors, the relevant costs for the decision are the **differential costs** between the two options.
- Possible pitfalls
  - **Classifying sunk costs and committed costs as relevant costs**
  - **Not learning all of the formulae required in order to carry out CVP analysis calculations!**

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13 Cost-volume-profit (CVP) analysis 43 mins

13.1 A company makes a single product and incurs fixed costs of $30,000 per month. Variable cost per unit is $5 and each unit sells for $15. Monthly sales demand is 7,000 units.

What is the breakeven point in terms of monthly sales units?

☐ units **(2 marks)**

13.2 **Which of the following describes the margin of safety?**

A Actual contribution margin achieved compared with that required to break-even
B Actual sales compared with sales required to break-even
C Actual verses budgeted net profit margin
D Actual verses budgeted sales **(2 marks)**

Data for questions 13.3 to 13.4

Information concerning K Co's single product is as follows.

	$ per unit
Selling price	6.00
Variable production cost	1.20
Variable selling cost	0.40
Fixed production cost	4.00
Fixed selling cost	0.80

Budgeted production and sales for the year are 10,000 units.

13.3 **What is the company's breakeven point, to the nearest whole unit?**

☐ units **(2 marks)**

13.4 **How many units must be sold if K Co wants to achieve a profit of $11,000 for the year?**

A 2,500 units
B 9,833 units
C 10,625 units
D 13,409 units **(2 marks)**

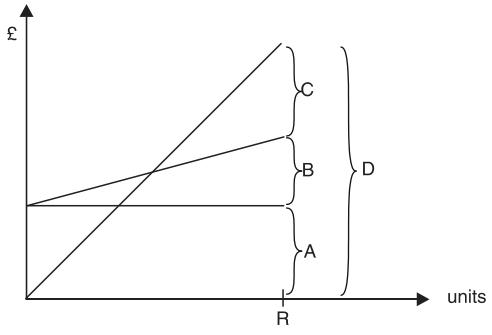
13.5 The following forecasts relate to a single-product business for a period:

Variable costs	$38,640
Fixed costs	$39,975
Sales revenue	$84,000
Sales unit	6,000

What sales revenue is required to achieve a profit of $12,000 in the period?

A $74,030
B $90,615
C $96,250
D $112,990 **(2 marks)**

13.6



On the above breakeven chart, which of the following shows the contribution at level of activity R?

A D less A
B D less B
C B + C
D A + B **(2 marks)**

13.7 A Co makes a single product which it sells for $10 per unit. Fixed costs are $48,000 per month and the product has a contribution to sales ratio of 40%.

Actual sales for the month were $140,000.

What was A Co's margin of safety (in units)?

A 2,000
B 12,000
C 14,000
D 20,000 **(2 marks)**

13.8 A single product company has a contribution to sales ratio of 40%. Fixed costs amount to $90,000 per annum.

How many units are required to break even?

A 36,000
B 150,000
C 225,000
D Impossible to calculate without further information **(2 marks)**

13.9 A company's breakeven point is 6,000 units per annum. The selling price is $90 per unit and the variable cost is $40 per unit.

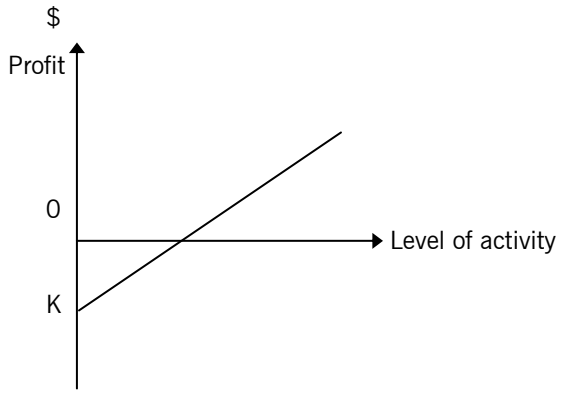
What are the company's annual fixed costs?

$ ☐ **(2 marks)**

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The following graph relates to questions 13.10 and 13.11



13.10 **What does point K on the graph indicate?**

A Semi-variable cost
B Total cost
C Variable cost
D Fixed cost **(2 marks)**

13.11 **What is the above graph known as?**

A Conventional breakeven chart
B Contribution breakeven chart
C Semi-variable cost chart
D Profit/volume chart **(2 marks)**

13.12 Windy Co manufactures a single product Q, data for which are as follows.

	$ per unit
Selling price	60
Direct material cost	14
Direct labour cost	12
Variable overhead cost	19
Fixed overhead cost	11
Profit	4

What is the contribution/sales ratio for product Q (to the nearest percent)?

[ ] % **(2 marks)**

13.13 E Co manufactures a single product, P. Data for the product are as follows.

	$ per unit
Selling price	20
Direct material cost	4
Direct labour cost	3
Variable production overhead cost	2
Variable selling overhead cost	1
Fixed overhead cost	5
Profit per unit	5

What is the contribution/sales ratio for product P?

A 25%
B 50%
C 55%
D 60%

(2 marks)

Data for questions 13.14 and 13.15

W Co sells one product for which data is given below:

	$ per unit
Selling price	10
Variable cost	6
Fixed cost	2

The fixed costs are based on a budgeted level of activity of 5,000 units for the period.

13.14 **How many units must be sold if W Co wishes to earn a profit of $6,000 for one period?**

A 1,500
B 1,600
C 4,000
D 8,000

(2 marks)

13.15 **What is W Co's margin of safety for the period if fixed costs prove to be 20% higher than budgeted and the actual activity is as per budget?**

A 29%
B 40%
C 50%
D 66%

(2 marks)

Data for questions 13.16 and 13.17

Sales units	128,000
Sales revenue	$640,000
Variable costs	$384,000
Fixed costs	$210,000

13.16 **What sales revenue is required to earn a profit of $65,000?**

A $458,333
B $590,000
C $687,500
D $705,000

(2 marks)

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13.17 **How many sales units are required to earn a profit of $52,000?**

A 52,400 units
B 87,333 units
C 131,000 units
D 160,500 units **(2 marks)**

13.18 **This question has been taken from the January – June 2016 examining team report.**

The following data relate to a single product business:

	$ per unit
Selling price	30.00
Prime costs	11.20
Production overheads	8.60 (15% variable)
Non-production overheads	5.70 (10% variable)
Net profit	4.50

What is the contribution/sales (C/S) ratio?

A 15.0%
B 34.0%
C 56.5%
D 58.4% **(2 marks)**

(Total = 36 marks)

14 Decision making 38 mins

14.1 In decision making, costs which need to be considered are said to be **relevant costs.**

Which of the following are characteristics associated with relevant costs?

(i) Future costs
(ii) Unavoidable costs
(iii) Incremental costs
(iv) Differential costs

A (i) and (iii) only
B (ii) only
C (i), (iii) and (iv) only
D (i), (ii), (iii) and (iv) **(2 marks)**

14.2 **Which of the following is NOT a relevant cost for decision making?**

A Differential cost
B Committed cost
C Opportunity cost
D Incremental cost **(2 marks)**

14.3 You are currently employed as a Management Accountant in an insurance company. You are contemplating starting your own business.

In considering whether or not to start your own business, what type of cost would your current salary level be considered to be?

A A sunk cost
B An incremental cost
C An irrelevant cost
D An opportunity cost **(2 marks)**

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14.4 Sue is considering starting a new business and she has already spent $5,000 on market research and intends to spend a further $2,000.

In the assessment of the relevant costs of the decision to set up the business, what are the market research costs considered to be?

A A sunk cost of $7,000
B A sunk cost of $5,000 and an incremental cost of $2,000
C A sunk cost of $2,000 and an incremental cost of $5,000
D An opportunity cost of $7,000 **(2 marks)**

14.5 A firm has some material which originally cost $45,000. It has a scrap value of $12,500 but if reworked at a cost of $7,500, it could be sold for $17,500. There is no other foreseen use for the material.

What is the relevant cost of using the material for a special job?

$ ☐ **(2 marks)**

14.6 Your company regularly uses material X and currently has in inventory 500 kg for which it paid $1,500 two weeks ago. If this were to be sold as raw material, it could be sold today for $2.00 per kg. You are aware that the material can be bought on the open market for $3.25 per kg, but it must be purchased in quantities of 1,000 kg.

You have been asked to determine the relevant cost of 600 kg of material X to be used in a job for a customer.

What is the relevant cost of the 600 kg?

A $1,325
B $1,825
C $1,950
D $3,250 **(2 marks)**

14.7 X Co has 500 kg of material K in inventory for which it paid $2,000. The material is no longer in use in the company and could be sold for $1.50 per kg.

X Co is considering taking on a single special order which will require 800 kg of material K. The current purchase price of material K is $5 per kg.

In the assessment of the relevant cost of the decision to accept the special order, what is the cost of material K?

A A sunk cost of $2,000
B A sunk cost of $2,000 and an incremental cost of $1,500
C An opportunity cost of $750 and an incremental cost of $1,500
D An incremental cost of $4,000 **(2 marks)**

14.8 A company is considering the use of Material X in a special order. A sufficient quantity of the material, which is used regularly by the company in its normal business, is available from inventory.

What is the relevant cost per kg of Material X in the evaluation of the special order?

A Cost of the last purchase
B Nil
C Replacement cost
D Saleable value **(2 marks)**

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14.9 MF Co manufactures three products, the selling price and cost details of which are as follows.

	Product M	*Product F*	*Product S*
	$ per unit	$ per unit	$ per unit
Selling price	129	137	141
Direct material ($8/kg)	32	16	40
Direct labour ($6/hour)	30	36	24
Variable overhead	10	12	8
Fixed overhead	15	20	14

In a period when direct labour is restricted in supply, which of the following ranks the products from the most profitable to the least profitable according to use of direct labour?

A Product M, Product F, Product S
B Product F, Product S, Product M
C Product S, Product M, Product F
D Product S, Product F, Product M **(2 marks)**

14.10 A company makes a single product for which cost details are as follows.

	$ per unit
Direct material ($8 per litre)	72
Direct labour ($7 per hour)	49
Production overhead	56
Total production cost	177

The product is perishable and no stocks (inventory) are held.

Demand for next period will be 2,000 units but only 16,000 litres of material and 15,000 hours of labour will be available.

Which of the following are limiting factors for the period?

A Material only
B Labour only
C Both material and labour
D Neither material nor labour **(2 marks)**

14.11 A company is launching a new product. In order to manufacture this new product, two types of labour are required – skilled and semi-skilled. The new product requires five hours of skilled labour and five hours of semi-skilled labour.

A skilled employee is currently paid $10 per hour. Should he/she work on the new product a replacement would have to be obtained at a rate of $9 per hour, for the work which would otherwise be done by the skilled employee. The current rate for semi-skilled workers is $5 per hour and an additional employee would be appointed for this work.

What is the relevant cost of labour to be used in making one unit of the new product?

$ ☐ **(2 marks)**

14.12 A company manufactures three products, details of which are as follows.

	Product J	*Product K*	*Product L*
	$ per unit	$ per unit	$ per unit
Selling price	140	122	134
Direct materials ($2/kg)	22	14	26
Other variable cost	84	72	51
Fixed cost	20	26	40

In a period when direct material is restricted in supply, which of the following ranks the products in terms of the most profitable use of the material?

A Product J, Product K, Product L
B Product J, Product L, Product K
C Product K, Product L, Product J
D Product L, Product K, Product J

(2 marks)

14.13 Brian Co produces three products which have the following unit contributions and labour requirements.

Product	*Unit contribution*	*Labour requirement*
	$	Hours
Scratch	6	2
Purr	7	3
Buzz	8	3

Due to industrial action only 2,600 labour hours are available in control period 13, when expected demand is 700 units of each product. Fixed costs are $1,700 for the period.

What is the profit-maximising product mix?

	Product Scratch	*Product Purr*	*Product Buzz*
	Units	Units	Units
A	0	166	700
B	700	0	400
C	700	700	700
D	1,300	0	0

(2 marks)

14.14 An engineering company has been offered the opportunity to bid for a contract which requires a special component. Currently, the company has a component in inventory, which has a net book value of $250. This component could be used in the contract, but would require modification at a cost of $50. There is no other foreseeable use for the component held in inventory. Alternatively, the company could purchase a new specialist component for $280.

What is the relevant cost of using the component currently held in inventory for this contract?

$ [ ]

(2 marks)

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14.15 A company manufactures and sells four products. The labour hours available for manufacture are restricted but any quantities of the products can be bought-in from other suppliers to satisfy sales demand. The following information is provided:

	Product			
	1	*2*	*3*	*4*
	per unit	per unit	per unit	per unit
Selling price ($)	6.00	7.50	9.00	15.00
Variable manufacturing costs ($)	5.00	4.00	6.50	8.50
Bought-in price ($)	7.50	6.75	8.50	12.00
Labour (hours)	1.5	3	2	2.5

Which is the best product to buy-in in order to maximise profit?

A Product 1
B Product 2
C Product 3
D Product 4

(2 marks)

14.16 **This question has been taken from the January – June 2019 examining team report**.

Machinery owned by a company is not currently in use but could be used on a short-term contract. The net book value of the machinery is $2,000. If it is not used on this new contract, the machinery could be sold for $2,200. After use on the contract, there would be no resale value and the cost of disposal would be $900.

What is the relevant cost of the machinery to the contract?

A $2,000
B $3,100
C $2,200
D $1,300

(2 marks)

(Total = 32 marks)

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Do you know? – Capital investment appraisal

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- The basic principle of ..... involves calculating the present value of an investment. The present value of an investment is the amount of money which must be invested now (for a number of years) in order to earn a future sum (at a given rate of interest).
- A constant sum of money received or paid each year for a given number of years is known as an ..... . If this constant sum lasts forever, then it is known as a ..... .
- Annuity × annuity factor = ...
- Annuity ÷ interest rate = ...
- The two main discounted cash flow methods — NPV; IRR
  - **Net present value (NPV) method**. If an investment has a ..... NPV then it is acceptable. An investment with a ..... NPV should be rejected.
  - **Internal rate of return (IRR) method**. This method determines the rate of interest at which the NPV of the investment = ... The project is viable if the IRR exceeds the minimum acceptable return.
- The IRR formula is as follows.

$$IRR = a\% + \left[\frac{A}{A-B} \times (b-a)\right]\%$$

Where a = ...

b = ...

A = ...

B = ...

- The time that is required for the cash inflows from a capital investment project to equal the cash outflows is known as the ..... .
- Possible pitfalls

  Write down the mistakes you know you should avoid.

Did you know? – Capital investment appraisal

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- The basic principle of **discounting** involves calculating the present value of an investment. The present value of an investment is the amount of money which must be invested now (for a number of years) in order to earn a future sum (at a given rate of interest).
- A constant sum of money received or paid each year for a given number of years is known as an **annuity**. If this constant sum lasts forever, then it is known as a **perpetuity**.
- Annuity × annuity factor = **present value of an annuity**
- Annuity ÷ interest rate = **present value of a perpetuity**
- The two main discounted cash flow methods — NPV; IRR
  - **Net present value (NPV) method**. If an investment has a **positive** NPV then it is acceptable. An investment with a **negative** NPV should be rejected.
  - **Internal rate of return (IRR) method**. This method determines the rate of interest at which the NPV of the investment = **zero**. The project is viable if the IRR exceeds the minimum acceptable return.
- The IRR formula is as follows.

$$IRR = a\% + \left[\frac{A}{A-B} \times (b-a)\right]\%$$

Where a = **one interest rate**

b = **the other interest rate**

A = **NPV at rate a**

B = **NPV at rate b**

- The time that is required for the cash inflows from a capital investment project to equal the cash outflows is known as the **payback period**.
- Possible pitfalls
  - **Not being able to calculate and distinguish between the nominal rate of interest and the effective annual rate of interest**
  - **Not being able to calculate the IRR of an investment, even when given the IRR formula (you must remember what the symbols in the formula mean so that you can use the correct figures in your calculations)**

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15 Capital investment appraisal 53 mins

15.1 A building society adds interest monthly to investors' accounts even though interest rates are expressed in annual terms. The current nominal rate of interest is 6% per annum.

An investor deposits $1,000 on 1 January.

How much interest will have been earned by 30 June?

A $30.00
B $30.38
C $60.00
D $300 **(2 marks)**

15.2 A one-year investment yields a return of 15%. The cash returned from the investment, including principal and interest, is $2,070.

How much interest has been earned?

A $250
B $270
C $300
D $310.50 **(2 marks)**

15.3 A single sum of $12,000 is invested at 8% per annum (nominal) with interest compounded quarterly.

What is the amount to which the principal will have grown by the end of year three (to the nearest $)?

A $15,117
B $9,528
C $15,219
D $30,924 **(2 marks)**

15.4 **Which is worth most, at present values, assuming an annual rate of interest of 8%?**

A $1,200 in exactly one year from now
B $1,400 in exactly two years from now
C $1,600 in exactly three years from now
D $1,800 in exactly four years from now **(2 marks)**

15.5 A bank offers depositors a nominal 4% per annum, with interest payable quarterly.

What is the effective annual rate of interest (to two decimal places)?

[ ]% **(2 marks)**

15.6 A project has an NPV of $22 at 9% and an NPV of –$4 at 10%.

What is the IRR for the project (to two decimal places)?

[ ]% **(2 marks)**

15.7 A sum of money was invested for 10 years at 7% per annum and is now worth $2,000.

What was the original amount invested (to the nearest $)?

A $1,026
B $1,016
C $3,937
D $14,048 **(2 marks)**

15.8 House prices rise at 2% per calendar month.

What is the annual effective rate of increase correct to one decimal place?

A 24%
B 26.8%
C 12.7%
D 12.2%

(2 marks)

15.9 **What is the present value of ten annual payments of $700 discounted at 8% per annum, with the first payment being made immediately?**

A $4,697
B $4,723
C $4,435
D $5,073

(2 marks)

15.10 A machine has an investment cost of $60,000 at time 0. The present values (at time 0) of the expected net cash inflows from the machine over its useful life are:

Discount rate	*Present value of cash inflows*
10%	$64,600
15%	$58,200
20%	$52,100

What is the internal rate of return (IRR) of the machine investment?

A Below 10%
B Between 10% and 15%
C Between 15% and 20%
D Over 20%

(2 marks)

15.11 An investment project has a positive net present value (NPV) of $7,222 when its cash flows are discounted at the cost of capital of 10% per annum. Net cash inflows from the project are expected to be $18,000 per annum for five years. The cumulative discount (annuity) factor for five years at 10% is 3.791.

What is the investment at the start of the project?

A $61,016
B $68,238
C $75,460
D $82,778

(2 marks)

15.12 **Which TWO of the following statements, relating to an investment project that has been discounted at rates of 10% and 20%, are true?**

☐ The discounted payback period at 10% will be longer than the discounted payback period at 20%.

☐ The discounted payback period at 20% will be longer than the discounted payback period at 10%.

☐ The non-discounted payback period will be longer than the discounted payback period.

☐ The non-discounted payback period will be shorter than the discounted payback period.

(2 marks)

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15.13 **Which of the following accurately defines the internal rate of return (IRR)?**

A The average annual profit from an investment expressed as a percentage of the investment sum.

B The discount rate (%) at which the net present value of the cash flows from an investment is zero.

C The net present value of the cash flows from an investment discounted at the required rate of return.

D The rate (%) at which discounted net profits from an investment are zero. **(2 marks)**

15.14 An investment project has the following discounted cash flows ($'000):

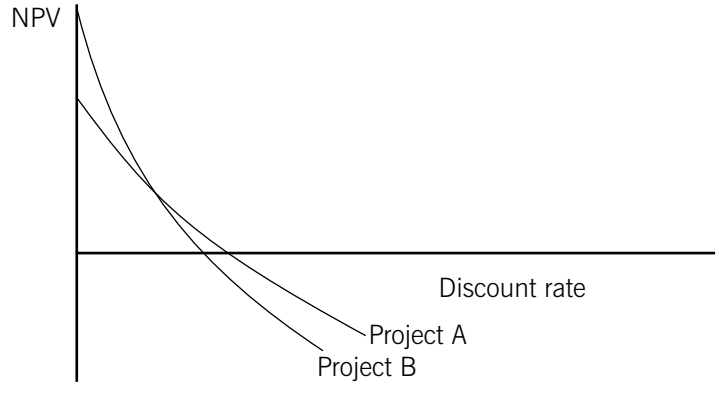
Year	*Discount rate*		
	0%	*10%*	*20%*
0	(90)	(90)	(90)
1	30	27.3	25.0
2	30	24.8	29.8
3	30	22.5	17.4
4	30	20.5	14.5
	30	5.1	(12.3)

The required rate of return on investment is 10% per annum.

What is the discounted payback period of the investment project?

A Less than three years
B Three years
C Between three years and four years
D More than four years **(2 marks)**

15.15 The following chart shows the discounted values of two investment projects:



On the basis of the chart, are the following statements true or false?

	True	*False*
Project A has a higher internal rate of return than Project B.	○	○
Project B has higher initial outlay than Project A.	○	○

(2 marks)

15.16 An investment project has net present values as follows:

Discount rate 10% per annum, net present value $24,760 positive

Discount rate 20% per annum, net present value $16,110 negative

Using the data above, what is the best estimate of the internal rate of return (IRR)?

A 10.6%
B 13.9%
C 16.1%
D 38.6% **(2 marks)**

15.17 A company has decided to lease a machine. Six annual payments of $8,000 will be made with the first payment on receipt of the machine. Below is an extract from an annuity table:

Year	*Annuity factor*
	10%
1	0.909
2	1.736
3	2.487
4	3.170
5	3.791
6	4.355

What is the present value of the lease payments at an interest rate of 10%?

A $30,328
B $34,840
C $38,328
D $48,000 **(2 marks)**

15.18 A company is considering an immediate investment in new machinery. The machinery would cost $100,000 with expected net cash inflows of $30,000 per year starting in Year 1. The disposal value of the machine after five years is expected to be $10,000. $15,000 has already been incurred on development costs.

What is the payback period of the investment based on future incremental cash flows?

A 3.0 years
B 3.3 years
C 3.5 years
D 3.8 years **(2 marks)**

15.19 **What is the present value of a perpetuity of $8,652 per year given an interest rate of 7% per year, assuming that the first cash flow occurs today?**

A $8,652
B $16,738
C $123,600
D $132,252 **(2 marks)**

15.20 **This question has been taken from the July – December 2014 examining team report**.

A new machine, costing $100,000, has an estimated realisable value of $25,000 after five years. The expected profit from investment in the machine is $25,000 per year, net of straight-line depreciation.

What is the payback period?

A 4.0 years
B 3.0 years
C 1.875 years
D 2.5 years **(2 marks)**

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15.21 **This question has been taken from the July – Dec 2015 examining team report**.

An investment project has a positive net present value of $33,274 when discounted at the cost of capital of 10% per annum. The estimated net cash inflows at the end of each year of the project's five-year life are $64,000. The annuity factor at 10% per annum for five years is 3.791.

What is the investment amount in Year 0?

A $275,898
B $286,726
C $273,350
D $209,350

(2 marks)

15.22 **This question has been taken from the January – June 2019 examining team report**.

Tubal Co has leased a fleet of vehicles for five years, paying $3,000 at the beginning of each year. You have been provided with the following discount factors:

Year	*Annuity factors 10%*
0	1
1	0.909
2	1.736
3	2.487
4	3.170
5	3.791

What is the present value of the lease payments?

A $12,510
B $11,373
C $9,510
D $15,000

(2 marks)

(Total = 44 marks)

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Do you know? – Cash management

Check that you can fill in the blanks in the statements below before you attempt any questions. If in doubt, you should go back to your BPP Interactive Text and revise first.

- Working capital is the difference between a company's …… and ……..
- The …… measures the period of time between cash outflows for materials and cash inflows from customers. It is also sometimes known as the ……..
- Cash accounting is different to accruals accounting because cash accounting records …….. whereas accruals accounting …… income and expenditure.
- …… is the corporate handling of all financial matters and includes the generation of external and internal funds, the management of currencies and cash flows, and the strategies, policies and procedures of corporate finance.
- Organisations produce cash budgets in order to …… whether there are likely to be …… or ……..
- An …… is a measure over a period of time of the average changes in the values (prices or quantities) of a group of items.
- Cash management should be conducted with three factors in mind: ……., ……. and ……
- …….. deposit accounts are either fixed term or notice accounts. The customer agrees to deposit money either for a fixed period, or a notice period and the money is then invested in ……. and ……
- ……. stocks may be issued by any size of authority from County Councils to Borough Councils.
- Gilts are securities offered by the ……..
- A …… is a certificate indicating that a sum of money has been deposited with a bank and will be repaid at a later date.
- When investing surplus funds, two factors must be considered: …… and ……, …… across a range of investments can reduce the overall risk to the investor.
- Possible pitfalls

  Write down the mistakes you know you should avoid.

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Did you know? – Cash management

Could you fill in the blanks? The answers are in bold. Use this page for revision purposes as you approach the exam.

- Working capital is the difference between a company's **current assets** and **current liabilities**.
- The **operating cycle** measures the period of time between cash outflows for materials and cash inflows from customers. It is also sometimes known as the **cash cycle**.
- Cash accounting is different to accruals accounting because cash accounting records **cash payments and receipts** whereas accruals accounting **matches** income and expenditure.
- **Treasury management** is the corporate handling of all financial matters and includes the generation of external and internal funds, the management of currencies and cash flows, and the strategies, policies and procedures of corporate finance.
- Organisations produce cash budgets in order to **forecast** whether there are likely to be **cash shortages** or **large surpluses**.
- An **index** is a measure over a period of time of the average changes in the values (prices or quantities) of a group of items.
- Cash management should be conducted with three factors in mind: **liquidity**, **safety** and **profitability**.
- **Money-market** deposit accounts are either fixed term or notice accounts. The customer agrees to deposit money either for a fixed period, or a notice period and the money is then invested in **stocks** and **shares.**
- **Local authority** stocks may be issued by any size of authority from County Councils to Borough Councils.
- Gilts are securities offered by the **UK government**.
- A **certificate of deposit** is a certificate indicating that a sum of money has been deposited with a bank and will be repaid at a later date.
- When investing surplus funds, two factors must be considered: **risk** and **return**, **diversification** across a range of investments can reduce the overall risk to the investor.
- Possible pitfalls

  Not understanding the working capital cycle
  Getting confused over cash accounting and accruals accounting
  Confusion over different types of cash forecasts
  Not understanding the different ways of investing cash surpluses

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16 Cash management 86 mins

16.1 **Which of the following statements about working capital is correct?**

A Working capital is the difference between a company's total assets and its total liabilities.
B Working capital is the difference between a company's total assets and its current liabilities.
C Working capital is the difference between a company's current assets and its total liabilities.
D Working capital is the difference between a company's current assets and its current liabilities.

(2 marks)

16.2 **Which of the following is NOT classed as working capital?**

A Overdraft
B Inventory
C Accruals
D Bank loan **(2 marks)**

16.3 A business has general overheads of $160,000 in September 20X9 but it is anticipated that these will increase by 1.75% per month for the next few months. Overheads are paid the month after they are incurred.

What is the cash outflow for overheads in the month of December 20X9?

A $162,800
B $165,600
C $168,548
D $165,649 **(2 marks)**

16.4 **Which of the following is a way of improving operational cash flows in a business?**

A Taking more credit from suppliers
B Giving more credit to customers
C Increasing inventories
D Taking advantage of early settlement discounts from suppliers **(2 marks)**

16.5 **Which of the following are functions carried out by a company's treasury department?**

(i) Working capital management
(ii) Preparation of annual financial statements
(iii) Exchange dealing, including futures and options

A (i) and (ii) only
B (ii) and (iii) only
C (i) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

16.6 **Which of the following are ways of funding a forecast cash deficit?**

(i) Leading and lagging
(ii) Borrowing from the bank
(iii) Selling short-term financial investments

A (i) and (ii) only
B (i) and (iii) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

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16.7 **Which of the following may be causes of cash flow problems?**

(i) Losses
(ii) Growth
(iii) Seasonal business

A (i) and (ii) only
B (i) and (iii) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

16.8 The revenue for Y Co is as follows:

Year	*Revenue*
	$
20X1	26,500
20X2	28,680
20X3	29,530
20X4	27,670
20X5	30,230

Using 20X1 as the base year, what is the index for 20X4 (to the nearest whole number)?

[ ] **(2 marks)**

16.9 **Which of the following statements about UK government securities are correct?**

(i) Most are fixed interest
(ii) Most have a face value of £100
(iii) The yield is higher than local authority stocks

A (i) and (ii) only
B (i) and (iii) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

16.10 **Which of the following statements about certificates of deposit is correct?**

A Certificates of deposit are issued by the UK government.
B Ownership of certificates of deposit is transferred by physical delivery from buyer to seller.
C Certificates of deposit are not negotiable.
D The deposit is invested in stocks and bonds. **(2 marks)**

16.11 **Which of the following uses of an overdraft is a bank most likely to approve of?**

A To purchase new machinery
B To cover a temporary cash shortfall
C To invest in a new company
D To expand overseas **(2 marks)**

16.12 **Which of the following describes exceptional cash transactions?**

A Relate to the long-term functioning of the business
B Capitalised in the accounts
C Occur at regular intervals
D Unusual costs such as the closure of part of a business **(2 marks)**

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16.13 **When cash and profit relates to the same period, which of the following transactions will affect the amount of cash in a business in the same way as it affects profit reported in the statement of profit or loss (income statement)?**

A The sale of a fleet of company vehicles
B The monthly rent accrual
C The payment of wages at the end of each month
D The purchase of new computers **(2 marks)**

16.14 **Which TWO of the following are important procedures to follow when handling cash payments?**

☐ Restricting access to cash and cheques
☐ Ensuring prompt banking of cash
☐ Ensuring all cash banked is reconciled
☐ Completing appropriate supporting documentation

(2 marks)

16.15 **Are the following statements true or false?**

	True	*False*
When economic conditions are unfavourable, businesses are likely to try to preserve cash balances.	○	○
When economic conditions are unfavourable, businesses are likely to be reluctant to borrow funds.	○	○

(2 marks)

16.16 **Cash forecasts are likely to show which TWO of the following?**

☐ How much cash is required
☐ Profit
☐ When cash is required
☐ The sources of funding available

(2 marks)

16.17 The trend figures for sales in $ for a business for the four quarters of last year and the seasonal variations are estimated as:

	Trend sales $	*Seasonal variation*
Quarter 1	160,000	+12,820
Quarter 2	164,500	+14,805
Quarter 3	169,000	–5,070
Quarter 4	173,500	–22,555

Assuming the trend continues, what are the forecast sales for Quarter 1 next year?

A $190,820
B $186,320
C $150,945
D $172,820 **(2 marks)**

16.18 Budgeted revenues for a business are shown below.

Month	$
January	60,000
February	70,000
March	84,000

Payment is expected to be received as follows:

55% received in the month of sale
28% received in the month following that of sale
17% will not be received at all

What are the budgeted cash receipts in February?

$ ☐ **(2 marks)**

16.19 **Which of the following is the least likely to be carried out by the treasury department?**

A Dealing in foreign exchange
B Involvement in business acquisitions and sales
C Preparing the corporate budget and business plan
D Negotiating arrangements with banks **(2 marks)**

16.20 **In the additive model, where A = Actual figure, T = Trend figure and S = Seasonal variation, how is seasonal variation calculated?**

A $S = A - T$
B $S = A \times T$
C $S = \frac{A}{T}$
D $S = A + T$ **(2 marks)**

16.21 **Which of the following procedures help to ensure that cash and cheques are not misappropriated?**

(i) Sending regular statements to customers showing transactions and cash received during the period

(ii) Checking to ensure that the bank statement is in agreement with the paying-in records

(iii) Ensuring that two persons are always present when the mail is opened

(iv) Conducting spot checks on the cash balance maintained at the company's premises

A (i), (iii) and (iv)
B (iii) and (iv) only
C (i) and (ii)
D (ii) and (iii) **(2 marks)**

16.22 **Which of the following should cheque signatories be responsible for?**

A Approving cheque requisitions
B Recording payments
C Preparing cheques
D None of the above **(2 marks)**

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16.23 A company is preparing the budget for a product and the following data has been provided:

	Month 1	*Month 2*	*Month 3*	*Month 4*
Planned sales (units)	2,000	2,000	2,500	2,800

No inventory of materials is held. Closing inventory (finished product) in each month must be 40% of the next month's sales. Suppliers are paid in the month following purchase. The standard cost of materials is $4.00 per unit.

What is the budgeted payment to materials suppliers in Month 3?

$ ☐ **(2 marks)**

16.24 A company's projected revenue for 20X6 is $342,000. It is forecast that 20% of revenue will occur in January and the rest will be equally spread among the remaining 11 months. All sales are on credit.

Payment is expected to be received as follows:

50% received in the month of sale

45% received in the following month

5% not received and written off as bad debts after two months

What are the budgeted cash collections for March?

A $24,873
B $23,629
C $22,386
D $27,075 **(2 marks)**

16.25 One of the main criteria in considering the investment of surplus cash is risk.

Which of the following investments is generally considered to carry the greatest risk?

A Preference shares
B Convertible loan notes
C Government securities
D Secured loans **(2 marks)**

16.26 **This question has been taken from the July – December 2014 examining team report**.

An overhead cost budget for the next calendar year includes:

- Depreciation charged on a straight-line basis at $11,200 per month;
- Machine maintenance of $5,900 per quarter, payable in advance in January, April, July and October.

The remaining overheads, of $207,600 for the year, are budgeted to be incurred at an even rate per month, payable one month in arrears. The expected accrued overhead, at the end of the calendar year prior to the budget year, is $16,800.

What amount of overhead should be included in the cash budget for January?

A $23,200
B $22,700
C $33,900
D $28,500 **(2 marks)**

16.27 **This question has been taken from the January – June 2015 examining team report.**

The following were the only transactions in the first trading period of a new business:

Credit sales: $140,900
Receipts from customers: $118,700
Credit purchases of goods: $96,600 (there was no inventory at the end of the period)
Payments to suppliers of goods: $72,000
Purchase of, and payment for, a machine: $20,000 (the machine is depreciated on a straight-line basis over 20 periods, assuming nil residual value)
Expenses incurred and paid for: $17,800

Comparing accrual accounting and cash accounting, what is the difference in the surplus for the period?

A $16,600
B $17,600
C $2,400
D $3,400 **(2 marks)**

16.28 **This question has been taken from the July – December 2015 examining team report.**

Which of the following will affect the working capital cycle?

1 An increase in the credit period agreed with customers

2 A reduction in inventory of raw materials calculated as a percentage of the amount used in production

3 Delayed payment to suppliers

4 An increase in the finished goods turnover period

A 1 and 3 only
B 2 and 4 only
C 1, 2 and 3 only
D 1, 2, 3 and 4 **(2 marks)**

16.29 **This question has been taken from the January – June 2016 examining team report.**

The sales revenue budget of X Co for the next three months is:

	$
June	145,000
July	156,000
August	130,000

5% of sales by value are cash sales. 25% of credit sales by value are received one month after sale with the remainder two months after sale.

What are the budgeted receipts in August (to the nearest $)?

A $146,863
B $152,088
C $154,250
D $136,925 **(2 marks)**

16.30 **What are government securities otherwise known as?**

A Certificates of deposit
B Bills
C Gilts
D Commercial paper **(2 marks)**

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16.31 **Which of the following best describes a money-market deposit?**

A A certificate giving its holder the right to ownership at a future date of money already held in a bank deposit

B A negotiable instrument which can be bought and sold

C A deposit account offered by a bank which invests in stocks and bonds

D A form of government stock **(2 marks)**

16.32 **Which of the following investments would NOT be appropriate for a company with a temporary cash surplus of approximately 1 week?**

A Local authority deposits
B Money-market lending arranged directly
C Treasury bills
D Option deposits **(2 marks)**

16.33 **This question has been taken from the January – June 2019 examining team report**.

A firm has sales of $300,000 and cost of sales of $200,000 for a period. During the period the following occurred:

(1) Receivables increased by $5,000
(2) Payables increased by $4,000
(3) Inventories increased by $6,000

What is the operational cash flow for the period?

A $93,000
B $107,000
C $97,000
D $103,000 **(2 marks)**

16.34 **Are the following statements true or false?**

	True	*False*
Where public money is invested by a public sector organisation, investments are restricted by law.	○	○
Local authorities are permitted to invest in long term, high interest accounts.	○	○

(2 marks)

16.35 **When is an investment by an organisation restricted by law?**

(1) Where tax payers' money is invested by a public sector organisation
(2) Where the money is invested by a company of behalf of personal investors for a pension
(3) In the case of trusts

A 1 only
B 2 only
C 1 and 3 only
D 1, 2 and 3 **(2 marks)**

16.36 **Are the following statements true or false?**

	True	*False*
Local authorities must choose investments that give easy access to funds.	○	○
Local authorities are restricted to investing in government securities and high street banks and building societies.	○	○

(2 marks)

(Total = 72 marks)

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17 Mixed bank 1 48 mins

17.1 The management accountant of X Co has written a report assessing the cost savings that could be made if the company was to invest in new technology.

In which area will the report primarily aid the management of X Co?

A Budgeting
B Control
C Decision-making
D Monitoring **(2 marks)**

17.2 **Which of the following only contains essential features of useful management information?**

A Accurate, clear, presented in report format
B Timely, reliable, supported by calculations
C Regular, complete, communicated in writing
D Clear, accurate, relevant for its purpose **(2 marks)**

17.3 **What is an interlocking bookkeeping system?**

A A single, combined system containing both cost accounting and financial accounting records
B A system combining cost accounting and management accounting
C A system supported by prime entry records
D A system where separate accounts are kept for cost accounting and for financial accounting **(2 marks)**

17.4 A company carries out production in accordance with the special requirements of each customer.

Which of the following costing methods would be MOST appropriate for the company?

A Batch costing
B Job costing
C Process costing
D Service costing **(2 marks)**

17.5 Total costs incurred by a business may be expressed as:

$y = a + bx$

when y represents the total costs
a represents the total fixed costs
b represents the variable costs per unit
x represents the number of units of output

A company has variable costs of $12.20 per unit and total costs, for output of 7,400 units in a period, of $156,980.

Using the above formula and information, what are the total fixed costs in the period?

$ ☐ **(2 marks)**

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17.6 A company currently produces 6,000 units of its single product each period, incurring total variable costs of $60,000 and fixed costs of $42,000. Production will increase to 8,000 units per period if the company expands capacity resulting in changes both to the variable costs per unit and to the total fixed costs. For production of 8,000 units per period total variable costs would be $76,000 and fixed costs $50,000.

What is the reduction in total cost per unit comparing the costs for 8,000 units per period with the unit costs currently being incurred?

A $0.50
B $0.75
C $1.25
D $2.08 **(2 marks)**

17.7 The following documents are used in accounting for raw materials:

(i) Goods received note
(ii) Materials returned note
(iii) Materials requisition note
(iv) Delivery note

Which of the documents may be used to record raw materials sent back to stores from production?

A (i) and (ii)
B (i) and (iv)
C (ii) only
D (ii) and (iii) **(2 marks)**

17.8 Material M is used by a manufacturer. Inventory of Material M at 1 May was valued at a cost of $3,302 (260 kg at $12.70 per kg). 500 kg were purchased on 7 May for $6,500. 410 kg of Material M were used in production during the month. The LIFO method is applied at the end of each month.

What is the cost accounting entry for the issues of Material M during the month?

	Debit		*Credit*		
A	Material inventory	$5,252	Work-in-progress	$5,252	
B	Work-in-progress	$5,252	Material inventory	$5,252	
C	Material inventory	$5,330	Work-in-progress	$5,330	
D	Work-in-progress	$5,330	Material inventory	$5,330	**(2 marks)**

17.9 **How is the re-order level calculated if stock-outs are to be avoided?**

A Maximum usage × Maximum lead time
B Maximum usage × Minimum lead time
C Minimum usage × Maximum lead time
D Minimum usage × Minimum lead time **(2 marks)**

17.10 The following information relates to a raw material inventory item:

Economic order quantity	800 units (established using the formula $\sqrt{\frac{2cd}{h}}$)
Demand	12,000 units per annum
Cost of holding inventory	$1.50 per unit per annum

What is the cost of placing an order?

$ ☐ **(2 marks)**

17.11 **Which TWO of the following labour records may be used to allocate costs to the various cost units in a factory?**

- ☐ Employee record card
- ☐ Attendance record card
- ☐ Time sheet
- ☐ Job card

(2 marks)

17.12 **How is the activity (production volume) ratio calculated?**

A Actual hours ÷ budgeted hours
B Budgeted hours ÷ actual hours
C Standard hours for actual output ÷ actual hours
D Standard hours for actual output ÷ budgeted hours **(2 marks)**

17.13 **Which of the following relates to capital expenditure?**

A Cost of acquiring or enhancing non-current assets
B Expenditure on the manufacture of goods or the provision of services
C Recorded as an asset in the statement of profit or loss (income statement)
D Recorded as a liability in the statement of financial position **(2 marks)**

17.14 Overheads in a factory are apportioned to four production cost centres (A, B, C and D). Direct labour hours are used to absorb overheads in A and B and machine hours are used in C and D. The following information is available:

	Production cost centre			
	A	*B*	*C*	*D*
Overhead expenditure ($)	18,757	29,025	46,340	42,293
Direct labour hours	3,080	6,750	3,760	2,420
Machine hours	580	1,310	3,380	2,640

Which cost centre has the highest hourly overhead absorption rate?

A Production Cost Centre A
B Production Cost Centre B
C Production Cost Centre C
D Production Cost Centre D **(2 marks)**

17.15 A company sold 56,000 units of its single product in a period for a total revenue of $700,000. Finished inventory increased by 4,000 units in the period. Costs in the period were:

Variable production	$3.60 per unit
Fixed production	$258,000 (absorbed on the actual number of units produced)
Fixed non-production	$144,000

Using absorption costing, what was the profit for the period?

A $82,000
B $96,400
C $113,600
D $123,200 **(2 marks)**

17.16 A company with a single product sells more units than it manufactures in a period.

Which of the following correctly describes the use of marginal costing in comparison with absorption costing in the above situation?

A Both profit and inventory values will be higher
B Both profit and inventory values will be lower
C Profit will be higher; inventory values will be lower
D Profit will be lower; inventory values will be higher **(2 marks)**

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17.17 A product has the following unit costs:

Variable manufacturing	$7.60
Variable non-manufacturing	$1.40
Fixed manufacturing	$3.70
Fixed non-manufacturing	$2.70

The selling price of the product is $17.50 per unit.

What is the contribution/sales ratio (to one decimal place)?

[ ]% **(2 marks)**

17.18 A company manufactures and sells four types of component. The labour hours available for manufacture are restricted but any quantities of the components can be brought-in from an outside supplier in order to satisfy sales demand. The following further information is provided:

	Component			
	A	*B*	*C*	*D*
	per unit	per unit	per unit	per unit
Selling price ($)	12.00	15.00	18.00	20.00
Variable manufacturing costs ($)	6.00	7.50	9.00	11.50
Bought-in price ($)	11.00	12.00	13.00	16.00
Labour (hours)	1.5	4	2	3

Which is the best component to buy-in in order to maximise profit?

A Component A
B Component B
C Component C
D Component D **(2 marks)**

17.19 An investment project has net present values as follows:

At a discount rate of 5%	$69,700 positive
At a discount rate of 14%	$16,000 positive
At a discount rate of 20%	$10,500 negative

Using the above figures, what is the BEST approximation of the internal rate of return (IRR) of the investment project?

A 17.6%
B 17.9%
C 18.0%
D 22.7% **(2 marks)**

17.20 **Which of the following is the correct description of a flexed budget?**

A A budget that cannot be adjusted
B A budget that is adjusted according to actual activity
C A budget that is adjusted according to planned activity
D A budget that cannot be used to calculate variances **(2 marks)**

(Total = 40 marks)

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18 Mixed bank 2 48 mins

18.1 **Which of the following describes the control process?**

A The action of monitoring something to keep it on course
B The choice between alternatives
C The development of strategies to achieve objectives
D The establishment of a plan for a future period **(2 marks)**

18.2 Consider the following statements in relation to management information:

(i) It should always be provided regardless of its cost.

(ii) It is data that has been processed in such a way as to be meaningful to the person who receives it.

(iii) It should not be provided until it is as detailed and accurate as possible.

Which of the above statements is/are true of good management information?

A (i) only
B (ii) only
C (i) and (iii)
D (ii) and (iii) **(2 marks)**

18.3 **Which of the following is a feature of cost accounting but not of financial accounting?**

A Control accounts
B Cost classification
C Cost units
D Periodic stocktaking **(2 marks)**

18.4 **Which of the following factors may affect the choice of computer output medium?**

(i) Whether a hard copy of the output is required
(ii) Whether the output requires further computer processing
(iii) Whether a large volume of output is to be used for reference purposes

A (i) and (ii) only
B (i) and (iii) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

18.5 Production units and total costs relating to the last three periods have been:

	Period 1	*Period 2*	*Period 3*
Production (units)	129,440	117,620	126,310
Total cost ($)	198,968	187,739	195,376

Using the high-low method, what is the estimated variable cost per unit of production (to two decimal places)?

$ ☐ **(2 marks)**

18.6 **What is the first-in first-out (FIFO) method used for?**

A Valuing wastage
B Valuing abnormal losses
C Valuing raw material issues from inventory
D Valuing raw material receipts into inventory **(2 marks)**

18.7 Wastage of a raw material during a manufacturing process is 20% of input quantity.

What input quantity of raw material is required per kg of output (to two decimal places)?

☐ kg **(2 marks)**

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18.8 **In the context of inventory control what is the maximum inventory control level?**

A The level below which inventory should not fall if usage is at the maximum expected
B The level below which inventory should not fall if average usage occurs
C The level that inventory should not exceed if usage is at the minimum expected
D The level that inventory should not exceed if average usage occurs **(2 marks)**

18.9 **Which TWO items will usually be included in production overheads?**

☐ Cost centre supervisors' wages
☐ Overtime hours of direct operatives at basic rate
☐ Overtime costs of indirect operatives
☐ Piecework payments to direct operatives **(2 marks)**

18.10 The following statements relate to the depreciation of a non-current asset:

(i) A higher expected disposal value at the end of the asset's useful life would result in a lower depreciation charge.

(ii) The use of the machine hour method may result in a different depreciation charge each period.

(iii) If the actual disposal value of the asset exceeds its net book value a further depreciation cost will arise.

Which of the statements is/are true?

A (i) only
B (i) and (ii) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

18.11 The overheads of two service departments (SCC1 and SCC2) in a factory require reapportionment to the two production departments (PCC1 and PCC2):

	Total overhead	*% to PCC1*	*% to PCC2*
SCC1	$32,170	35	65
SCC2	$24,850	65	35

What is the total reapportionment to production department PCC2?

A $19,957
B $27,412
C $29,608
D $37,063 **(2 marks)**

18.12 A company with a single product manufactured 10,200 units in a period in which 10,300 units were sold.

Are the following statements true or false in relation to the situation described?

	True	*False*
The profit for the period would be higher using absorption costing (compared with marginal costing).	○	○
Inventory values would be higher using absorption costing (compared with marginal costing).	○	○

(2 marks)

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18.13 'A deposit account offered by a bank which invests in stocks and bonds'.

What sort of an investment is this a description of?

A Gilt edged security
B Money-market deposit
C Certificate of deposit
D Local authority loan **(2 marks)**

18.14 **For which costing method is the concept of equivalent units relevant?**

A Batch costing
B Job costing
C Process costing
D Service costing **(2 marks)**

18.15 Costs incurred in a process totalled $61,600 for a period. 22,000 units of finished product were manufactured, of which 440 units were rejected. This is the normal level of rejects for the process. Rejected units are sold for $1.80 per unit.

What was the cost per unit of good output (to the nearest cent)?

A $2.76
B $2.80
C $2.82
D $2.86 **(2 marks)**

18.16 One of the key principles that a business should base its cash management policy on is liquidity.

Which of the following is an example of this principle?

A Ensuring the business has investments which are easily convertible into cash
B Managing investments carefully to minimise costs
C Ensuring short-term investments are protected from heavy losses
D Ensuring cash and other liquid assets are secure from theft

(2 marks)

18.17 Products X and Y are joint products. The joint process costs for a period, during which 10,000 units of X and 15,000 units of Y were manufactured, were $87,500.

The sales value method is used to apportion the joint process costs. Selling prices are $6.00 per unit for Product X and $8.00 per unit for Product Y.

What is Product X's share of the joint process costs?

A $29,167
B $35,000
C $37,500
D $43,750 **(2 marks)**

18.18 A hotel has 60 available rooms. Room occupancy was 80% during a 90 day period during which total costs incurred were $104,976.

What was the cost per occupied room per night in the period?

A $12.44
B $15.55
C $19.44
D $24.30 **(2 marks)**

18.19 **Which of the following is NOT a factor that should affect a decision as to whether to investigate a variance?**

A Controllability of variance
B Cost of investigation
C Personnel involved
D Trend of variance **(2 marks)**

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18.20 A capital investment project requires a cash outflow of $81,000 at the start of the project. Annual cash inflows are forecast to be constant for four years (Years 1 to 4).

The net present value (NPV) of the project at a discount rate of 12% per annum is $8,683 (positive). The internal rate of return of the project is 17%. Annuity factors (Years 1 to 4) at 12% and 17% are 3.037 and 2.743 respectively.

What is the forecast annual cash inflow?

A $23,810
B $26,670
C $29,530
D $32,695 **(2 marks)**

(Total = 40 marks)

19 Mixed bank 3 48 mins

19.1 **Which of the following best describes a profit centre?**

A Part of a business where management makes investment decisions
B Part of a business that provides a service to other parts of the business
C Part of a business where finished products are manufactured
D Part of a business where management is responsible for revenues and costs **(2 marks)**

19.2 **In a large company, which of the following activities may be the responsibility of a trainee accountant?**

A Calculating cost variances
B Making capital investment decisions
C Approving budgets
D Allocating warehouse space **(2 marks)**

19.3 **Which TWO of the following are characteristics of management accounting information?**

☐ Non-financial as well as financial
☐ Used by all stakeholders
☐ Concerned with cost control only
☐ Not legally required **(2 marks)**

19.4 **When production has been completed what double-entry would be made in a cost accounting system?**

	Debit	*Credit*
A	Cost of sales	Finished goods
B	Finished goods	Work-in-progress
C	Finished goods	Cost of sales
D	Work-in-progress	Finished goods

(2 marks)

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19.5 The following is an extract from the list of accounts of a motor vehicle manufacturer:

	Cost codes
Direct materials	1000 – 1999
Indirect materials	2000 – 2999
Direct labour	3000 – 3999
Indirect labour	4000 – 4999

Which of the following is coded INCORRECTLY?

	Code	*Description*
A	4262	Wages of materials stores personnel
B	4131	Wages of canteen supervisor
C	1008	Metal for vehicle body
D	1361	Cleaning materials

(2 marks)

19.6 The following shows the cost per unit of an item of expense at different levels of activity:

Activity (units)	*Cost per unit*
	$
1	10,000
50	200
100	120
150	80

What is the correct behavioural classification for the expense item?

A Fixed cost
B Semi-variable cost
C Stepped-fixed cost
D Variable cost **(2 marks)**

19.7 **Which of the following would be classified as a fixed cost in the operation of a motor vehicle?**

A Oil change every 10,000 kilometres
B Petrol
C Insurance
D Tyre replacement **(2 marks)**

19.8 A particular cost is classified as being semi-variable.

What is the effect on the cost per unit if activity increases by 10%?

A Decrease by 10%
B Decrease by less than 10%
C Increase by less than 10%
D Remain constant **(2 marks)**

19.9 The raw materials issued to a job were overestimated and the excess is being sent back to the materials store.

What document is required?

A Stores credit note
B Stores debit note
C Materials returned note
D Materials transfer note **(2 marks)**

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19.10 Analysis of the gross wages in a factory reveals:

	Direct operatives	*Indirect operative*
	$	$
Productive hours at basic rate	41,200	17,600
Overtime premium	1,100	450
Idle time	760	
Group bonuses	2,780	
Total gross pay	45,840	18,050

What amount would NORMALLY be accounted for as production overhead?

A $18,050
B $18,810
C $21,590
D $22,690 **(2 marks)**

19.11 **Which of the following are aspects of payroll systems?**

(i) Attendance records
(ii) Calculation of bonuses
(iii) Employee tax codes
(iv) Apportionment of wages to cost centres

A (i), (ii) and (iii) only
B (iii) and (iv) only
C (i), (ii) and (iv) only
D All four items **(2 marks)**

19.12 The direct labour capacity ratio for a period was 104%.

What could have caused this?

A Actual hours worked being greater than budgeted hours
B Actual hours worked being less than budgeted hours
C Standard time for actual output being greater than budgeted hours
D Standard time for actual output being less than budgeted hours **(2 marks)**

19.13 The following may occur depending upon how overhead absorption rates are set and used:

(i) Delay in the establishment of job costs
(ii) Change in unit costs reflecting seasonal activity
(iii) Overhead over or under recovery

Which of the above may result from the use of predetermined rates set for a year rather than actual rates recalculated every three months?

A (i) only
B (ii) only
C (iii) only
D None of the above **(2 marks)**

19.14 A company manufactures and sells 4,000 units of a product each month at a selling price of $22 per unit. The prime cost of the product is $11.60 per unit and the monthly overheads are:

	$
Variable production	7,200
Variable selling and administration	5,200
Fixed production	16,400
Fixed selling and administration	6,800

What is the product's gross profit margin (to one decimal place)?

[ ]% **(2 marks)**

19.15 A product has the following costs:

	$/unit
Variable production costs	4.80
Total production costs	7.50
Total variable costs	5.90
Total costs	10.00

11,400 units of the product were manufactured in a period during which 11,200 units were sold.

What is the profit difference using absorption costing rather than marginal costing?

A The profit for the period is $540 lower
B The profit for the period is $540 higher
C The profit for the period is $820 lower
D The profit for the period is $820 higher **(2 marks)**

19.16 A job cost estimate includes 630 productive labour hours. In addition, it is anticipated that idle time will be 10% of the total hours paid for the job. The wage rate is $12 per hour.

What is the total estimated labour cost for the job?

A $6,804
B $7,560
C $8,316
D $8,400 **(2 marks)**

19.17 Costs incurred in a process totalled $216,720 for a period. 24,000 units of finished product were manufactured including 1,200 units which were rejected on inspection and disposed of. The level of rejects in the period was normal. Rejects are sold for $2.00 per unit.

What was the cost per unit for the process (to two decimal places)?

$ ☐ **(2 marks)**

19.18 **Which of the following will affect profit in the same way as it affects the amount of cash in a business for a period?**

A Purchase of a new computer for a business
B Accrual of the electricity bill
C Payment of the monthly wages at the end of the month
D Sale of a company car **(2 marks)**

19.19 **Are the following statements about joint product cost apportionment true or false?**

	True	*False*
Using the sales value method of cost apportionment, and where there is no further processing, the gross profit margin of each product will be the same.	○	○
Using the units of output method of cost apportionment, the joint cost per unit will be the same for all joint products.	○	○

(2 marks)

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19.20 The following extracts are from Sarah Co's financial statements.

	$
Non-current assets	250,000
Inventory	56,000
Receivables	12,000
Overdraft	2,000
Payables	15,000
Accruals	1,500

What is Sarah Co's working capital?

A $49,500
B $51,000
C $53,500
D $299,500 **(2 marks)**

(Total = 40 marks)

20 Mixed bank 4 48 mins

20.1 Sources of useful data may be:

(i) External
(ii) Internal
(iii) Financial
(iv) Non-financial

Which of the above sources may be used by a trainee accountant?

A (i) and (ii) only
B (ii), (iii) and (iv) only
C (iii) only
D (i), (ii), (iii) and (iv) **(2 marks)**

20.2 **Which TWO of the following statements about cost and management accounting are true?**

☐ Cost accounting cannot be used to provide inventory valuations for external financial reporting.
☐ There is a legal requirement to prepare management accounts.
☐ The format of management accounts may vary from one business to another.
☐ Management accounting provides information to help management make business decisions.

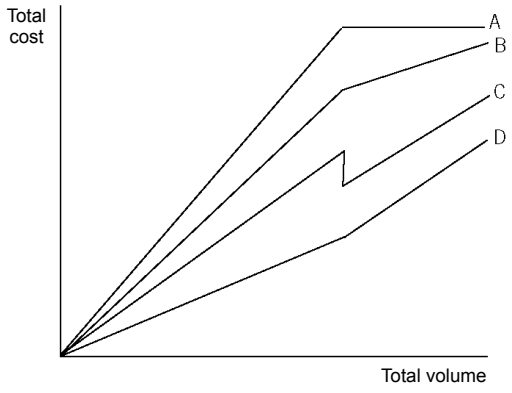
(2 marks)

20.3 **Which of the following are features of an efficient and effective cost coding system?**

(i) Codes need to be complex to include all items.
(ii) Each code must have a combination of alphabetic and numeric characters.
(iii) Codes for a particular type of item should be consistent in length and structure.

A (i) only
B (iii) only
C (i) and (ii)
D (ii) and (iii) **(2 marks)**

20.4 Four cost behaviour patterns are demonstrated on the chart below.



Which line on the chart represents the behaviour of total raw material costs where a volume discount applies to all purchases in a period once a required level is reached?

A Line A
B Line B
C Line C
D Line D **(2 marks)**

20.5 Production costs have been estimated at two levels of output:

	50,000 units	*55,000 units*
Prime costs	$430,000	$473,000
Overheads	$330,000	$339,000

Using the high-low method, what are the estimated production costs per unit at an output level of 54,000 units (to two decimal places)?

$ ☐ **(2 marks)**

20.6 A particular cost is classified as 'semi-variable'.

What effect would a 15% reduction in activity have on the unit cost?

A Increase by less than 15%
B Increase by 15%
C Reduce by less than 15%
D Remain constant total volume total cost **(2 marks)**

20.7 The inventory record of a raw material has the following details for a week:

Day	*Cost ($ per unit)*	*Receipts (units)*	*Issues (units)*
2	260	18	
3	270	12	
4			10
6			14

The first-in first-out (FIFO) method is used for pricing issues. There was no raw material at the start of Day 1.

Which was the value of the inventory on Day 5?

A $5,200
B $5,220
C $5,320
D $5,400 **(2 marks)**

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20.8 Average usage of a raw material is 200 kg per day, the average ordering lead time is five days, the reorder level is 1,600 kg and the reorder quantity is 2,800 kg.

What is the average raw material inventory?

☐ kg **(2 marks)**

20.9 The costs associated with labour turnover can be classified as 'preventative' costs or 'replacement' costs.

Which of the following is a preventative cost?

A Provision of leisure facilities for employees
B Lower productivity of new employees
C Increased wastage of raw materials
D Training costs for new employees **(2 marks)**

20.10 Consider the following statements, regarding the reapportionment of service cost centre overheads to production cost centres, where reciprocal services exist.

Which TWO statements are correct?

☐ The direct method results in costs being reapportioned between service cost centres.

☐ If the direct method is used, the order in which the service cost centre overheads are reapportioned is irrelevant.

☐ The step down method results in costs being reapportioned between service cost centres.

☐ If the step down method is used, the order in which the service cost centre overheads are reapportioned is irrelevant. **(2 marks)**

20.11 A firm uses job costing. Details of the three jobs worked on during a period are:

	Job BA	*Job DC*	*Job FE*
	$	$	$
Opening work-in-progress	22,760	3,190	–
Direct materials in the period	4,620	11,660	14,335
Direct labour in the period	12,125	10,520	7,695

Overheads are absorbed at 40% of prime cost in each period. Jobs DC and FE remained incomplete at the end of the period.

What is the value of the closing work-in-progress?

A $61,894
B $65,084
C $66,360
D $68,952 **(2 marks)**

20.12 Costs totalling $4,250 were incurred in a process in a period. 80 units of output were rejected and destroyed in the period, 20 units more than allowed for as a normal loss, leaving 420 units of good production to be transferred to finished goods.

What is the amount written off as abnormal loss (to the nearest $)?

A $170
B $177
C $193
D $202 **(2 marks)**

20.13 Consider the following statements relating to process costing:

Statement 1: Normal losses are credited to the process account at the cost per unit incurred on normal production.

Statement 2: Abnormal gains are debited to the process account at the cost per unit incurred on normal production.

Which statement(s) is/are true?

A Both statements are true
B Neither statement is true
C Statement 1 only is true
D Statement 2 only is true **(2 marks)**

20.14 **When communicating information, which of the following determine(s) the choice of method used?**

(i) Comparative cost
(ii) Degree of confidentiality
(iii) Speed of delivery

A (i) only
B (iii) only
C (i) and (ii) only
D (i), (ii) and (iii) **(2 marks)**

20.15 **What is a by-product?**

A A product that has insignificant saleable value compared with the joint products
B A product that has no saleable value
C A product that can be further processed
D A waste product that has to be disposed of at a cost **(2 marks)**

20.16 5,400 units of a company's single product were sold for a total revenue of $140,400. Fixed costs in the period were $39,420 and net profit was $11,880.

What was the contribution per unit?

A $7.30
B $9.50
C $16.50
D $18.70 **(2 marks)**

20.17 A company manufactures and sells four products. Details are as follows:

	Product			
	P	*Q*	*R*	*S*
	$	$	$	$
Contribution per unit	16.0	14.5	17.6	19.0
Net profit per unit	4.6	4.8	5.2	5.0
Contribution per machine hour	5.0	4.8	4.4	3.8
Net profit per machine hour	1.4	1.6	1.3	1.0

Machine hours available in the next period will not be sufficient to meet production requirements. There are no product-specific fixed costs.

What should be the order of priority for production in order to maximise profit?

A Product P, Product Q, Product R, Product S
B Product Q, Product P, Product R, Product S
C Product R, Product S, Product Q, Product P
D Product S, Product R, Product P, Product Q **(2 marks)**

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20.18 A company has incurred development costs of $25,000 to date on a proposed new product. Further costs of $18,000 would be required to complete the development of the product.

In deciding whether to continue with the new product development which of the following is correct regarding development costs?

	Sunk cost	*Incremental cost*
A	$0	$43,000
B	$18,000	$25,000
C	$25,000	$18,000
D	$43,000	$0

(2 marks)

20.19 A company is proposing to launch a new product. Incremental net cash inflows of $36,000 per annum for five years are expected, starting at Time 1.

An existing machine, with a net book value of $85,000, would be used to manufacture the new product. The machine could otherwise be sold now, Time 0, for $60,000. The machine, if used for the manufacture of the new product, would be depreciated on a straight-line basis over five years, starting at Time 1.

What are the relevant amounts that should be used, at Time 0 and Time 1, in the discounted cash flow appraisal of the project?

	Time 0	*Time 1*
A	$0	$19,000
B	$0	$24,000
C	($60,000)	$36,000
D	($85,000)	$36,000

(2 marks)

20.20 **Which of the following could cause differences between a company's cash flow forecast and its actual cash flows?**

(i) Poor forecasting techniques
(ii) Unexpected inflation
(iii) Loss of a major customer
(iv) Interest rate changes

A (i) and (iii) only
B (i), (ii) and (iv) only
C (ii), (iii) and (iv) only
D (i), (ii), (iii) and (iv) **(2 marks)**

(Total = 40 marks)

21 Mixed bank 5 48 mins

21.1 **Which TWO of the following are characteristics of management accounting information?**

☐ Forward looking
☐ Legally required
☐ Concerned with cost control
☐ Follows clearly defined standards **(2 marks)**

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21.2 **Which of the following best describes an investment centre?**

A Part of a business that uses non-current assets
B Part of a business that provides a service for other parts of the business
C Part of a business where management is only responsible for investment costs
D Part of a business where management is responsible for capital investment as well as profit

(2 marks)

21.3 Consider the following four accounts:

(i) Cost ledger control
(ii) Financial ledger control
(iii) Receivables control
(iv) Work-in-progress control

Which of the accounts are features of an integrated accounting system?

A (i) and (ii) only
B (i), (ii) and (iii)
C (ii), (iii) and (iv)
D (iii) and (iv) only **(2 marks)**

21.4 **When goods are sold, what double-entry would be made to record the transfer of costs?**

	Debit	*Credit*	
A	Finished goods account	Cost of sales account	
B	Sales account	Cost of sales account	
C	Cost of sales account	Sales account	
D	Cost of sales account	Finished goods account	**(2 marks)**

21.5 An extract from the list of accounts of a chemical processor follows:

	Cost codes
Direct materials	001 to 099
Direct labour	100 to 199
Indirect materials	200 to 299
Indirect labour	300 to 399

Which of the following items is coded INCORRECTLY?

	Description	*Code*	
A	Chemical C6 used in process	061	
B	Wages of process supervisor	106	
C	Chemicals used for cleaning	229	
D	Wages of maintenance engineer	345	**(2 marks)**

21.6 **Which of the following may be used for capturing or storing management accounting data by computer?**

(i) Scanner
(ii) Printer
(iii) CD
(iv) Bar code reader

A (i) and (ii) only
B (iii) and (iv) only
C (i), (iii) and (iv)
D (i), (ii), (iii) and (iv) **(2 marks)**

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21.7 **Which of the following are the key factors a company should consider when considering the investment of surplus funds?**

A Safety, exposure and profitability
B Safety, exposure and liquidity
C Exposure, profitability and liquidity
D Safety, profitability and liquidity **(2 marks)**

21.8 The cost per unit of an expense item at different levels of activity is shown below:

Activity (units)	*Cost per unit*
	$
200	12.00
300	8.00
400	7.00
500	5.60

What is the behavioural classification of the expense item?

A Fixed cost
B Semi-variable cost
C Stepped-fixed cost
D Variable cost **(2 marks)**

21.9 A particular cost is classified as being semi-variable.

What is the effect on the TOTAL COST if activity increases by 20%?

A Stays the same
B Decreases by less than 20%
C Increases by 20%
D Increases by less than 20% **(2 marks)**

21.10 Analysis of the gross wages in a factory reveals the following:

	Direct operatives	*Indirect operatives*
	$	$
Normal hours worked at basic rate	36,260	14,320
Overtime hours at basic rate	4,112	1,760
Overtime premium	1,028	440

Overtime working is a usual aspect of running the business.

What amount would be recorded as a direct cost?

$ ☐ **(2 marks)**

21.11 The production volume ratio for a period was 95%.

What could have caused this?

A Actual hours worked being greater than budgeted hours
B Actual hours worked being less than budgeted hours
C Standard time for actual output being greater than budgeted hours
D Standard time for actual output being less than budgeted hours **(2 marks)**

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21.12 A cost centre is charged with the following actual overhead costs for a period:

Allocated costs	$28,720
Apportioned costs	$10,260

Overheads were absorbed in the cost centre over the period on 1,760 actual labour hours at a predetermined absorption rate of $21.50 per hour. Actual labour hours worked in the period were 90 hours above budget.

What was the overhead over/under absorption in the cost centre (to the nearest whole number)?

$ [ ] **(2 marks)**

21.13 A company sells more than it manufactures in a period.

Which of the following explains the difference in profit between absorption and marginal costing in the above situation?

A Absorption costing profit is higher because of the difference in inventory levels.
B Absorption costing profit is lower because of the difference in inventory levels.
C Absorption costing profit is higher because of overhead over-absorption.
D Absorption costing profit is lower because of overhead under-absorption. **(2 marks)**

21.14 The production cost of Job J6 was $12,600. Administration costs are charged to jobs at 30% of production cost. The amount charged to the customer is calculated to provide a **GROSS PROFIT MARGIN** of 40%.

What is the net profit on Job J6?

A $1,260
B $4,620
C $10,920
D $15,120 **(2 marks)**

21.15 The following information relates to a process for a period:

Input costs	$45,705
Output passing inspection	9,600 units
Normal inspection loss	300 units (sold for $2.00 per unit)
Abnormal inspection loss	100 units (sold for $2.00 per unit)

What was the cost per unit (to two decimal places of $)?

$ [ ] **(2 marks)**

21.16 A process produces two joint products, X and Y, with selling prices of $10 per litre and $20 per litre respectively. In a period, joint costs were $56,000 and finished output was:

Product X	5,000 litres
Product Y	2,000 litres

The sales value method is used to apportion joint costs.

What amount of joint costs should be apportioned to Product Y?

A $16,000
B $24,889
C $31,111
D $37,333 **(2 marks)**

21.17 **What is the effective annual rate of interest of 4.3% compounded every six months?**

A 8.60%
B 8.78%
C 9.25%
D 10.88% **(2 marks)**

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21.18 Net cash flows, estimated for a capital investment project, have been discounted at four discount rates with the following results:

	Discount rate			
	5%	*10%*	*15%*	*20%*
Net present value ($'000)	92.9	39.1	(4.8)	(40.9)

What is the best estimate of the IRR using only the above data as appropriate?

A 13.6%
B 14.5%
C 15.4%
D 15.7% **(2 marks)**

21.19 A project, investing in new machinery, has an estimated five year life. The cost of capital is 10% per annum.

Estimated cash flows are:

Time	*Cash flows*
0	(cost) ($186,000)
1 to 5	(inflows) $56,000 per annum
5	(residual value) $10,000

The cumulative discount factor at 10% for Time 1 to 5 is 3.79. The discount factor at 10% for Time 5 is 0.62.

What is the net present value of the project?

A $16,240
B $20,040
C $32,440
D $36,240 **(2 marks)**

21.20 A capital investment project requires an initial investment sum. The investment returns are expected to be a constant amount in each year of the life of the investment.

How is the payback period for the investment calculated?

A Investment sum / net cash inflow per annum
B Investment sum / net profit per annum
C (Investment sum + residual value) / net cash inflow per annum
D (Investment sum + residual value) / net profit per annum **(2 marks)**

(Total = 40 marks)

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22 Mixed bank 6 48 mins

22.1 Consider the following incomplete statements relating to features of management information:

(i) Communicated in writing
(ii) Presented in report format
(iii) Supported by calculations
(iv) Timely and clear to the user

Which of the above are necessary features of useful management information?

A (i) and (iv) only
B (i), (ii) and (iii)
C (iv) only
D (ii) and (iii) only **(2 marks)**

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22.2 **Which of the following performance measures would be appropriate for an investment centre rather than a profit centre or a cost centre?**

A Contribution/sales ratio
B Cost per unit
C Labour efficiency ratio
D Residual income **(2 marks)**

22.3 W**hat are government bonds also known as?**

A Certificates of deposit
B Bills
C Gilts
D Securities **(2 marks)**

22.4 The cost accounts of a business are kept separate from the financial accounts but the two sets of accounts are reconciled each period.

What accounting system is being described?

A Cost control accounts
B Cost ledger accounts
C Integrated accounts
D Interlocking accounts **(2 marks)**

22.5 Cost Z is fixed in total for a period.

If the level of activity in the period is increased by 50% what change would occur in Cost Z per unit of activity?

A Decrease by a third
B Decrease by a half
C Increase by a third
D Increase by a half **(2 marks)**

22.6 In a factory, a team of six maintenance staff are paid a guaranteed weekly wage.

Which of the following is the most appropriate cost classification for their wages?

A Direct labour cost
B Indirect labour cost
C Semi-variable cost
D Variable overhead cost **(2 marks)**

22.7 Production costs incurred in the manufacture of 2,400 units of a product in a period are:

	$
Direct costs	19,680
Variable overheads	3,120
Fixed overheads	14,640

What would be the expected total cost of manufacturing 2,300 units of the product in a period?

$ ☐ **(2 marks)**

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22.8 The following documents are used in the process of purchasing and using raw materials:

Which TWO of the documents would be used to update the stores ledger accounts?

- ☐ Dispatch note
- ☐ Goods received note
- ☐ Materials requisition
- ☐ Purchase order
- ☐ Purchase requisition

(2 marks)

22.9 A company produced 6,200 units of a product in a period. The product used 80 kg of material per 100 units of output. The inventory holding of the material reduced by 380 kg in the period.

What quantity of material was purchased in the period?

A 4,580 kg
B 4,960 kg
C 5,340 kg
D 7,370 kg

(2 marks)

22.10 A firm has used the economic order quantity (EOQ) formula to arrive at an EOQ for Component C1 of 400 units. Annual demand for Component C1 is 12,000 units and the cost of placing an order for the component is $40.

What is the cost of holding one unit of Component C1 in inventory for one year?

A $0.17
B $0.33
C $3.00
D $6.00

(2 marks)

22.11 'Materials Stores' is one of the service cost centres in a factory.

What would be the most appropriate basis for the reapportionment of the overheads of Materials Stores to the cost centres it serves?

A Number of materials requisitions
B Number of purchase requisitions
C Reorder level of each material
D Value of materials inventory

(2 marks)

22.12 Production cost centre X absorbs overheads on the basis of machine hours and has the following budgeted and actual figures:

	Budget	*Actual*
Overheads	$20,290	$19,110
Machine hours	560	514

What is the predetermined production overhead absorption rate in production cost centre X (to two decimal places)?

A $34.13
B $36.23
C $37.18
D $39.47

(2 marks)

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22.13 There are two production cost centres in a factory. Production overhead absorption rates are:

Cost centre A	$10.60 per direct labour hour
Cost centre B	$36.20 per machine hour

Product P requires the following hours per unit of finished product in each cost centre:

	Cost centre	
	A	B
Direct labour hours	3.0	0.5
Machine hours	0.2	1.3

What is the total production overhead cost per unit of Product P?

A $78.86
B $91.40
C $99.08
D $234.00

(2 marks)

22.14 **Are the following statements true or false?**

	True	*False*
The difference between the profit reported by absorption costing and that reported by marginal costing is due to over or under absorption of overhead.	○	○
Absorption costing profit will be higher than marginal costing profit if sales units exceed production units.	○	○

(2 marks)

22.15 **In which of the following organisations would job costing be most appropriate?**

A College
B Hospital
C Car repairer
D Chemical manufacturer

(2 marks)

22.16 Manufacturing process costs total $179,070 for a period. 9,000 kg of raw materials were processed with the following result:

Completed good output	8,100 kg
Normal loss	1,200 kg
Abnormal gain	300 kg

What was the cost per kg (to two decimal places)?

$ ☐

(2 marks)

22.17 Consider the following two descriptions:

(1) A product which is incidental to the main purpose of a process
(2) A product which has an insignificant value relative to other products from a process

Do the above descriptions apply to a by-product?

	Description (1)	*Description (2)*
A	No	No
B	No	Yes
C	Yes	No
D	Yes	Yes

(2 marks)

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22.18 **Which of the following statements is true of service costing?**

A A composite cost unit may be used.
B Indirect costs normally represent a small proportion of total costs.
C Output is often tangible.
D The cost of direct materials tends to be high in relation to other costs. **(2 marks)**

22.19 Budgeted sales of a company's single product in a period are 20,000 units, producing a total contribution of $180,000 at a selling price of $24 per unit. Fixed costs are $6 per unit based on the budgeted sales quantity.

What is the budgeted variable cost per unit?

A $3
B $9
C $15
D $18 **(2 marks)**

22.20 A firm makes a single product. Budgets have been prepared for the year ahead and include production and sales of 60,000 units with a break-even point of 45,000 units.

What is the margin of safety ratio?

A 25%
B 33%
C 75%
D 133% **(2 marks)**

(Total = 40 marks)

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Answers

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1 Management information and information technology

1.1 D This is the correct definition of a management information system.

1.2 B Management accounts are produced for the internal managers of an organisation. The other groups of people would use the financial accounts of an organisation.

1.3 D Electronic point of sale (EPOS) devices act as both cash registers and as terminals connected to a main computer. Fully itemised, accurate and descriptive receipts can be produced for customers who will also benefit from faster moving queues at the checkout.

1.4 D Laser printers print a whole page at a time, rather than line by line. They are more expensive than other types of printer and can print up to 500 pages per minute.

1.5 C Marginal costing is a common feature of cost accounting but not financial accounting.

1.6 The correct answers are:

- Clear to the user
- Relevant for purpose

Management information should be clear to the user and relevant for its purpose. It doesn't necessarily need to be detailed or completely accurate in order to be useful. Sometimes estimates are sufficient. The benefits of providing management information should, however, outweigh the costs of providing it.

1.7 The correct answers are:

- Icons
- Pull-down menu

Graphical user interfaces are the means by which humans communicate with machines and feature windows, icons, a mouse and pull-down menus.

1.8 C Management information is used for planning, control and decision-making.

1.9 B Management accounting data can be captured with bar codes and stored on hard disks and DVD. Printers are used for the output of data.

1.10 The correct answers are:

- Relevant
- Timely

Good management information should be relevant and timely (remember mnemonic ACCURATE).

1.11 C In a faceted code, each digit of the code gives information about the item.

1.12 D MICR is Magnetic Ink Character Recognition and involves the recognition by a machine of special formatted characters printed in magnetic ink.

1.13 A In a large business the trainee accountant is likely to be responsible for coding invoices.

1.14 C The financial statements of competitors are an external source of management information.

2 Cost classification and cost behaviour

2.1 C A cost unit is a unit of product or service.

2.2 C Option A is the definition of a cost unit.

Option B describes the **cost** of an activity or cost centre.

Option D describes an investment centre.

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2.3 B Prime cost is the total of direct material, direct labour and direct expenses. Therefore the correct answer is B.

Option A describes total production cost, including absorbed production overhead. Option C is only a part of prime cost. Option D is an overhead or indirect cost.

2.4 A Option A is a part of the cost of direct materials.

Options B and D are production overheads. Option C is a selling and distribution expense.

2.5 A Depreciation is an indirect cost because it does not relate directly to a specific cost unit produced.

Items (ii) and (iii) can be traced directly to specific cost units, therefore they are direct expenses.

2.6 A A direct expense

The royalty cost can be traced in full to units of the product, therefore it is a direct expense.

2.7 The correct answers are:

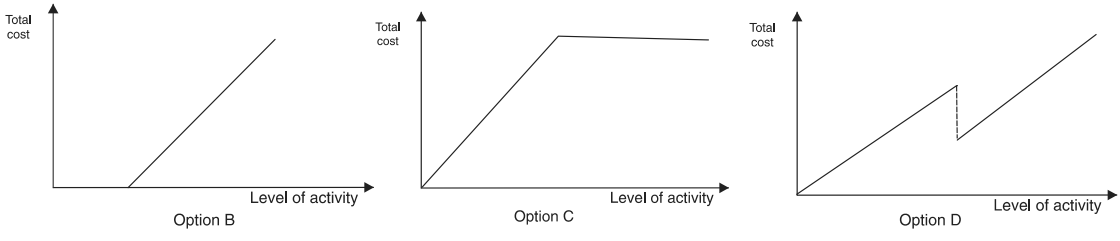
- Cheque received and processed
- Customer account

Both the telephone expense and the credit control manager's salary are costs for the department, not potential cost units. A cost unit is a unit of product or service which has costs attached to it.

2.8 B The stores assistant's wages cannot be charged directly to a product, therefore the stores assistant is part of the indirect labour force.

2.9 A The depicted cost has a basic fixed element which is payable even at zero activity. A variable element is then added at a constant rate as activity increases. Therefore the correct answer is A.

Graphs for the other options would look like this.



2.10 A The cost described will increase in steps, remaining fixed at each step until another supervisor is required. Graph 1 depicts a step cost therefore the correct answer is A.

2.11 D The salary is part fixed ($650 per month) and part variable (5 cents per unit). Therefore it is a semi-variable cost and answer D is correct.

If you chose options A or B you were considering only part of the cost.

Option C, a step cost, involves a cost which remains constant up to a certain level and then increases to a new, higher, constant fixed cost.

2.12 $0.50

Variable cost for 340 guest-nights	= $570 – $400 = $170
Variable cost per guest-night	= $170/340 = $0.50

2.13 B Graph 2

2.14 A Graph 1

2.15 C A semi-variable or mixed cost is a cost which contains both fixed and variable components and so is partly affected by changes in the level of activity.

2.16 A Graph 1

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2.17 $10.65

	Production costs	*Output*
	$	$
Highest	254,554	14,870
Lowest	230,485	12,610
	24,069	2,260

$$\text{Variable production costs per unit} = \frac{\$24,069}{2,260 \text{ units}}$$

$$= \$10.65 \text{ per unit}$$

2.18 A Increased storage requirements

2.19 B As activity levels rise, the fixed production cost per unit will fall. This is the situation that is shown in the graph given in the question.

2.20 C Suppose fixed costs for a period = $10,000 and activity in period = 10,000 units.

$$\text{Fixed cost per unit} = \frac{\$10,000}{10,000 \text{ units}} = \$1 \text{ per unit}$$

If the activity level in the period falls by 50%, ie to 5,000 units then:

$$\text{Fixed cost per unit} = \frac{\$10,000}{5,000 \text{ units}} = \$2 \text{ per unit}$$

A reduction in activity of 50% therefore causes the fixed cost per unit to increase from $1 to $2, ie an increase in cost per unit of 100%.

2.21 The correct answers are:

- Direct
- Production

Direct and production could be applied to the cost of raw materials used by a company in the manufacture of its range of products.

2.22 B A semi-variable cost is a cost which contains both fixed and variable components and is partly affected by changes in the level of activity. Therefore, if activity increases by 10%, total variable costs will increase by 10% and fixed costs will remain constant. The total costs will therefore increase, but **not** in proportion to the change in activity.

2.23 D Tyre replacement

2.24 D The cost of the tools is a **direct cost** of the job because it can be specifically identified with the job.

2.25 B Profit centres, revenue centres and investments centres are all examples of responsibility centres within an organisation.

2.26 The correct answers are:

- A car manufactured in a car factory
- A pizza sold by a pizza restaurant

A cost centre is something that incurs costs as it operates (for example a car factory or a pizza restaurant). A cost unit is the ultimate product or service to which the cost centre costs are allocated (such as a car or a pizza). A hairdresser is a cost centre (their cost unit would be the hairdressing service that they provide). An accounts department is a cost centre (their cost unit would be the software sold by their company).

2.27 B

ACCA examining team comments

Only a few candidates selected the correct answer in this question as B. The variable costs are \$7.80 and \$1.60 (from the semi-variable \$5.30 – \$3.70 = \$1.60) if you multiply these two figures by the 1,200 units, this will total \$11,280 for variable costs. Then add the fixed cost as \$10,200 which is \$6.50 and \$3.70 from the semi-variable, multiplied by the original budgeted 1,000 units. Therefore the total cost is \$21,480 which is \$11,280 plus \$10,200. Most candidates selected A which incorrectly takes all costs as variable, so calculating \$19.60 × 1,200 units which would give \$23,520. Others selected \$21,160 which is \$11.80 per unit for fixed cost made up of the \$6.50 given in the question as fixed plus all of the semi-variable \$5.30 taken as fixed in error. Thus 1,000 units at \$11.80 is \$11,800 plus \$7.80 as variable cost per unit multiplied by 1,200 units is \$9,360 giving a total cost of \$21,160. Those that chose \$21,900 had calculated variable as \$13,800 which is \$7.80 plus \$3.70 multiplied by 1,200 and added fixed cost of \$8,100 which is \$6.50 plus \$1.60 in error multiplied by 1,000. Thus many candidates displayed a complete lack of understanding of treatment of variable and fixed costs.

2.28 \$23 per unit

ACCA examining team comments

	\$
Variable cost per unit: 15 × 25,000	375,000
Fixed cost per unit: 10 × 20,000	200,000
	575,000
divided by 25,000 units	23 per unit

3 Information for comparison

3.1 C A flexed budget is a budget that reflects the actual activity level.

3.2 A Flexed budget = \$100,500 × 14,600/15,000
= \$97,820

3.3 C Flexed budget = \$84,600 × 13,900/12,500
= \$94,075

3.4 D All three options provide meaningful comparisons for revenue analysis.

3.5 C Exception reporting is the reporting of variances which exceed a certain amount.

3.6 C Profit centres A and B have lower sales in December 20X5 compared to December 20X4.

3.7 The correct answers are:

- Favourable variances are always good for an organisation – false
- Variance reporting is the comparison of the actual results with budgeted results – true

Favourable variances may not always be good. For example, a favourable materials variance might be achieved by buying poorer quality material which means that the labour force have to spend much longer working on the material leading to an adverse labour variance.

Variance reporting can include the reporting of differences between the actual results and the flexed budget or the original budget.

3.8 \$5,190

	\$	
Flexed budget 5,300 units × \$12.30	65,190	
Actual cost	60,000	
Variance	5,190	favourable

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3.9 A

	$	
Flexed budget 12,200 units × $6.80	82,960	
Actual cost	85,200	
Variance	2,240	adverse
Percentage of budget 2,240/82,960 × 100	2.7%	adverse

3.10 A

	$	
Flexed budget 3,700 units × $6.00	22,200	
Actual cost	25,600	
Variance	3,400	adverse
Percentage of budget 3,400/22,200 × 100 =	15.3%	adverse

3.11 2.1%

	$	
12,600 units × $15.50	195,300	
Actual cost	199,400	
Variance	4,100	adverse
Percentage of budget 4,100/195,300 × 100 =	2.1%	adverse

3.12 B Responding to an anticipated fall in sales revenue is a feedforward control (not a feedback control) as it happens in advance. The other options all occur after the event when the problems have already occurred, and are therefore feedback controls.

3.13 A

Direct materials	2,589/24,000 × 100	= 10.8%
Direct labour	2,137/16,200 × 100	= 13.2%
Direct expenses	72/2,100 × 100	= 3.4%

3.14 C

	$	
Revenue should have been $2.50 × 8,000	20,000	
Revenue actually was	19,968	
	32	adverse

3.15 B

Budgeted sales volume	8,000	units
Actual sales volume	8,320	units
Activity variance in units	320	units (favourable)
× Budgeted selling price per unit	× $2.50	
Activity variance	$800	favourable

3.16 A

	$	
Sales revenue from 8,320 units should have been (× $2.50)	20,800	
But was	19,968	
	832	adverse

3.17 A

	$	
Fixed budget $3.20 × 60,000	192,000	
Did cost	187,200	
	4,800	favourable

3.18 B

Budget production volume	60,000	units
Actual production volume	62,400	units
Activity variance in units	2,400	units A
× budgeted cost per unit	× $3.20	
Activity variance	$7,680	adverse

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3.19 C

	$	
Material cost for 62,400 units should have been (× $3.20)	199,680	
But was	187,200	
Direct materials variance	12,480	favourable

3.20 $6,773

	$	
Sales revenue should have been (521 × $300)	156,300	
But was (521 × $287)	149,527	
Sales price variance	6,773	(A)

3.21 B Statement (i) is consistent with an adverse material price variance. Higher quality material is likely to cost more. Statement (ii) is consistent with an adverse material price variance. Removal of bulk discounts would result in a higher material price.

Statement (iii) is not consistent with an adverse material price variance. **Favourable** variances would result if the budget was set too high.

Therefore the correct answer is B.

3.22 C Statement (i) is not consistent with a favourable labour efficiency variance. Employees of a lower skill level are likely to work less efficiently, resulting in an **adverse efficiency variance**.

Statement (ii) is consistent with a favourable labour efficiency variance. **Time would be saved in processing** if the material was easier to process.

Statement (iii) is consistent with a favourable labour efficiency variance. **Time would be saved in processing** if working methods were improved.

Therefore the correct answer is C.

3.23 A A wage rate increase could result in an **adverse direct labour rate variance**, it is not a cause of an adverse labour efficiency variance. Therefore Reason A is not consistent with the variance and the correct answer is A.

Options B, C and D could all result in a loss of active production time and therefore reduced labour productivity.

3.24 A All of the statements are consistent with a favourable labour rate variance. Therefore the correct answer is A.

Employees of a lower grade (statement (i)) are likely to be paid a **lower hourly rate**.
An unrealistically high budget (statement (ii)) would result in **favourable rate variances**.
If a pay increase did not occur (statement (iii)) this would lead to **savings in labour rates**.

3.25 The correct answer is: Both statements are consistent with the variances.

The direct material price variance is $2,000 adverse ($800 adverse – $1,200 favourable).

Both statements are consistent with the variances, because both situations would lead to a **higher price** for materials (adverse material price variance) and **lower usage** (favourable material usage variance).

3.26 B Foreseeing a variance and taking corrective action is a feedforward control as it happens in advance. The other options all occur after the event when the problems have already occurred.

3.27 D The personnel involved should have no bearing on the decision whether or not to investigate a variance.

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3.28 C

	Units	$
High activity	14,000	10,500
Low activity	8,000	7,200
Increase	6,000	3,300

Variable cost per unit = $3,300/6,000 = $0.55

Fixed cost, substituting in high activity = $10,500 – (14,000 × 0.55)
= $2,800

Budget cost allowance to distribute 12,000 units:

	$
Variable cost (12,000 × $0.55)	6,600
Fixed cost	2,800
	9,400

If you selected **option A** you did not include an allowance for fixed cost, and if you selected **option B** or **D** you calculated the answer on a pro rata basis from the data given. This does not take account of the fixed element of the distribution cost.

3.29 A $4,755 (A)

The activity variance is the increase in cost resulting from a change in the volume of activity, ie the difference between the original budget and the flexed budget.

Activity variance = $126,100 – $130,855
= $4,755 (A)

3.30 D $2,725 (A)

The rate/efficiency variance is the difference between the flexed budget and the actual results.

Rate/efficiency variance = $130,855 – $133,580
= $2,725 (A)

4 Reporting management information

4.1 A All three factors would influence the method of communication used.

4.2 B Yours sincerely.

4.3 The correct answers are:

- Email can be used for confidential information – true
- Email should not be used where a signature is necessary – false

Provided that passwords are used and protected, email can be used for confidential information. Email can be used with an attachment that has a signature.

4.4 C In this instance, a letter is the most appropriate form of communication.

4.5 D Options A and B are both data users. Option C is the definition of personal data.

4.6 B The independent variable is shown on the x axis. A line graph shows a relationship between two variables and there is always a dependent and an independent variable.

4.7 D Line graph

4.8 A Pie chart

4.9 D This should be referred as you do not know whether the request for information is actually for business purposes.

4.10 C C would be more relevant if it referred to sent messages.

4.11 C Email is the most suitable method to send a message to multiple recipients.

4.12 B An index is not used in a formal report.

4.13 A This is not an objective of house style, documents should have a consistent look under house style.

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4.14 D Option A is correct as the report will 'propose changes needed to bring it into line with current practice', which means that in the future the manual will be applicable to actual practice.

Option B and C are accurate descriptions of the report – commissioned by the Board and conducted by an external firm of management consultants.

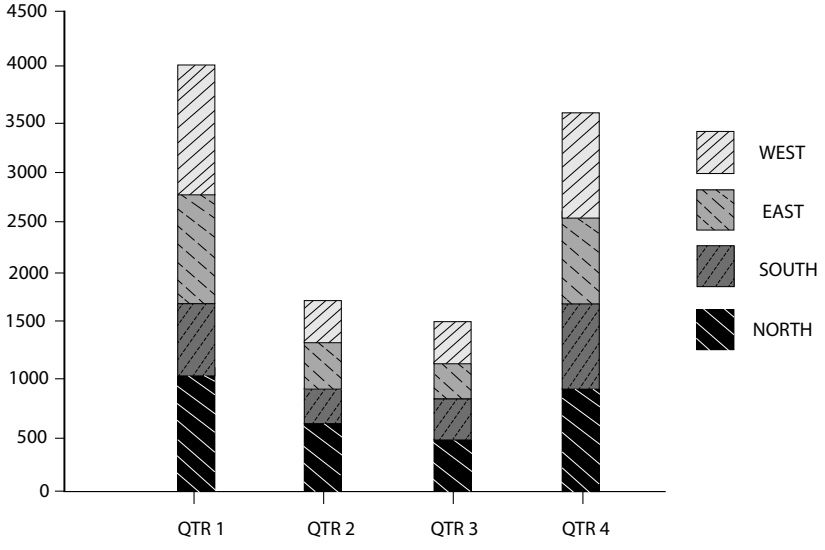
Option D is the correct answer, as the report says nothing about unnecessary changes that have been made in the past.

4.15 B

ACCA examining team comments

With a line graph the dependent variable is on the Y-axis and the independent variable on the X axis. An independent variable is the variable you have control over, you choose to manipulate and test to see if they are the cause. The dependent variable is the one that gets changed when the independent variable is changed.

Here is an example of a component column chart to aid with understanding the question:



Therefore statement 1 is correct. In a component column chart, the width of each column will be the same.

The default settings for a component column chart mean the width of the column will be the same for each column when it is created. Very few candidates correctly selected B; most choose A that both statements were correct and a few selected D that both statements were incorrect by mistake.

5 Materials

5.1 C The lead time, sometimes called the delivery period, is the time between sending a purchase order to the supplier and receiving the goods into stores. Therefore the correct answer is C. None of the other time periods described are in common use in the calculation of inventory control levels.

5.2 D When inventory reaches the predetermined reorder level a replenishment order should be placed with the supplier.

Option A describes the reorder quantity which should be ordered when inventory reaches the reorder level. Option B is the maximum level and Option C is the minimum level.

5.3 880

Average inventory = safety inventory + ½ reorder quantity
= 280 + (0.5 × 1,200)
= 880

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5.4 300 units

The formula for the economic order quantity (EOQ) is:

$$EOQ = \sqrt{\frac{2cd}{h}}$$

with c = \$10
d = 5,400 ÷ 12 = 450 per month
h = \$0.10

$$EOQ = \sqrt{\frac{2 \times \$10 \times 450}{\$0.10}}$$

$$= \sqrt{90{,}000}$$

= 300 units

5.5 2,400 kg

Maximum inventory control level = reorder level + reorder quantity – (minimum usage × minimum lead time)

∴ Maximum inventory control level = 1,600 + 1,400 – (600 × 1)
= 3,000 – 600
= 2,400 kg

5.6 C Minimum inventory control level = reorder level – (average usage × average lead time)

∴ Minimum inventory control level = 1,600 – (700 × 1.5)
= 1,600 – 1,050
= 550 kg

5.7 A Buffer inventory is the inventory that is kept in reserve to cope with fluctuations in demand and with suppliers who cannot be relied upon to deliver the right quality and quantity of materials at the right time. The introduction of buffer inventory would result in the increase of average inventory levels. The introduction of buffer inventory would not have an effect on holding costs, ordering costs nor the EOQ.

5.8 C

Average price of inventory on 23 March:

Units		\$
2,400	× \$6	14,400
4,000	× \$6.20	24,800
2,000	× \$6.86	13,720
8,400		52,920

Average price per component = \$52,920/8,400 = \$6.30
Value of inventory on 23 March = (8,400 – 5,100) × \$6.30
= \$20,790

5.9 B \$31,140

The FIFO method uses the price of the oldest batches first:

Units		\$
2,400	× \$6	14,400
2,700	× \$6.20	16,740
5,100		31,140

5.10 D FIFO uses the price of the oldest items in inventory. When prices are rising this will be the items with the lowest prices. Consequently costs are lower and profits are higher.

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5.11 3,400 units

Reorder level = maximum usage × maximum lead time
= 600 × 4 = 2,400 units

Maximum level = reorder level + reorder quantity – (min. usage × min. lead time)
= 2,400 + 1,900 – (300 × 3)
= 3,400 units

5.12 D The FIFO method prices issues of raw materials in the order that they were received into inventory. The prices paid for the oldest material in inventory are therefore used to price issues of raw material to production. Option B describes the LIFO method of inventory valuation. Option C describes what is likely to be physically happening (any good storekeeper will issue goods in the order that they were received) but this does not have any effect on the cost accounting procedures adopted. It is unlikely that the situation described in Option A would happen. If the last material received were the first issued to production then there would be a greater risk of the oldest items left in inventory becoming obsolete.

5.13 D The inventory count revealed 21 fewer units of Z99 and the **same number** more units of Z100 than expected. Since the two types of bracket are very similar, the most likely explanation of the difference is that some brackets were mistakenly put in the incorrect storage racks.

5.14 B When material prices are falling, FIFO will give rise to **higher production costs** than LIFO. If production costs are higher, then the cost of sales will be higher also. If the cost of sales is higher, then the **profits will be lower.**

5.15 D The economic order quantity is calculated as follows.

$$Q = \sqrt{\frac{2cd}{h}}$$

where h = the cost of holding one unit of inventory for one year
c = the cost of ordering a consignment
d = the annual demand
Q = the economic order quantity

Holding costs (h) include the cost of storing materials (i), the cost of interest (ii) and the cost of insuring materials (iv). The cost of ordering materials (iii) relates to (c) in the EOQ equation. Therefore items (i), (ii), (iii) and (iv) would be needed in order to calculate the economic order quantity.

5.16 A Statement 1 is true but Statements 2 and 3 are false.

If prices are rising then the LIFO method will charge the more recent, higher prices to production costs. Therefore Statement 1 is true, and Statement 2 is false because higher production costs will result in lower profits with LIFO. Inventory valuations with FIFO use the most recent, higher prices therefore Statement 3 is false.

5.17 $375

Average inventory held $= \frac{\text{Order quantity}}{2}$

$= \frac{250 \text{ units}}{2}$

= 125 units

Inventory holding cost = $3 per unit per annum
= $3 × 125 units
= $375

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5.18 B The formula for the economic order quantity is as follows.

$$Q = \sqrt{\frac{2cd}{h}}$$

Where h = inventory holding cost
c = ordering cost
d = demand (usage)

The purchase price is therefore not relevant to the calculation of the economic order quantity of a raw material.

5.19 C

Average inventory held = safety inventory + (order quantity ÷ 2)
= 1,200 + 1,000
= 2,200 kg

Total annual stock holding cost = average inventory held × inventory holding cost per kg
= 2,200 kg × $1.20
= $2,640

5.20 C [(1,100/200 × 60) + 840] ÷ (150 + 60)

5.21 Both statements are true.

5.22 A The minimum inventory control level is calculated as the re-order level minus the average usage in average lead time.

5.23 D Warehouse rent, interest on inventory investment and inventory theft are all examples of holding costs.

5.24 B A goods received note (GRN) will be completed by goods inwards staff on the basis of a physical check, which involves counting the items received and seeing that they are not damaged.

5.25 C The number of quality rejects as a percentage of total output is a typical measure of output wastage.

5.26 C A perpetual inventory system ensures that there is a continuous record of the balance of each item of inventory.

5.27 A

ACCA examining team comments

Because of the 10% wastage, the amount of materials required to be input to achieve 1.6 litres of output is 1.6 ÷ 0.9 = 1.7 recurring. This required input amount is then multiplied by the number of units to be produced in the period. This is the expected sales of the product less the planned reduction in finished goods inventory. Thus the correct answer is [(5,700 – 200 units) × 1.7 recurring per unit] = 9,778 litres (Option A).

The most popular answer, selected by majority of the candidates, was Option D. Candidates selecting this option made the mistake of thinking that the materials input per unit that was required to allow for the 10% wastage was 1.6 × 1.1 = 1.76 litres. This in fact represents only a 9.09% wastage allowance {[(1.76 – 1.6) ÷ 1.76] × 100} thus understating the materials required for production. The incorrect usage per unit of output was then multiplied by the correct production units. For Option C, the answer was based on the correct allowance for wastage but the materials input required was then multiplied by the sales units given in the question, instead of the production units which needed to be calculated. It is the production units that determine material usage requirements. A minority of candidates selected Option B which was also based on the correct allowance for wastage. Here, the candidates made an attempt to calculate the production units but incorrectly added, rather than deducted, the 200 units inventory change when adjusting the sales units.

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5.28 A

ACCA examining team comments

Option A is the correct answer. The minimum control level is defined as the level below which inventory should not fall in average circumstances. It is calculated as the inventory level at which a replenishment order is placed with a supplier (based on maximum expected usage and lead time) less the average usage and lead time. If the minimum control level is reached this is highlighted enabling the situation to be reviewed and action to be taken, if necessary, to avoid a stockout.

Option B is incorrect as the reorder level is the level of inventory at which a replenishment order is raised not the amount purchased. Option C describes a perpetual inventory system not a continuous stocktaking system. Option D describes continuous stocktaking not periodic stocktaking.

5.29 Yes, Yes

ACCA examining team comments

The EOQ is found at the point where total costs (holding + ordering) are at a minimum. This is the same point at which holding costs are equal to ordering costs. The following is formula allows us to calculate the EOQ accurately:

$$Q = EOQ = \sqrt{\frac{2C_oD}{C_h}}$$

Where:

D = Demand per annum

C_o = Cost of placing one order

C_h = Cost of holding one unit for one year – which would be the finance cost of holding a unit for one year

Q = Reorder quantity (EOQ)

Thus, both are correct.

6 Labour

6.1 B The only direct costs are the wages paid to direct workers for ordinary time, plus the basic pay for overtime.

$25,185 + $5,440 = $30,625.

If you selected Option A you forgot to include the basic pay for overtime of direct workers, which is always classified as a direct labour cost.

If you selected Option C you have included overtime premium and bonus payments, which are usually treated as indirect costs. However, if overtime and bonus payments are incurred specifically for a particular cost unit, then they are classified as direct costs of that cost unit. There is no mention of such a situation here.

Option D includes sick pay, which is classified as an indirect labour cost.

6.2 D The personnel manager's salary would be incurred regardless of the level of labour turnover.

Options A, B and C are all costs which would be incurred as a result of employees leaving and being replaced.

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6.3 C Group bonus schemes are useful to reward performance when production is integrated so that all members of the group must work harder to increase output, for example in production line manufacture. Statement (i) is therefore true.

Group bonus schemes are not effective in linking the reward to a particular individual's performance. Even if one individual makes a supreme effort, this can be negated by poor performance from other members of the group. Therefore Statement (ii) is not true.

Non-production employees can be included in a group incentive scheme, for example when all employees in a management accounting department must work harder to produce prompt budgetary control reports. Statement (iii) is therefore true, and the correct Option is C.

6.4 B **Efficiency ratio** $= \frac{(54{,}000 \times 8) \text{ hours}}{480{,}000} \times 100\% = 90\%$

Capacity utilisation ratio $= \frac{480{,}000 \text{ hours}}{400{,}000 \text{ hours}} \times 100\% = 120\%$

The correct answer is therefore B.

If you selected Option A, you calculated the efficiency ratio correctly but calculated the capacity utilisation ratio as 400,000 hours/480,000 hours instead of the other way round.

If you selected Option C, you calculated both ratios incorrectly because you mixed up the numerators and denominators.

If you selected Option D, you calculated the capacity utilisation ratio correctly but calculated the efficiency ratio as 480,000 hours/(54,000 × 8 hours) instead of the other way round.

6.5 B Overtime premium is always classed as production overheads unless it is: worked at the specific request of a customer to get his order completed; or worked regularly by a production department in the normal course of operations, in which case it is usually incorporated into the direct labour hourly rate.

6.6 $416

Product X (6 × 2 hours)	12 hours
Product Z (10 × 4 hours)	40 hours
	52 hours

Therefore, employee's pay = 52 hours × $8 = $416 for the week.

6.7 C

Day	*Units produced*	*Piecework earnings @ $0.20 per unit*	*Guaranteed minimum earnings*	*Total daily earnings*
		$	$	$
Monday	90	18	–	18
Tuesday	70	14	15	15
Wednesday	75	15	–	15
Thursday	60	12	15	15
Friday	90	18	–	18
				81

6.8 D Idle time may arise for many reasons. Sometimes it may be due to **uncontrollable external factors**, such as a world shortage of material supply. Therefore **Options B and C** are incorrect. However idle time can also arise due to **controllable internal factors** such as inefficient production scheduling or inadequate machine maintenance leading to machine breakdowns. Therefore **Option A** is incorrect.

6.9 C Production volume ratio = capacity utilisation ratio × efficiency ratio

$\therefore$ 1.035 = capacity ratio × 0.9

$\therefore$ Capacity ratio $= \frac{1.035}{0.9} = 1.15$, or 115%

6.10 A

	Basic pay	*Overtime premium*	*Total*
	$	$	$
Direct labour		300 (W1)	300
Indirect labour	2,200 (W2)	150 (W3)	2,350
	2,200	450	2,650

Workings

1 *Overtime premium – direct labour*

20 × 6 hours × $2.50 per hour = $300

2 *Basic pay – indirect labour*

10 × 44 hours × $5 per hour = $2,200

3 *Overtime premium – indirect labour*

10 × 6 hours × $2.50 per hour = $150

6.11 106.32%

$$\text{Production volume ratio} = \frac{\text{Expected hours to make actual output}}{\text{Hours budgeted}} \times 100\%$$

$$= \frac{3{,}455.40 \text{ hours}}{3{,}250 \text{ hours}} \times 100\%$$

= 106.32% (to 2 decimal places)

Workings

Expected time to produce 5,316 units = 0.65 hours × 5,316 units
= 3,455.40 hours

6.12 B

Direct wages	$
36 hours @ $3.60	129.60
2 hours @ $3.60 (special job basic)	7.20
2 hours @ $1.80 (special job overtime)	3.60
Total direct wages	140.40

6.13 B

1,065 units should take (× 2.4 minutes) = 2,556 minutes

$$= \frac{2{,}556}{60}$$

= 42.6 hours

∴ Productivity bonus = (42.6 – 37.5) hours × ($8.50 ÷ 3)
= 5.1 hours × $2.8333
= $14.45

Basic salary = 37.5 hours × $8.50
= $318.75

∴ Total earnings = basic salary + productivity bonus
= $318.75 + $14.45
= $333.20

6.14 The correct answers are:

- Training of direct operatives
- Normal idle time in the factory

6.15 A

		$
620 units @ a basic rate of $0.50 per unit	=	310
100 units @ $0.05 per unit premium	=	5
20 units @ $0.10 per unit premium	=	2
		317

6.16 B A direct cost is a cost that can be traced in full to saleable cost units that are being costed.

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6.17 A

ACCA examining team comments

The key aspect of this question concerns the charging of the costs of overtime in a situation where the requirement for overtime working is driven by sales demand in general rather than by the urgent requirement of a particular customer.

Where the overall sales demand, at any particular time, necessitates overtime working, it is regarded as unfair to charge the premium paid to workers over and above the normal rate to those products that happen to be scheduled for production in the overtime hours. In other words the premium is not charged directly to those particular products (even though the cost can be identified) but instead indirectly to (and thus shared by) all products as part of production overhead.

The correct answer is Option A. The amount of labour costs charged to production overhead comprise the gross wages of the indirect operatives plus the premium paid to direct operatives for the overtime hours worked. Thus the calculation is [$14,300 + $1,650 + $3,200 ($9,600 × 50/150)] = $19,150.

However, Options B and C were each chosen by a significantly larger proportion of candidates than Option A.

Option B included the gross wages of the indirect operatives only ($14,300 + £1,650 = $15,950). It is important to realise that there may be a difference between the wages classified according to type of operative and the wages classified as to whether they are going to be charged as direct costs or as indirect costs to production.

Option C charged the whole of the overtime wages of direct operatives, rather than just the overtime premium, to production overhead as well as the wages of the indirect operatives ($14,300 + $1,650 + $9,600 = $25,550). It should be realised that the cost of the overtime hours at the normal rate would be charged directly to the appropriate products.

Option D, which was selected by a very small proportion of candidates, simply charged the total hours of indirect operatives at the normal rate to production overhead [$14,300 + $550 ($1,650 × 50/150)]. As a result the whole of the overtime premium for both direct and indirect operatives would be charged as a direct cost (which is unlikely to have been practically feasible anyway).

7 Expenses

7.1 The correct answers are:

- Purchase of a building
- Extension to a building

Costs associated with fixing broken windows and replacing missing roof tiles are examples of **revenue expenditure**.

7.2 B

	$
Cost of new machinery	40,000
Cost of extension	20,000
Capital expenditure	60,000

If you selected Option A, you forgot to include the cost of the extension as capital expenditure.

If you selected Option C, you incorrectly included maintenance costs of $4,000 as capital expenditure.

If you selected Option D, you incorrectly included both machinery maintenance costs and office repainting costs as capital expenditure. These are examples of revenue expenditure.

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7.3 B Using the straight line method, annual depreciation charge = ($25,000 – $5,000)/4 years = 5,000.

After three years, depreciation charge = $5,000 × 3 = $15,000

∴ Net book value of asset = $25,000 – $15,000 = $10,000

7.4 C Capital expenditure is **not** charged to the statement of profit or loss (income statement) as an expense. A **depreciation charge** is instead charged to the statement of profit or loss (income statement) in order to write off the capital expenditure over a period of time. Non-current assets are capitalised on the statement of financial position and they provide benefit over a number of accounting periods.

7.5 The correct answers are:

- Administration expenses
- Plant maintenance costs

The purchase of a building (new factory) or a vehicle (even if it is second-hand) are both items of capital expenditure.

7.6 A

$$\text{Machine-hour rate} = \frac{\text{Cost} - \text{residual value}}{\text{Useful life}}$$

$$= \frac{\$200,000 - \$10,000}{40,000 \text{ hours}}$$

$$= \frac{\$190,000}{40,000 \text{ hours}}$$

$$= \$4.75 \text{ per machine hour}$$

Depreciation charge (Year 1) = 3,000 hours × $4.75

= $14,250

7.7 C Suppose the machine cost $10,000

$$\text{Straight-line depreciation} = \frac{\$10,000}{5 \text{ years}}$$

= $2,000 per annum

Reducing balance depreciation is as follows.

	$
	10,000
25% – Year 1	2,500
	7,500
25% – Year 2	(1,875)
	5,625
25% – Year 3	(1,406)
	4,219
25% – Year 4	(1,055)
	3,164
25% – Year 5	(791)
	2,373

Therefore, depreciation would be greater in Year 1 but less in Year 5 if the reducing balance method, rather than the straight-line method, was used.

7.8 D Chart D shows the unit cost behaviour of straight-line depreciation costs.

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7.9 B If the asset's initial value is $1000, depreciation each year would be:

Year	*Straight line*	*20% Reducing balance*	*25% Reducing balance*	*15% Reducing balance*	*18% Reducing balance*
1	100	200	250	150	180
2	100	160	188	128	148
3	100	128	141	108	121

Using 20% reducing balance, depreciation will eventually be lower than the straight line method.
Depreciation is not lower in each year using 15% reducing balance.
Depreciation is not lower in Year 2 using 18% reducing balance.

7.10 A Using the reducing balance method, depreciation will fall each year so, if output stays the same, unit costs will fall.

Using the straight-line method, depreciation stays the same each year, so if output increases, costs per unit will decline.

8 Overheads and absorption costing

8.1 B Overhead absorption (Option A) is the final process of absorbing the total cost centre overheads into product costs. Overhead allocation (Option C) is the allotment of whole items of overhead costs to a particular cost centre or cost unit. Overhead analysis (Option D) is the general term used to describe all of the tasks of processing overhead cost data.

8.2 D Number of employees in packing department = 2 direct + 1 indirect = 3

Number of employees in all production departments = 15 direct + 6 indirect
= 21

Packing department overhead

Canteen cost apportioned to packing department = $\frac{\$8,400}{21} \times 3$

= $1,200

Original overhead allocated and apportioned = $8,960

Total overhead after apportionment of canteen costs = $10,160

If you selected Option A you forgot to include the original overhead allocated and apportioned to the packing department. If you selected Option B you included the four canteen employees in your calculation, but the question states that the basis for apportionment is the number of employees in each **production** cost centre.

If you selected Option C you based your calculations on the direct employees only.

8.3 B From the four options available, a basis relating to space occupied would seem to be most appropriate. This eliminates Options C and D. Since heating is required to warm the whole of the space occupied, from floor to ceiling, the volume of space is most appropriate. Therefore the correct answer is B.

8.4 A Description B could lead to under-absorbed overheads if actual overheads far exceeded both budgeted overheads and the overhead absorbed. Description C could lead to under-absorbed overheads if overhead absorbed does not increase in line with actual overhead incurred. Description D could also lead to under absorption if actual overhead does not decrease in line with absorbed overheads.

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8.5 $4.08

	$
Actual overheads	295,000
Under-absorbed overheads	9,400
Overheads absorbed for 70,000 hours at budgeted absorption rate (x)	285,600

70,000x = $285,600
x = $285,600/70,000
= $4.08

8.6 D Overhead apportionment involves sharing overhead costs as fairly as possible over a number of cost centres. Apportionment is used when it is not possible to allocate the whole cost to a single cost centre.

8.7 $14.00

	$
Actual overheads	640,150
Over-absorbed overheads	35,000
Overheads absorbed for 48,225 hours at budgeted overhead absorption rate (x)	675,150

$$48{,}225x = 675{,}150$$

$$x = \frac{675{,}150}{48{,}225}$$

$$x = \underline{\underline{\$14}}$$

8.8 A

$$\text{Budgeted hours} = \frac{\text{Budgeted overheads}}{\text{Budgeted overhead absorption rate}}$$

$$= \frac{\$475{,}200}{\$32}$$

$$= \underline{\underline{14{,}850}}$$

8.9 C Overhead absorption rates are used to charge overheads to products.

8.10 C The number of machine hours budgeted for the cutting department is much higher than the budgeted number of labour hours. Therefore the activity is **machine intensive** and the type of overhead incurred is likely to be **machine related,** for example power costs, maintenance and depreciation. The most appropriate absorption rate in the cutting department is therefore one based on **machine hours.**

In the finishing department the budgeted number of labour hours is much higher than the budgeted number of machine hours. Therefore the most appropriate absorption rate in the finishing department is one based on **labour hours.**

8.11 D Under-or over-absorbed overhead

= actual overhead incurred – overhead absorbed
= actual overhead incurred – (actual hrs worked × machine hr rate)

Therefore the calculation of the under or over absorbed overhead requires three pieces of information:

- Actual overhead incurred
- Actual hours worked
- Pre-determined machine hour rate

8.12 A It is necessary to have data available relating to the budgeted and actual activity levels (machine hours or direct labour hours). Without such data it is not possible to determine the overhead over/under-absorption.

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8.13 D Under/over-absorption = Actual overheads – absorbed overheads
= $126,740 – $125,200
= $1,540

Under-absorption has occurred because actual overheads incurred were greater than the overheads absorbed.

8.14 C Total reapportionment to Cost Centre Y

	$
60% × $42,000	25,200
45% × $57,600	25,920
	51,120

8.15 A

	$
Actual overheads	158,000
Absorbed overheads (9,800 hrs × $16.40)	160,720
Over-absorbed overheads	2,720

$$\text{Overhead absorption rate per hour} = \frac{\text{Budgeted overheads}}{\text{Budgeted labour hours}}$$
$$= \frac{\$164,000}{10,000 \text{ hours}}$$
$$= \$16.40 \text{ per hour}$$

8.16 A From the information given, it cannot be determined whether Statements 1 and 2 are true or false.

8.17 D Number of employees is the most appropriate basis for reapportioning the cost of personnel services.

8.18 The correct answers are:

- To establish the profitability of different products
- To value closing inventory in accordance with accounting principles

The main reasons for using absorption costing are for inventory valuations and establishing the profitability of different products. **Marginal** costing is based on the concept of contribution and aids decision making.

8.19 $25.60

	$
Direct materials	7.15
Direct labour	8.25
Assembly overhead ($1.76 × 3 hours)	5.28
Finishing overhead ($3.28 × 1.5 hours)	4.92
Full production cost per unit	25.60

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8.20 A

ACCA examining team comments

Answers to this question were fairly evenly spread across the four options. This demonstrates a widespread, but varied, misunderstanding of the overhead absorption process. The key to understanding the process that results in over-absorption or under-absorption of overhead is recognising that production overheads are absorbed using a rate established in advance of an accounting period based on expectations both of activity and expenditure. Inevitably activity and/or overhead expenditure differ(s) from expectation and it is this that causes over-absorption or under-absorption of the overhead.

The over-absorption or the under-absorption is calculated as the difference between the actual overhead incurred (not the expected amount to be incurred), which is charged to the production overhead account, and the amount of overhead absorbed which is transferred out of the overhead account. The amount absorbed is calculated by multiplying the actual activity (not the expected activity) by the predetermined production overhead absorption rate. If the actual overhead expenditure exceeds the overhead absorbed then the overhead is under-absorbed and the balancing amount is debited against profit. If the overhead absorbed exceeds the actual expenditure then the overhead is under-absorbed and the balancing amount is credited to profit.

Only a minority of candidates selected the correct answer, Option A. Both expenditure below budget and activity (direct labour hours in this question) above budget will result in over-absorption. Option B was the reverse of Option A thus causing under-absorption. Options C and D were both incorrect because absorption based on actual expenditure will exactly absorb the amount spent regardless of whether actual activity is above or below budget. The absorption rate is not predetermined but instead is established after the event and therefore calculated to fully absorb the overhead incurred based on the actual activity. More than half of the candidates selected Option C or Option D indicating that there was a widespread and fundamental misunderstanding of overhead absorption.

8.21 A

ACCA examining team comments

This question tested candidates' understanding of overhead absorption, using actual data for a period to calculate the absorption rate that had been applied in the production cost centre. Overhead over or under absorption occurs when overheads are absorbed into product costs for a period using an absorption rate, determined before the period commenced, based on budgeted data.

In this question, the fact that overheads were under absorbed in the period indicates that the actual overhead expenditure was greater than the amount of overhead absorbed (actual direct labour hours × overhead absorption rate). The overhead absorbed was, therefore, $84,424 ($86,920 – $2,496). It follows that the overhead absorption rate in the production cost centre was $8.65 ($84,424 absorbed ÷ 9,760 actual direct labour hours worked) and thus Option A is the correct answer.

Each of the three incorrect options were selected by a significant proportion of candidates. All of the absorption rates in the options included the actual overhead expenditure but the calculations added (rather than subtracted) the under absorbed amount and/or used budgeted (rather than actual) direct labour hours. Option B incorrectly added the under absorbed amount to the actual overhead expenditure. Option C incorrectly divided the correct total amount by the budgeted direct labour hours. Option D made both of the errors included in Options B and C.

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8.22 C

ACCA examining team comments

The following should aid candidates to understand how to calculate the answer for this question.

Actual overhead incurred	$74,960
Overhead under-absorption	$3,762
Thus budgeted must have been less	$71,198
If actual hours are 3,670	
Then budgeted OAR per hour is ($71,198/3,670)	19.4
Thus budgeted expenditure of	$73,820/19.4
Budgeted hours are	3,805

Those candidates that selected option D treated the absorption adjustment as an over-absorption. They did this by taking $74,960 then adding the $3,762 under absorption to get $78,722. Then $78,722 is divided by the actual labour hours of 3,670 to get an overhead absorption rate of $21.45 per hour. As budgeted expenditure is $73,820, this is divided by the overhead absorption rate of $21.45 to get 3,441 hours. Candidates who chose B would have calculated the overhead absorption rate by ignoring the under-absorption. They did this by taking the actual of $74,960 and dividing it by 3,670 actual hours to give an overhead absorption rate of $20.425 per hour. As budgeted expenditure is $73,820 divide this by $20.425 to give 3614 hours. Those who chose A calculated the overhead absorption rate by ignoring the under-absorption. They did this by taking the actual of $74,960 and dividing it by 3,670 actual hours to give an overhead absorption rate of $20.425 per hour. Then adjusting budgeted expenditure of $73,820 by the under absorption of $3,762 which gives $70,058 divided by $20.425 to give 3,430 hours.

9 Marginal costing and absorption costing

9.1 C Difference in profit = change in inventory level × fixed overhead per unit
= (200 – 250) × ($2.50 × 3)
= $375

The absorption costing profit will be greater than the marginal costing profit because inventory has increased.

If you selected Option A you calculated the correct profit difference but the absorption costing profit would be greater because fixed overheads are carried forward in the increasing inventory levels.

If you selected Option B you multiplied the inventory difference by the direct labour – hour rate instead of by the total overhead cost per **unit**, which takes three hours.

If you selected Option D you based the profit difference on the closing inventory only (250 units × $2.50 × 3).

9.2 B Sales volume exceeded production volume by 500 units, therefore inventory reduced. The absorption costing profit will be lower than the marginal costing profit because fixed overheads were 'released' from inventory.

Profit difference = inventory reduction in units × fixed overhead per unit
= 500 × $5 = $2,500
Absorption costing profit = $60,000 – $2,500 = $57,500

If you selected Option A you based your calculation of the profit difference on the closing inventory of 2,500 units. Option C is calculated as $7 profit per unit × 8,500 units sold, however, this takes no account of the actual level of fixed overhead cost.

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If you selected Option D you calculated the correct profit difference but you added it to the marginal costing profit instead of subtracting it.

9.3 $9.00

Decrease in inventory levels = 48,500 – 45,500 = 3,000 units

Difference in profits = $315,250 – $288,250 = $27,000

Fixed overhead per unit $= \frac{\$27,000}{3,000} = \$9$ per unit

9.4 C Argument (i) is correct. Accounting standards require closing inventory values to include a share of fixed production overhead. Argument (ii) is not correct. Opponents of absorption costing argue that under and over absorption are the biggest problems with the use of the method. Argument (iii) is correct. Absorption costing carries fixed overheads forward in inventory to be matched against sales as they arise.

Therefore the correct answer is B.

9.5 D Under marginal costing, closing inventory will be valued **lower** than under absorption costing. If production is greater than sales the inventory level has increased during the month. Absorption costing would therefore produce a higher profit than marginal costing.

9.6 C

	Litres
Opening inventory	8,500
Closing inventory	6,750
Change in inventory	1,750
× overhead absorption rate	$2
Profit difference	3,500

Since inventory reduced during the period the absorption costing profit would be lower than the marginal costing profit. Absorption costing profit = $27,400 – $3,500 = $23,900.

9.7 The correct answers are:

- Inventory value will always be lower than when using absorption costing – true.
- Profit will always be higher than when using absorption costing – false.

In absorption costing, fixed production costs are absorbed into the cost of units and are carried forward in inventory to be charged against sales for the next period. Inventory values using absorption costing are therefore greater than those calculated using marginal costing. If there has been an increase in inventory during the period then absorption costing will report a higher profit than marginal costing. Therefore the second statement is false.

9.8 A Opening inventory = nil

Closing inventory = (97,000 – 96,000) units
= 1,000 units

∴ Profit difference = 1,000 units × $1.40 per unit
= $1,400

During the period, inventory levels have increased, and therefore marginal costing profits will be less than absorption costing profits.

Therefore, the correct answer is: '$1,400 less using marginal costing'.

9.9 D Statement 1 is true.

Inventory value at the end of the period would be higher than at the beginning of the period because 2,000 more units were manufactured than sold. These 'extra' units will give rise to an increase in inventory volume, and hence inventory value, during the period.

Statement 2 is true.

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Inventory values both at the beginning and at the end of the period would be higher using absorption rather than marginal costing. Absorption costing inventory valuations include a share of fixed production costs, whereas marginal costing inventory valuations do not. Absorption costing inventory valuations will be higher, therefore, than marginal costing inventory valuations.

9.10 C Inventory valuation using absorption costing includes a share of all production costs.

9.11 A In absorption costing, closing inventory valuation is higher, so profits are higher than if marginal costing is used. Only production overheads are absorbed into the cost of the product.

9.12 B Contribution is the difference between sales value and the marginal cost of sales.

10 Cost bookkeeping

10.1 B The question describes interlocking accounts, where the cost accounts are distinct from the financial accounts.

With integrated accounts, Option D, a single set of accounting records provides both financial and cost accounts.

Therefore the correct answer is B.

10.2 Both statements are true.

The first statement is correct because **only one set of accounts is kept in an integrated system.** The second statement is correct because **profit differences do not arise with an integrated system.**

10.3 B The credit balance on the wages control account indicates that the amount of wages incurred and analysed between direct wages and indirect wages was **higher** than the wages paid through the bank. Therefore there was a $12,000 balance of **wages owing** at the end of February and statement B is not correct. Therefore the correct option is B.

Statement A is correct. $128,400 of wages was paid from the bank account.

Statement C is correct. $79,400 of direct wages was transferred to the work in progress control account.

Statement D is correct. $61,000 of indirect wages was transferred to the production overhead control account.

10.4 C Statement (i) is not correct. A debit to stores with a corresponding credit to work in progress (WIP) indicates that **direct materials returned** from production were $18,000.

Statement (ii) is correct. **Direct costs of production** are 'collected' in the WIP account.

Statement (iii) is correct. **Indirect costs of production or overhead** are 'collected' in the overhead control account.

Statement (iv) is correct. The purchases of materials on credit are credited to the payables account and debited to the material stores control account.

Therefore the correct answer is C.

10.5 D Production overhead control account.

10.6 B Over-absorbed overhead means that the overhead charged to production was too high therefore there must be a credit to the statement of profit or loss (income statement).

10.7 C Indirect wages are 'collected' in the overhead control account, for subsequent absorption into work in progress.

10.8 D Items appearing in the cost accounts but not in the financial accounts are infrequent, but usually relate to notional costs.

10.9 C Indirect materials are treated as a production overhead. Therefore, when indirect materials are issued, the material inventory account is credited and the production overhead account is debited with the value of the indirect materials.

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10.10 D *Debit* Work in Progress Account *Credit* Production Overhead Account

10.11 D Direct labour costs are debited to the work-in-progress account and credited to wages control.

10.12 B The correct answer is *Debit* materials control and *Credit* cost ledger control.

11 Job, batch and service costing

11.1 A Job costing is a costing method applied where work is **undertaken to customers' special requirements.** Option B describes process costing, C describes service costing and D describes absorption costing.

11.2 $904.00

Workings

	Job LJ1 $
Direct material	154.00
Department X Direct Labour (20 × $3.80)	76.00
Department Y Direct Labour (12 × $3.50)	42.00
Total direct cost	272.00
Department X Overhead (20 × $12.86)	257.20
Department Y Overhead (12 × $12.40)	148.80
Total cost	678.00
Profit*	226.00
Total selling price	904.00

*The gross profit margin is given as a percentage on selling price. 25% on selling price is the same as $33^{1}/_{3}$% (25/75) on cost.

11.3 C Since wages are paid on a piecework basis they are a variable cost which will increase in line with the number of binders. The machine set-up cost and design costs are fixed costs for each batch which will not be affected by the number of binders in the batch.

For a batch of 300 binders:

	$
Direct materials (30 × 3)	90.00
Direct wages (10 × 3)	30.00
Machine set up	3.00
Design and artwork	15.00
Total production cost	138.00
Profit (25% gross margin = $33^{1}/_{3}$% of cost)	46.00
Selling price for a batch of 300	184.00

11.4 $103.75

Production overhead absorption rate = $240,000/30,000 = $8 per labour hour

Job B124	$
Direct materials (3 kg × $5)	15.00
Direct labour (4 hours × $9)	36.00
Production overhead (4 hours × $8)	32.00
Total production cost	83.00
Profit margin: 20% of sales (× $^{20}/_{80}$)	20.75
Price to be quoted	103.75

11.5 B In service costing it is difficult to identify many attributable direct costs. Many costs must be **shared over several cost units**, therefore characteristic (i) does apply. Composite cost units such as tonne-mile or room-night are often used, therefore characteristic (ii) does apply. Equivalent units are more often used in **costing for tangible products**, therefore characteristic (iii) does not apply, and the correct answer is B.

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11.6 C Cost per tonne-kilometre (i) is appropriate for cost control purposes because it **combines** the distance travelled and the load carried, **both of which affect cost.**

The fixed cost per kilometre (ii) is not particularly useful for control purposes because it varies with the number of kilometres travelled.

The maintenance cost of each vehicle per kilometre (iii) can be useful for control purposes because it focuses on a particular aspect of the cost of operating each vehicle. Therefore the correct answer is C.

11.7 D All of the activities identified would use service costing, except the light engineering company which will be providing **products not services**.

11.8 B The most appropriate cost unit is the **tonne-kilometre**. Therefore the cost per unit =

$$\frac{\$562{,}800}{375{,}200} = \$1.50$$

Option A is the cost per kilometre travelled. This is not as useful as the cost per tonne-kilometre, which **combines** the distance travelled and the load carried, **both of which affect cost**.

Option C is the cost per hour worked by drivers and D is the cost per driver employed. Costs are more likely to be incurred in relation to the distance travelled and the load carried.

11.9 D Bed occupied/night. This is the key determinant of costs. Not all guests will have meals. Guest stays will be of varying lengths, generating varying costs.

11.10 C (2) only. Manufacturing overhead is absorbed at a predetermined rate.

11.11 A

ACCA examining team comments

Option A is the correct answer. The production costs of the job are $1,457 (direct costs $1,070 + production overheads 45 direct labour hours × $8.60 per hour). An amount equivalent to 22% of sales revenue (10% + 12%) is required on top of the production costs to cover the fixed non-production overheads and provide the required net profit. This means that the production costs are 78% (100% – 22%) of the selling price of the job. The selling price is, therefore, $1,457 ÷ 0.78 = $1,868. A larger proportion of candidates, incorrectly selected Option B. This answer of $1,778 adds 22% to the total production costs with the effect that the percentage of sales is only 18% [(22 ÷ 122) × 100], not 22%, thus understating the required selling price of the job. Adding 22% is assuming that this is the required mark-up % not the required margin % of sales. The remaining options, C and D, also make incorrect uplifts to the production costs of the job to determine the selling price which in each case understates the required amount. Option C, incorrectly adds 10% to the production costs to understate the total costs but then correctly divides the resulting figure by 0.88 to provide the profit. Option D correctly divides the total production costs by 0.9 for the non-production costs but then incorrectly adds 12% to the correct total costs to understate the required profit.

11.12 D

ACCA examining team comments

This is calculated by taking the costs for the period and then dividing by the number of beds actually occupied in the period. Cost per patient per day = $194,400 / (20 beds × 90% × 30 days) = $360.

Some candidates chose option A and incorrectly divided the cost by the number of days available in the period as $6,480 (=$194,400 / 30 days).

Whilst others wrongly divided the cost by the total bed capacity available, not what was actually occupied so they took $194,400 / (20 beds × 30 days) = $324 and thus selected Option B.

A few chose Option C, they incorrectly adjusted the 30 days for occupancy by dividing the 30 by 0.9 giving the cost per day as = $194,400 / (30 days / 0.9) which gave $5,832 in error.

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11.13 4,800 passenger-kilometres

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The bus operates between two towns which are 25 kilometres apart thus this is a 50 kilometre journey, run three times a day with 40 passenger seats but only at 80% occupancy.

So 50 kilometres × 3 journeys = 150 kilometres in a day

With potentially 40 seats available but only 80% in use so:

150 kilometres × 40 seats × 80% capacity = 4,800.

12 Process costing

12.1 B Joint products are two or more products produced by the same process and separated in processing, each having a sufficiently high saleable value to merit recognition as a main product.

A joint product may be subject to further processing, as implied in Option A, but this is not the case for all joint products.

12.2 D A by-product is output of some value produced in manufacturing something else (the main product).

Option A is incorrect because a by-product has some value.

Option B is incorrect because this description could also apply to a joint product.

Option C is incorrect because the value of the product described could be relatively high, even though the output volume is relatively low.

12.3 D The abnormal loss units are valued at their **full production cost** and **credited** to the process account, so that their occurrence does not affect the cost of good production. Therefore the correct answer is D.

Options A and C are incorrect because the scrap value of the abnormal loss is debited to the **scrap account** and credited to the **abnormal loss account**, it has no impact on the process account.

12.4 $100

Normal loss = 10% × input

= 10% × 5,000 kg
= 500 kg

When scrap has a value, normal loss is valued at the value of the scrap ie 20c per kg.

Normal loss = $0.20 × 500 kg
= $100

12.5 C The **point of separation,** also referred to as the split-off point, is the point in a process where **joint products** become separately identifiable. Costs incurred prior to this point are common or **joint costs**.

12.6 A Abnormal loss units are valued at the same cost per unit as completed output. The cost per unit of output and the cost per unit of abnormal loss are based on expected output.

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12.7 D Conversion costs are production costs excluding direct materials.

12.8 D Let

	x	= material input to process
	0.1x	= normal loss
	0.05x	= abnormal loss
∴ Output		= x – 0.1x – 0.05x
340 litres		= x – 0.15x
340 litres		= 0.85x
x		$= \frac{340 \text{ litres}}{0.85}$
		= 400 litres

12.9 D Debited at a cost per unit based on total production cost divided by normal output.

12.10 D $30{,}000 \times \frac{(4{,}000 \times 18)}{(4{,}000 \times 18) + (2{,}000 \times 12)} = \$22{,}500$

	Product		
	A	*B*	*Total*
	$	$	$
Total sales value	(2,000 × $12)	(4,000 × $18)	
	24,000	72,000	96,000

Joint process costs – product B $= \frac{72{,}000}{96{,}000} \times \$30{,}000$

$= \$22{,}500$

12.11 The correct answers are:

- The higher the net realisable value of losses the lower will be the cost per unit of normal output – true
- The higher the abnormal losses the higher will be the cost per unit of normal output – false

If the net realisable value of losses increases, the actual process costs are reduced by these amounts. If total process costs are reduced, the cost per unit of normal output is also reduced (or lowered).

12.12 A Normal loss is an estimated figure. The actual loss may be more or less than the normal loss. If the actual loss is more than the normal loss, the difference is treated as an abnormal loss.

If the actual loss is less than the normal loss, the difference between the actual loss and normal loss is known as an abnormal gain. An abnormal gain is therefore a reduction in loss and not an actual gain in itself.

12.13 100 litres

	Litres
Input	800
Less: 25% Normal loss	(200)
Normal output	600
Less: Actual output	(500)
Abnormal loss	100

12.14 D

	Product X	*Product Y*	*Total*
	15,000 litres	7,500 kg	
	$	$	$
Final sales value	345,000	123,750	
Further processing costs	120,000	0	
Net realisable value	225,000	123,750	348,750
Pre-separation costs	116,129*	63,871	180,000

* $225,000/$348,750 × $180,000 = $116,129

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12.15 D

ACCA examining team comments

This question tests item C4h in the Study Guide on the subject of accounting for joint products. The question specifically requires the net realisable value method to be used to apportion the joint process costs, rather than final sales value (an alternative method also based on relative sales values) or weight of output (a method which ignores relative sales values and instead bases the cost apportionment on relative output volume). Despite this, apportionment based on weight of output (Option B) has been the option most frequently chosen by candidates [(3,000 ÷ 8,000 kg) × 100 = 37.5%].

Net realisable value is the final sales value of the output less the further processing costs. This provides an estimated market value of the incomplete products at the split off point in a situation where the products cannot be sold at that point. The correct answer, using net realisable value, is Option D {[($60,000 – $15,000) ÷ ($120,000 – $20,000)] × 100 = 45.0%}. Many candidates, however, simply used the relative final sales values (Option A), which fails to adjust for the value added by further processing [($60,000 ÷ $120,000) × 100 = 50.0%], or the relative selling prices (Option C) which in addition ignores the weighting of the differing output of the two products ([$20 ÷ $32) × 100 = 62.5%].

12.16 A

ACCA examining team comments

All four options were selected by a significant proportion of students. Options B and C were equally popular, with Option A (the correct answer) and Option D slightly less so.

In this question, further processing occurs after the split-off point in the joint process. Where the common process costs (ie those incurred before the split-off point) are to be apportioned on the basis of sales value then it is relative sales values of the joint products at the split-off point, rather than the final sales values after further processing, that are relevant. Also, sales value means the total sales values of the output of the products (X $20,000: Y $34,500) not just the selling prices (X $8.00: Y $15.00) in order to determine the appropriate weighting between the products.

Option A is the correct answer. The apportionment of common process costs to Product X is $14,165 [$38,600 × ($20,000 ÷ $54,500)].

Option B ($14,533) is incorrect because it uses final sales values, i.e. after further processing [$38,600 × ($25,000 ÷ $66,400)]. Option D ($13,426) is incorrect because the apportionment is based on the relative selling prices at the split-off point [$38,600 × ($8.00 ÷ $23.00)]. Option C ($20,104) is incorrect because it ignores sales prices/values entirely and instead bases the common cost apportionment on output quantities [$38,600 × (2,500 units ÷ 4,800 units)].

12.17 A

ACCA examining team comments

Only a few candidates correctly chose $40 per kg after deducting the further processing cost for product A as follows.

	kg	$ per kg	$	$	**Correct Answer is Option A**
A	1,000	40	40,000	**5,200**	40,000/120,000*15,600
B	4,000	20	80,000		
	5,000		120,000		

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Some candidates chose option C which used the kilograms as the basis of apportionment by mistake:

	kg	$	
A	1,000	**3,120**	1,000/5,000*15,600
B	4,000		
	5,000		

Some candidates chose option B where they used the $50 per kg for product A, ignoring the further processing cost in error:

	kg	$ per kg	$	$	
A	1,000	50	50,000	**6,000**	50,000/130,000*15,600
B	4,000	20	80,000		

A select few chose option D where they used the total for product B as the basis of apportionment by mistake:

	kg	$ per kg	$	$	
A	1,000	40	40,000	**10,400**	80,000/120,000*15,600
B	4,000	20	80,000		
	5,000		120,000		

13 Cost-volume-profit (CVP) analysis

13.1 3,000 units

$$\text{Breakeven point} = \frac{\text{Fixed costs}}{\text{Contribution per unit}} = \frac{\$30{,}000}{\$(15-5)} = 3{,}000 \text{ units}$$

If you selected Option A you divided the fixed cost by the selling price, but remember that the selling price also has to cover the variable cost. Option C is the margin of safety, and if you selected option D you seem to have divided the fixed cost by the variable cost per unit.

13.2 B The margin of the safety is the difference in units between the expected sales volume and breakeven sales volume and it is sometimes expressed as a percentage of the expected sales volume.

13.3 10,909 units

$$\text{Breakeven point} = \frac{\text{Fixed costs}}{\text{Contribution per unit}}$$

$$= \frac{10{,}000 \times (\$4.00 + 0.80)}{(\$6.00 - (\$1.20 + \$0.40))} = \frac{\$48{,}000}{\$4.40} = 10{,}909 \text{ units}$$

13.4 D

Contribution required for target profit	=	fixed costs + profit
	=	$48,000 + $11,000
	=	$59,000
÷ Contribution per unit (from qu 3)	=	$4.40
∴ Sales units required	=	13,409 units

If you selected Option A you divided the required profit by the contribution per unit, but the fixed costs must be covered before any profit can be earned. If you selected Option B you identified correctly the contribution required for the target profit, but you then divided by the selling price per unit instead of the contribution per unit. Option C ignores the selling costs, which must be covered before a profit can be earned.

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13.5 C (39,975 + 12,000) ÷ 0.54

13.6 B The contribution is the difference between the sales value and the variable cost incurred at activity level R.

13.7 A Breakeven point $= \frac{\$48{,}000}{0.4} = \$120{,}000$ sales value

Margin of safety = \$140,000 – \$120,000 = \$20,000 sales value

(÷ \$10) = 2,000 units

Option B is the breakeven point and Option C is the actual sales in units. If you selected Option D you calculated the margin of safety correctly as 20,000 but you misinterpreted the result as the sales **volume** instead of the sales **value**.

13.8 D Breakeven quantity $= \frac{\text{Fixed costs}}{\text{Contribution per unit}}$

Since we do not know the contribution per unit, and we cannot determine it from the information available, it is not possible to calculate the breakeven point in terms of units. Therefore the correct answer is D.

We can determine the **value** of breakeven sales as \$90,000/0.4 = \$225,000, but this does not tell us the number of units required to break even. If you selected Option C you probably performed this calculation.

13.9 \$300,000

Contribution per unit = \$90 – \$40 = \$50. The sale of 6,000 units just covers the annual fixed costs, therefore the fixed costs must be \$50 × 6,000 = \$300,000.

13.10 D The profit line on a profit/volume chart cuts the y-axis at the point representing **the loss incurred at zero activity.** This is the fixed cost which must be paid **even if no units are sold.**

13.11 D The chart shows a single line depicting **the profit for a range of levels of activity.** Therefore it is a profit volume chart.

All of the other options would depict cost lines rather than profit lines, and the first two options would also include a sales revenue line.

13.12 25%

The contribution/sales ratio (C/S ratio) is another term used to describe the profit/volume ratio (P/V ratio).

$$\text{C/S ratio} = \frac{\text{Contribution per unit}}{\text{Selling price per unit}}$$

$$= \frac{\$(60-14-12-19)}{\$60} \times 100\%$$

$$= 25\%$$

13.13 B $\text{C/S ratio} = \frac{\text{Contribution per unit}}{\text{Selling price per unit}}$

$$= \frac{\$(20-4-3-2-1)}{\$20} \times 100\% = 50\%$$

If you selected Option A you calculated profit per unit as a percentage of the selling price per unit. Option C excludes the variable selling costs from the calculation of contribution per unit and Option D excludes the variable production overhead cost, but **all variable costs must be deducted from the selling price to determine the contribution.**

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13.14 C

	$
Target profit	6,000
Fixed costs (5,000 × $2)	10,000
Target contribution	16,000
Contribution per unit ($10 – $6)	$4
Units required to achieve target profit	4,000

If you selected Option A you divided $6,000 target profit by the $4 contribution per unit, but **the fixed costs must be covered before any profit can be earned**. If you selected Option B you divided by the selling price, but the variable costs must also be taken into account. If you selected Option D you divided by the profit per unit instead of the contribution per unit, but the fixed costs are taken into account in the calculation of the target contribution.

13.15 B

Fixed costs ($10,000 × 120%)	$12,000
Units required now to break even (÷ $4 contribution)	3,000
Budgeted units of sales	5,000
Margin of safety (units)	2,000

In percentage terms, margin of safety $= \frac{2,000}{5,000} \times 100\% = 40\%$

Option A increases the **variable** cost by 20% and Option C increases the **activity** by 20%. If you selected Option D you calculated the margin of safety as a percentage of the breakeven volume, but it is **usually expressed as a percentage of budgeted sales.**

13.16 C Sales revenue required = (Fixed costs + target profit) / C/S ratio

fixed costs + target profit = $210,000 + $65,000
= $275,000

C/S ratio = ($640,000 - $384,000) / $640,000
= 0.4

∴ Sales revenue required = $275,000 / 0.4
= $687,500

13.17 C Required contribution = fixed costs + target profit
= $210,000 + $52,000
= $262,000

Required sales $= \frac{\$262,000}{\text{Contribution per unit}}$

$= \frac{\$262,000}{\$2\,*}$

= 131,000 units

* Contribution per unit $= \frac{\text{Sales revenue} - \text{variable costs}}{\text{Sales units}}$

$= \frac{\$640,000 - \$384,000}{128,000 \text{ units}}$

= $2 per unit

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13.18 C

ACCA examining team comments

Options A, B and C were each selected by a significant proportion of students indicating a widespread lack of knowledge in this area. Many students calculated the gross profit % of sales or the net profit % of sales rather than the contribution.

Contribution is calculating by deducting both production and non-production variable costs from sales revenue. It is the contribution towards covering the fixed costs and providing profit. The variable costs comprise:

Prime costs	$11.20 per unit
Variable production overhead	$1.29 per unit (8.60 × 0.15)
Variable production overhead	$0.57 per unit (5.70 × 0.10)
	$13.06

The contribution is $16.94 per unit (selling price $30.00 – variable costs $13.06) and the contribution to sales (C/S) ratio is 56.5% [(16.94 ÷ 30.00) × 100]. Option C is the correct answer.

Option A calculates the net profit to sales ratio [(4.50 ÷ 30.00) × 100] = 15%. Option B calculates the gross profit to sales ratio. Gross profit is sales less production cost of sales which includes not only variable but also fixed production costs whilst at the same time disregarding the variable non-production costs. The gross profit is $10.20 per unit [30.00 – (11.20 + 8.60)] and the gross profit percentage of sales is 34.0% [(10.20 ÷ 30.00) × 100].

Finally, Option D used variable production costs only (i.e. ignoring the variable non-production costs) as the relevant costs in the calculation of contribution [(17.51 ÷ 30.00) × 100] = 58.4%. This option was selected by a smaller proportion of candidates.

14 Decision making

14.1 C A decision is about the future, therefore relevant costs are future costs (i). If a cost is unavoidable then any decision taken about the future will not affect the cost, therefore unavoidable costs are not relevant costs (ii). Incremental costs are extra costs which will be incurred in the future therefore relevant costs are incremental costs (iii). Differential costs are the difference in total costs between alternatives and they are therefore affected by a decision taken now and they are associated with relevant costs (iv).

14.2 B Only future costs that will be affected by the decision are relevant for decision making.

Committed costs **cannot be altered by a decision** taken now therefore they are not relevant and the correct answer is B.

A **differential cost** (Option A) is the difference in total costs between alternatives and is therefore **affected by a decision** and is a relevant cost.

An **opportunity cost** (Option C) is the value of the benefit sacrificed when one course of action is chosen, in preference to an alternative cost to be incurred in the future and is therefore relevant.

Incremental costs (Option D) are **extra costs which will be incurred in the future** and they are therefore relevant costs.

14.3 D An opportunity cost is the amount lost by choosing one option over another.

A sunk cost (Option A) is a **past cost** which is not relevant to the decision. An incremental cost (Option B) is an **extra cost** to be incurred in the future as the result of a decision taken now. The salary cost forgone is certainly relevant to the decision therefore Option C is not correct.

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14.4 B $5,000 has been spent on market research already and is therefore a sunk cost and irrelevant to the decision. The further $2,000 will only be spent if Sue continues with the project, therefore it is an incremental (relevant) cost of the decision to go ahead.

The cost is not an opportunity cost (Option D) because Sue has not forgone an alternative use for the resources.

14.5 $12,500

The original cost of $45,000 is a non-relevant sunk or past cost. The material would not be reworked, since its value would increase by only $5,000 ($17,500 – $12,500) for a cost of $7,500.

The relevant cost of using the material for the special job is therefore the opportunity cost of the $12,500 scrap sale forgone.

14.6 C The **material is in regular use** and so 1,000 kg will be purchased. 500 kg of this will replace the 500 kg in inventory that is used, 100 kg will be purchased and used and the remaining 400 kg will be kept in inventory until needed. The relevant cost is therefore 600 × $3.25 = $1,950.

If you selected option A you valued the inventory items at their resale price. However, the items are in regular use therefore they would not be resold.

Option B values the inventory items at their original purchase price, but this is a sunk or past cost. Option D is the cost of the 1,000 kg that must be purchased, but since the material is in regular use the excess can be kept in inventory until needed.

14.7 C **Relevant cost of material K**

		$
500 kg in inventory:	If not used on this order, would be sold ($1.50 × 500)	750
300 kg extra required:	Need to be purchased at current price ($5 × 300)	1,500
		2,250

The 300 extra kg which must be purchased represent an extra or incremental cost. If the 500 kg were not used on this job they would be sold, therefore the revenue forgone is an **opportunity cost** of $750. Therefore the correct answer is C.

Although the original $2,000 paid for the 500 kg in inventory is a **sunk cost**, in this situation there is an **alternative use for the material**, therefore the opportunity cost becomes relevant and Options A and B are not correct.

Option D is not correct because there is no need to incur an incremental cost of $4,000 since some of the material required is in inventory.

14.8 C The relevant cost of Material X in this situation is the current replacement cost because Material X is used regularly by the company in its normal business, and is available from inventory.

14.9 D

	Product M	*Product F*	*Product S*
Contribution per unit	$57	$73	$69
Hours required per unit	5	6	4
Contribution per hour of labour	$11.40	$12.17	$17.25
Ranking	3rd	2nd	1st

14.10 A

	Quantity per unit	*Quantity required*	*Quantity available*
Material ($72 ÷ $8)	9 litres (× 2,000)	18,000 litres	16,000 litres
Labour ($49 ÷ $7)	7 hours (× 2,000)	14,000 hours	15,000 hours

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14.11 $70

The rate of $10 per hour currently paid to the skilled employee is not relevant because it would be paid anyway. The relevant hourly rate is the incremental cost of $9 per hour.

	$ per unit
Skilled labour 5 hours × $9	45
Semi-skilled labour 5 hours × $5	25
	70

14.12 C

	Product J $ per unit	*Product K* $ per unit	*Product L* $ per unit
Selling price	140	122	134
Variable cost	106	86	77
Contribution	34	36	57
Kg of material	11	7	13
Contribution per kg	$3.09	$5.14	$4.39
Ranking	3	1	2

14.13 B We begin by calculating the contribution per unit of limiting factor.

			Priority ranking
Scratch	= $6/2	= $3 per labour hour	1
Purr	= $7/3	= $2.33 per labour hour	3
Buzz	= $8/3	= $2.67 per labour hour	2
			Hours
Production priorities:		1st Scratch (700 units × 2 hours)	1,400
		2nd Buzz (400 units × 3 hours)	1,200
			2,600
Production mix (units)		700 Scratch, 400 Buzz, 0 Purr	

If you selected Option A you allocated the available hours according to the contribution earned per unit of product. However, this does not take account of the **limiting factor**. Option C is the maximum demand for each product, but there are insufficient labour hours available to manufacture this volume of output. Option D allocates all the available hours to Scratch, the product which earns the highest contribution per hour. However, the maximum demand for Scratch is 700 units. Once that demand has been met the remainder of the available hours must be allocated to Buzz, the next product in the priority ranking.

14.14 $50

The net book value is not relevant; it is a **sunk cost**. The company would not purchase a new component because it would be cheaper to modify the existing component in inventory incurring an incremental cost of $50.

14.15 B

	Product 1 $	*Product 2* $	*Product 3* $	*Product 4* $
Variable manufacturing costs	5.00	4.00	6.50	8.50
Bought-in price	7.50	6.75	8.50	12.00
Difference	2.50	2.75	2.00	3.50
Labour hours/unit	1.5 hrs	3 hrs	2 hrs	2.5 hrs
Extra cost of buy-in (savings from in-house manufacture) per unit of scare resource (labour hours)	1.67	0.92	1.00	1.40

Product 2 has the lowest extra cost of buying in. This would therefore have the smallest effect on profit.

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14.16 B

ACCA examining team comments

Choice	$	Answer	
1	2,000	Incorrect	Just takes the NBV of machinery and ignores the cost of disposal incorrectly.
2	3,100	Correct	Takes the NBV of machinery and correctly adds the cost of disposal as a relevant cost.
3	2,200	Incorrect	Just considers the sales proceeds as the machine is not in use, but If used on the contract this is irrelevant as it will not be sold.
4	1,300	Incorrect	This takes the sales proceeds as the machine is not in use and then deducts the cost of disposal incorrectly.

15 Capital investment appraisal

15.1 B Current rate is 6% pa payable monthly

$\therefore$ Effective rate is 6/12% = ½% compound every month

$\therefore$ In the six months from January to June, interest earned =

($1,000 × [1.005]^6) – $1,000 = $30.38

Option A is incorrect since it is simply 6% × $1,000 = $60 in one year, then divided by 2 to give $30 in six months.

Option C represents the annual interest payable (6% × $1,000 = $60 pa).

Option D is also wrong since this has been calculated (incorrectly) as follows.

0.05 × $1,000 = $50 per month
Over six months = $50 × 6
= $300 in six months

15.2 B $2,070 = 115% of the original investment

$\therefore$ Original investment $= \frac{100}{115} \times \$2,070$

$= \$1,800$

$\therefore$ Interest $= \$2,070 - \$1,800$

$= \$270$

Option D is calculated (incorrectly) as follows.

$\frac{x}{\$2,070} = 15\%$

$\therefore x = 0.15 \times \$2,070$
$= \$310.50$

Make sure that you always tackle this type of question by establishing what the original investment was first.

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15.3 C We need to calculate the effective rate of interest.

8% per annum (nominal) is 2% per quarter. The effective annual rate of interest is $[1.02^4 - 1] = 0.08243 = 8.243\%$.

Now we can use $S = X(1 + r)^n$

$S = 12{,}000\ (1.08243)^3$

$S = \$15{,}218.81$

∴ The principal will have grown to approximately \$15,219.

Option A is incorrect because this represents $\$12{,}000 \times (1.08)^3 = \$15{,}117$. You forgot to calculate the effective annual rate of interest of 8.243%.

Option B represents the present value of \$12,000 at the end of three years, ie \$12,000 × 0.794 (from present value tables) at an interest rate of 8%.

Option D represents the cumulative present value of \$12,000 for each of the next three years at an interest rate of 8%.

15.4 D

		\$
PV of \$1,200 in one year	= \$1,200 × 0.926 =	1,111.20
PV of \$1,400 in two years	= \$1,400 × 0.857 =	1,199.80
PV of \$1,600 in three years	= \$1,600 × 0.794 =	1,270.40
PV of \$1,800 in four years	= \$1,800 × 0.735 =	1,323.00

15.5 4.06%

Effective quarterly rate = 1% (4% ÷ 4)

Effective annual rate = $[(1.01)^4 - 1]$

= 0.0406 = 4.06% pa

15.6 9.85%

The formula to calculate the IRR is $a\% + \left[\frac{A}{A - B} \times (b - a)\right]\%$

where a = one interest rate

b = other interest rate

A = NPV at rate a

B = NPV at rate b

$\text{IRR} = 9\% + \left[\frac{22}{22 + 4} \times 1\right]\%$

= 9 + 0.85 = 9.85%

15.7 B The discount factor for 10 years at 7% is 0.508.

∴ Original amount invested = \$2,000 × 0.508

= \$1,016

If you selected Option A, you mixed up the rate and the period in the present value tables and therefore found the discount factor for 7 years at 10%.

If you selected Option C, you divided \$2,000 by the discount factor instead of multiplying it.

If you selected Option D, you used the cumulative present value tables instead of the simple present value tables. Always check your answers for common sense – an investment **growing** at 7% per annum is unlikely to fall in value!

15.8 B If house prices rise at 2% per calendar month, this is equivalent to

$(1.02)^{12} = 1.268$ or 26.8% per annum

If you selected Option A, you forgot to take the effect of compounding into account, ie 12 × 2% = 24%.

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If you chose Option C, you correctly calculated that $(1.02)^{12} = 1.268$ but then incorrectly translated this into 12.68% instead of 26.8%.

If you selected Option D, you forgot to raise 1.02 to the power 12, and instead you multiplied it by 12.

15.9 D

Annuity = \$700

Annuity factor = 1 + 6.247 (cumulative factor for 9 years, first payment is **now**)
= 7.247

$$\text{Annuity} = \frac{\text{PV of annuity}}{\text{Annuity factor}}$$

$$\$700 = \frac{\text{PV of annuity}}{7.247}$$

\$700 × 7.247 = PV of annuity

PV of annuity = \$5,073 (to the nearest \$)

If you selected option A, you have calculated the present value of an annuity from Years 1–10 instead of from Time 0–9.

If you selected option B, you have calculated the present value of an annuity from Years 0–8 instead of from Time 0–9.

If you selected Option C, you appear to have mixed up the interest rate and periods and obtained an annuity factor for Years 0–8 at 10% = \$4,435.

15.10 B

	\$
Investment	(60,000)
PV of cash inflow	64,600
NPV @ 10%	4,600

	\$
Investment	(60,000)
PV of cash inflow	58,200
NPV @ 15%	(1,800)

The IRR of the machine investment is therefore between 10% and 15% because the NPV falls from \$4,600 at 10% to –\$1,800 at 15%. Therefore at some point between 10% and 15% the NPV = 0. When the NPV = 0, the internal rate of return is reached.

15.11 A

Let x = investment at start of project.

Year	*Cash flow*	*Discount factor*	*Present value*
	\$	10%	\$
0	x	1.000	(x)
1–5	18,000	3.791	68,238
			7,222

$\therefore -x + \$68{,}238 = \$7{,}222$

$x = \$68{,}238 - \$7{,}222$

$x = \$61{,}016$

15.12 The correct answers are:

- The discounted payback period at 20% will be longer than the discounted payback period at 10%.
- The non-discounted payback period will be shorter than the discounted payback period

15.13 B IRR is the discount rate at which the net present value of the cash flows from an investment is zero.

15.14 C At the end of Year 3, \$74,600 has been 'paid back'. The remaining \$15,400 for payback will be received during Year 4.

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15.15 The correct answers are:

Project A has a higher internal rate of return than Project B – true.

Project B has higher initial outlay than Project A – false.

15.16 C 10 + 10(24,760/40,870)

15.17 C Present value = \$8,000 + (\$8,000 × 3.791) = \$38,328

15.18 B

Year	*Cash inflow*	*Cumulative cash inflow*
	\$	\$
1	30,000	30,000
2	30,000	60,000
3	30,000	90,000
4	30,000	120,000

Payback: $3 + \left(\frac{(100{,}000 - 90{,}000)}{(120{,}000 - 90{,}000)} \times 12 \text{ months}\right) = 3.3$ years.

The \$15,000 already incurred is not a relevant cost.

15.19 D

$$\text{PV of a perpetuity} = \frac{\text{annuity}}{\text{interest rate}} = \frac{\$8{,}652}{0.07} = \$123{,}600 + \$8{,}652 = \$132{,}252$$

15.20 D

ACCA examining team comments

This question tests item D2f in the Study Guide. The question covers the payback period, one of the methods used to appraise the viability of capital investment (long-term) projects. Candidates' answers demonstrated confusion about how payback is calculated. Fundamentally, the payback method uses cash flows, rather than accounting profits, to determine how long it will take to recover the capital initially invested in a long-term project.

The correct answer (Option D) requires depreciation, which is not a cash flow, to be added back to the expected annual profit figure provided in the question. Straight-line depreciation is \$15,000 per annum [(\$100,000 investment – \$25,000 residual value) ÷ 5 years] and so the annual cash inflow is \$40,000 (\$25,000 profit + \$15,000 depreciation). The payback period is 2.5 years (\$100,000 ÷ \$40,000).

A much more popular, but incorrect, answer has been Option A which simply divides the investment amount by the annual profit net of depreciation (\$100,000 ÷ \$25,000 = 4.0 years). Another popular, but incorrect, answer is Option B which not only uses the annual profit, rather than cash flow, but also uses the net investment amount (\$100,000 investment – \$25,000 residual value = \$75,000). Thus \$75,000 ÷ \$25,000 = 3.0 years. Using the net investment figure fails to recognise that the residual value will not be received until the end of Year 5. Finally, Option C, the least popular answer chosen by candidates, is based correctly on cash flows but uses the net investment amount (\$75,000 ÷ \$40,000 = 1.875 years).

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15.21 D

ACCA examining team comments

All four options were popular.

The correct answer is Option D. The annuity factor at 10% per annum over a period of 5 years (3.791) is based on the same net cash flow arising at the end of each of the 5 years. It is multiplied by the figure of $64,000 to determine the present value (PV) of the net cash inflows. The resulting figure is $242,624. Because the net present value (NPV) of the investment project (ie PV of the net cash flows compared with the investment amount) is positive $33,274 then the investment amount will be less than the PV of the net cash inflows by that amount. Thus the investment amount is $209,350 ($242,624 – $33,274).

Option A is incorrect because the positive NPV of $33,274 has been added to (rather than deducted from) the PV of the net cash inflows ($242,624 + $33,274 = $275,898). This assumes that the NPV is negative (ie that the investment amount is greater than the PV of the net cash inflows). Option B is incorrect because a factor of 5 (for the five years) has been applied to the annual net cash inflows [($64,000 × 5) – £33,274 = $286,726]. This assumes no discounting. Option C is incorrect because a factor of 4.791 (rather than 3.791) has been applied to the annual net cash inflows [($64,000 × 4.791) – $33,274 = $273,350]. This incorrectly assumes that a further $64,000 is received as soon as the investment is made.

15.22 A

ACCA examining team comments

Option

1	12,510	3000 + (3.170 × 3000).	This correctly takes the first year as 3000 ignoring discounting as it is at the beginning of the year followed by the next four years discounted at 10%.

Incorrect Answers

Option

2	11,373	3.791 × 3000	This ignores the fact that the cashflow starts at the beginning of the year meaning no discounting on the first cashflow.
3	9,510	3.170 × 3000	This only considers 4 years discounted cashflow and ignores the 5th incorrectly.
4	15,000	5 × 3000	This option takes 5 years cashflows but ignores discounting.

It is imperative that candidates study and prepare well for all topics in the syllabus and not just a select few. Candidates must bear in mind that questions in the examination will include questions from all topics of the syllabus. Equipping themselves with adequate knowledge of all topics will certainly maximise and improve the performance of candidates in future examinations. Thus, candidates are advised to plan their revision timetable so that they have sufficient time to revise all the topics in the syllabus. Candidates are also reminded to try and attempt all questions.

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16 Cash management

16.1 D Working capital is the current assets less the current liabilities for a company.

16.2 D A bank loan would not be classed as working capital as it is not a current liability.

16.3 D December cash payment is November overheads.
November overheads $160,000 × 1.0175 × 1.0175 = $165,649

16.4 A Taking more credit from suppliers will improve operational cash flows. The other options will cause a deterioration in operational cash flows.

16.5 C The treasury department is not responsible for preparing the annual financial statements of a company.

16.6 D A forecast cash deficit could be funded by bank borrowing or leading and lagging. It could also be funded by selling any short-term financial investments.

16.7 D All these factors are legitimate causes of cash flow problems.

16.8 104

The index is calculated by taking the revenue for 20X4 and dividing it by the revenue of the base year, ie 27,670/26,500 × 100% = 104.

16.9 A Most UK government securities are fixed interest and have a value of £100. UK government securities generally offer a lower return than local authority stocks.

16.10 B Ownership of certificates of deposit is transferred by physical delivery from buyer to seller.

16.11 B A bank is most likely to approve an overdraft to cover a temporary cash shortfall as the associated risk is relatively low in comparison to the other options.

16.12 D Unusual costs, such as the closure of parts of a business.

16.13 C Of the four options, the payment of wages is the only one which affects the cash of the business in the same way as reported profits. This is because wages are charged to the statement of profit or loss (income statement) and paid in the same period.

16.14 The correct answers are:

- Restricting access to cash and cheques
- Completing appropriate supporting documentation

Cash handling procedures over payments include restricting access to cash and cheques and completing appropriate supporting documentation. The other options are procedures likely to be applied to cash receipts.

16.15 Both statements are true. When economic conditions are unfavourable, a business could face a fall in cash receipts. It therefore will wish to try to maintain the cash balances it already has for safety reasons, and will be reluctant to take on further commitments, such as commitments to pay interest and repay monies borrowed.

16.16 The correct answers are:

- How much cash is required
- When cash is required

Cash forecasts will not show profit and are unlikely to show the sources of funding that are available.

16.17 A

	Trend sales	*Seasonal variation*	*Forecast sales*
	$		$
Quarter 1	178,000 (173,500 + 4,500)	+12,820	190,820

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16.18 $55,300

	January	*February*
	$	$
From January's revenue	33,000	16,800
From February's revenue		38,500
		55,300

16.19 C The treasury department is highly unlikely to be involved in preparing the corporate budget and business plan. This is the function of the Finance and Management Accounts department.

16.20 A Seasonal variation is calculated as S = A – T.

16.21 A Checking to ensure the bank statement agrees with the paying-in records (ii) will not identify misappropriation of cash before monies are paid in.

16.22 D Cheque signatories are responsible for adhering to their limit on signing cheques. They are not responsible for any of the other activities which are usually carried out by finance or administration staff.

16.23 $8,800

Month 2 purchases × $4 = $8,800

	Month 1	*Month 2*	*Month 3*	*Month 4*
Opening inventory	800	800	1,000	1,120
Purchases (balancing figure)	2,000	2,200	2,620	
Closing inventory	(800)	(1,000)	(1,120)	
Sales	2,000	2,000	2,500	2,800

16.24 B

January revenue = $68,400
February – December = $24,873 pcm

Therefore March receipts = 45% of February revenue + 50% of March
= $23,629

16.25 A Preference shares are considered to be the most risky investment.

16.26 B

ACCA examining team comments

This question tests item E3c in the Study Guide on the preparation of cash budgets. For cash budgeting, the relevant amounts relating to non-current assets is when the investment expenditure takes place not when, or how, accounting depreciation is applied. Considerable confusion amongst candidates has again been demonstrated.

Thus, in this question, the depreciation of $11,200 per month is irrelevant for the cash budget. Also irrelevant, for the cash budget for January, are the remaining overheads of $207,600 for the year because they do not become payable until February (ie payable one month in arrears). The relevant data for the cash budget are the accrued overhead from the previous year of $16,800 and the quarterly machine maintenance of $5,900 because they will both be paid in January. The correct answer, which has not been the most popular option, is Option B ($16,800 + $5,900 = $22,700).

The answer most frequently selected by candidates is Option C which also includes depreciation ($16,800 + $5,900 + $11,200 = $33,900). Options A and D have also been commonly chosen by candidates. Option A includes one twelfth of the remaining overheads for the budget year rather than the accrued overhead from the previous year [($207,600 ÷ 12) + $5,900 = $23,200]. Option D incorrectly includes budget year remaining overheads, rather than the accrued overhead from the previous year, but also adds depreciation and ignores the machine maintenance [($207,600 ÷ 12) + $11,200 = $28,500].

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16.27 A

ACCA examining team comments

This question tested knowledge of accrual accounting and cash accounting and required a calculation of the difference, in the surplus for an accounting period in a given scenario, between the two methods. All four options were selected by a significant proportion of candidates.

Using accrual accounting, transactions are recognised when they occur, which is not generally coincident with a cash flow taking place. The objectives are the matching of expenditure with income in order to determine profit or loss for a period and to prepare a statement of assets and liabilities. Cash accounting compares cash receipts and payments in order to determine the net cash flow for a period.

The accrual accounting profit for the first trading period of the new business is $25,500 [sales $140,900 – (cost of goods sold $96,600 + expenses $17,800 + depreciation $1,000)]. The net cash inflow is $8,900 [sales receipts $118,700 – (payments to suppliers of goods and for expenses $89,800 + payment for machine $20,000)]. The difference in the surplus for the period, comparing accrual accounting with cash accounting, is $16,600 ($25,500 – $8,900) and, therefore, Option A is the correct answer. This could alternatively have been calculated directly, using the differences between the methods, as unpaid sales $22,200 + NBV of machine $19,000 – unpaid suppliers $24,600 = $16,600.

Option B is incorrect because it fails to deduct $1,000 for depreciation in calculating the accrual accounting profit. Option D is incorrect because it fails to deduct the payment for the machine in the calculation of net cash flow. Option C is incorrect because it fails to include both the depreciation (accrual accounting) and the payment for the machine (cash accounting).

16.28 D

ACCA examining team comments

Options B, C and D were all selected by a similar proportion of students but a significantly greater proportion chose Option A which was not the correct answer.

The working capital cycle is the period of time that elapses between the point at which cash begins to be expended on the production of a product and the collection of cash from the purchaser of the product.

The correct answer is Option D because all four of the items listed in the question will affect the working capital cycle.

An increase in the credit period agreed with customers (item 1) will mean that customers would take longer to pay for their purchases this increasing the working capital cycle of the seller.

A reduction in inventory of raw materials (item 2) may not in itself reduce the working capital cycle if it simply results from a reduction in production activity. However, the fact that the question states that raw materials inventory is reduced as a percentage of the amount used in production means that the materials will be held in inventory for a shorter period of time than previously before they are used.

Delayed payment to suppliers (item 3) will also reduce the working capital cycle because it would mean that cash is going out of the business later than hitherto.

An increase in the finished goods turnover period (item 4) means that finished goods will be held for a longer period of time than hitherto thus increasing the working capital cycle.

The most popular option, Option A, failed to recognise that the holding of raw materials and finished goods inventories are important elements of the working capital cycle, as well as supplier/customer payments. Option B failed to recognise that supplier/customer payments are important elements of the working capital cycle, as well as raw materials and finished goods inventories. Option C excluded finished goods inventory.

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16.29 A

ACCA examining team comments

All options were popular with students and seemed to indicate that failure to read the question carefully is a widespread problem.

The correct answer is Option A. The budgeted cash receipts in August comprise:

$6,500 which is the cash sales for August (5% of the $130,000 total revenue budget for August); + $37,050 which is 25% of the credit sales budget for July ($156,000 × 0.95 × 0.25) ie 25% received one month after the credit sale;

+ $103,313 which is 75% of the credit sales budget for June ($145,000 × 0.95 × 0.75) ie 75% received two months after the credit sale.

$6,500 + $37,050 + $103,313 = $146,863

Almost as many students selected Option C. The error here was in calculating the 25% credit sales receipts one month after sales and the 75% credit sales receipts two months after sale on the total sales revenue budget figures rather than on 95% of the total which represents the proportion of credit sales.

Options B and D were also popular. The error in Option B was taking 75% (rather than 25%) of the credit sales made in the month prior and 25% (rather than 75%) of the credit sales made two months prior. Option D confused the month of June with the month of August calculating 5% of the sales in June (rather than August) and 75% of the credit sales in August (rather than June).

16.30 C Gilts

Government securities are otherwise known as gilts or gilt-edged securities.

16.31 C A deposit account offered by a bank which invests in stocks and bonds

16.32 D Option deposits

Option deposits are for time periods ranging from 2 to 7 years.

16.33 A

ACCA examining team comments

This question required candidates to identify the cash flow from Operating Activities which is calculated as net income + noncash expenses + changes in working capital

	$
Sales	300,000
Cost of sales	200,000
Net Income (a)	100,000
Receivables	5,000
Inventories	6,000
Payables	(4,000)
Changes in working capital (b)	7,000
Operational cashflow (a) – (b)	93,000

(**BPP note:** Increases in receivables and inventory are effectively negative changes in working capital as they reflect cash outflows and they are therefore deducted from net income.)

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16.34 The first statement is true and the second statement is false.

Where public money is invested by a public sector organisation, investments are restricted by law – true. Local authorities are permitted to invest in long term, high interest accounts – false. One example of legislation is *The Local Authorities (Capital Finance) (Approved Investments) Regulations 2015*, which govern the investment policy of **local authorities**. These regulations ensure that local authorities choose investments that give **safe and easy access** to cash rather than high interest.

16.35 D All three are examples of special cases where an organisation's investments are restricted by law.

16.36 Statement 1 is true and statement 2 is false.

Local authorities must choose investments that give safe and easy access, so statement 1 is true.

Statement 2 is false because local authorities can use both a deposit facility run by the treasury and commercially run money market funds with the highest level of creditworthiness.

17 Mixed bank 1

17.1 C The report will aid management in decision-making.

17.2 D Useful management information is clear, accurate and relevant for its purpose.

17.3 D An interlocking bookkeeping system is where separate accounts are kept for cost accounting and for financial accounting.

17.4 B Job costing is most appropriate when production is carried out in accordance with the special requirements of each customer.

17.5 $66,700

$y = a + bx$

$156,980 = a + (12.20 \times 7,400)$
$156,980 = a + 90,280$
$a = 156,980 - 90,280$
$a = 66,700$

17.6 C Total cost of 6,000 units = 42,000 + 60,000 = 102,000
Cost per unit = 102,000/6,000 = $17

Total cost of 8,000 units = 50,000 + 76,000 = 126,000
Cost per unit = 126,000/8,000 = $15.75

Reduction in total cost per unit = 17 – 15.75 = $1.25

17.7 C A materials returned note is used to record materials sent back.

17.8 D The inventory used in production is valued at $6,500/500 = $13 per kg
410 kg is valued at 410 × $13 = $5,330
Work-in-progress is debited and material inventory credited when material is issued.

17.9 A Re-order level is calculated as maximum usage × maximum lead time to avoid stock-outs.

17.10 $40

$$EOQ = \sqrt{\frac{2cd}{h}}$$

$$800 = \sqrt{\frac{2 \times c \times 12,000}{1.5}}$$

$$640,000 = \frac{2 \times c \times 12,000}{1.5}$$

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960,000 = 24,000 × c

c = 40

17.11 The correct answers are:

- Time sheet
- Job card

Time sheets and job cards may be used to allocate costs to cost units.

17.12 D The production volume ratio is calculated as standard hours for actual output ÷ budgeted hours.

17.13 A Capital expenditure is the cost of acquiring or enhancing non-current assets.

17.14 D

	A	*B*	*C*	*D*
Overhead expenditure	18,757	29,025	46,340	42,293
Direct labour hours	3,080	6,750		
Machine hours			3,380	2,640
Overhead absorption rate	$6.09	$4.30	$13.71	$16.02

17.15 C Production cost per unit = $3.60 + ($258,000/60,000) = $7.90
Profit = 700,000 – (56,000 × 7.90) – 144,000 = $113,600

17.16 C Inventory levels have increased so marginal costing will result in higher profits and lower inventory values than absorption costing.

17.17 48.6%

Contribution = 17.50 – (7.60 + 1.40) = $8.50 per unit
Contribution/sales ratio = 8.5/17.5 × 100 = 48.6%

17.18 B

	A	*B*	*C*	*D*
	$	$	$	$
Variable manufacturing costs	6.00	7.50	9.00	11.50
Bought-in price	11.00	12.00	13.00	16.00
Difference	5.00	4.50	4.00	4.50
Labour hours/unit	1.5 hrs	4 hrs	2 hrs	3 hrs
Extra cost of buy-in (savings from in-house manufacture) per unit of scare resource (labour hours)	3.33	1.125	2.00	1.50

Component B has the lowest extra cost of buying in. This would therefore have the smallest effect on profit.

17.19 A $\text{IRR} = a\% + [\frac{A}{A-B} \times (b-a)]\%$

where a is one interest rate
b is the other interest rate
A is the NPV at rate a
B is the NPV at rate b

$$\text{IRR} = 14\% + \left[\frac{16{,}000}{(16{,}000+10{,}500)} \times (20-14)\right]\%$$

= 14% + 3.6%

= 17.6%

17.20 B A budget is 'flexed' according to actual activity levels in order to make meaningful comparisons.

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18 Mixed bank 2

18.1 A The control process is the action of monitoring something to keep it on course.

18.2 B Statement (i) is not true as the benefit and cost of management information should be compared. Statement (iii) is not true because the cost of providing very detailed and accurate information may exceed the potential value of that information.

18.3 C The concept of cost units is a feature of cost accounting but not of financial accounting.

18.4 D All three factors.

18.5 $0.95

Low production is in Period 2. High production is in Period 1. A change in production units must lead to the change in costs ie variable costs.

$$\frac{198{,}968 - 187{,}739}{129{,}440 - 117{,}620} = \frac{11{,}229}{11{,}820} = 0.95$$

18.6 C FIFO (LIFO or weighted average) is used to value issues from inventory and also to calculate the value of inventory at a period end.

18.7 1.25 kg

(1.0/0.8) = 1.25 kg

18.8 C The maximum inventory control level is the level that inventory should not exceed if usage is at the minimum expected.

18.9 The correct answers are:

- Cost centre supervisors' wages
- Overtime costs of indirect operatives

Indirect production costs or production overheads are costs that are incurred in the course of making a product or providing a service, but which cannot be traced directly and in full to the product or service.

18.10 B Statement (iii) is not true as a balancing charge, which is the opposite to a balancing allowance, would be made.

18.11 C 65% of SCC1 Total Overheads and 35% of SCC2 overheads

($32,170 × 0.65) + ($24,850 × 0.35) = $29,608

18.12 The profit for the period would be higher using absorption costing – false

Inventory values would be higher using absorption costing – true

The company manufactures less than it sells therefore inventory levels are falling.

18.13 B Money-market deposit

18.14 C Process costing

18.15 C

	$
Costs incurred in the process	61,600
less: scrap proceeds (440 × 1.80)]	(792)
	60,808

Per unit (/ 21,560) = $2.82 (to the nearest cent)

18.16 A Ensuring the business has investments which are easily convertible into cash.

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18.17 A Total joint costs = $87,500

Calculation of share for product X:

		$
Sales value of X	$6 × 10,000 =	60,000
Sales value of Y	$8 × 15,000 =	120,000
		180,000

(60/180) × $87,500 = $29,167

18.18 D Total costs divided by total number of occupied rooms

$104,976 / (60 × 0.8 × 90) = $24.30

18.19 C The personnel involved is not a valid factor when considering whether to investigate a variance.

18.20 C

	$
Initial Investment	81,000
NPV	8,683
	89,683

Divided by annuity factor

for Yrs 1–4 (3.037) = $29,530

OR:

Initial investment	81,000
Divided by annuity factor	

At IRR rate (2.743) = $29,530

19 Mixed bank 3

19.1 D Part of a business where management is responsible for revenues and costs

19.2 A Calculating cost variances

19.3 The correct answers are:

- Non-financial as well as financial
- Not legally required

Management accounting information can be non-financial as well as financial and is not legally required.

19.4 B *Debit* Finished goods *Credit* Work in progress

19.5 D Cleaning materials have been coded incorrectly based on the cost codes.

19.6 C Stepped-fixed cost

19.7 C Insurance

19.8 B Decrease by less than 10%

19.9 C Materials returned note

19.10 D ($17,600 + 450 + 760 + 2,780 + 1,100)

19.11 A Apportionment of wages to cost centres is not part of the payroll system.

19.12 A Actual hours worked being greater than budgeted hours could have caused a direct labour capacity ratio of 104%.

19.13 C Overhead over- or under-recovery

19.14 20.5%

{$22/unit – 11.6 – [($7,200 + 16,400) ÷ 4,000 units]} × 100 ÷ 22

19.15 B [200 units × ($7.5 – 4.8)/unit]

19.16 D [(630 ÷ 0.9 hours) × $12/hour)]

19.17 $9.40

{[$216,720 – (1,200 units × $2/unit)] ÷ (24,000 – 1,200 units)}

19.18 C This is the only one which affects profit in the same way as the cash of the business as the wages are charged to profit and paid in the same period.

19.19 Both statements are true.

19.20 A Working capital is current assets less current liabilities so 56,000 + 12,000 – 2,000 – 15,000 – 1,500 = $49,500.

20 Mixed bank 4

20.1 D Sources of useful data may be external, internal, financial or non-financial.

20.2 The correct answers are:

- The format of management accounts may vary from one business to another.
- Management accounting provides information to help management make business decisions.

20.3 B Codes for a particular type of item should be consistent in length and structure.

20.4 C Line C

20.5 $14.84

($52,000 ÷ 5,000 units) + {[($760,000 – (50,000 units × $10.40/unit)] ÷ 54,000 units}

20.6 A Increase by less than 15%

20.7 C [(8 units × $260/unit) + (12 units × $270/unit)]

20.8 2,000 kg

[(3,400 kg + 600 kg) ÷ 2]

20.9 A Provision of leisure facilities for employees

20.10 The correct answers are:

- If the direct method is used, the order in which the service cost centre overheads are reapportioned is irrelevant.
- The step down method results in costs being reapportioned between service cost centres.

20.11 B [($44,210 × 1.4) + $3,190]

20.12 C {[$4,250 ÷ (420 units good output + 20 units abnormal loss)] × 20 units}

20.13 D Statement 2 only is true.

20.14 D All three items are important.

20.15 A A by-product is a product that has insignificant saleable value compared with the joint products.

20.16 B [($39,420 + $11,880) ÷ 5,400 units]

20.17 A Product P, Product Q, Product R, Product S.

20.18 C Sunk cost = $25,000 Incremental cost = $18,000

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20.19 C Time 0 = ($60,000)
Time 1 = $36,000

20.20 D Poor forecasting techniques, unexpected inflation, the loss of a major customer and interest rate changes could all cause differences between a company's cash flow forecast and its actual cash flows.

21 Mixed bank 5

21.1 The correct answers are:

- Forward looking
- Concerned with cost control

Management accounting information is forward looking and is concerned with cost control.

21.2 D An investment centre is part of a business where management is responsible for capital investment as well as profit.

21.3 D The work-in-progress control account and the receivables control account are features of an integrated accounting system.

21.4 D *Debit* Cost of sales account *Credit* Finished goods account

21.5 B Wages of the process supervisor are coded incorrectly based on the cost codes.

21.6 C Scanner, CD, bar code reader

21.7 D Safety, profitability and liquidity

21.8 C Stepped-fixed cost

21.9 D Increases by less than 20%

21.10 $40,372

($36,260 + $4,112)

21.11 D Standard time for actual output being less than budgeted hours could have caused a production volume ration of 25%.

21.12 $1,140

[($28,720 + $10,260) – (1,760 units × $21.50 per unit)]

21.13 B Absorption costing profit is lower because of the difference in inventory levels.

21.14 B [($12,600 ÷ 0.6) – ($12,600 × 1.3)]

21.15 $4.65

{[$45,705 – (300 units × $2.00 per unit)] ÷ 9,700}

21.16 B [$56,000 × ($40,000 ÷ $90,000)]

21.17 B $[(1.043)^2 - 1]$

21.18 B {10% + [(15 – 10%) × (39.1 ÷ 43.9)]}

21.19 C [($56,000 × 3.79) + ($10,000 × 0.62) – $186,000]

21.20 A Investment sum / net cash inflow per annum.

22 Mixed bank 6

22.1 C Management information should be timely and clear to the user.

22.2 D Residual income

22.3 C Government bonds are also known as gilts.

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22.4 D Interlocking accounts

22.5 A Decrease by a third

22.6 B Indirect labour cost

22.7 $36,490

(22,800 × 23/24) + 14,640

22.8 The correct answers are:

- Goods received note
- Materials requisition

Goods received notes and materials requisitions are both used in the process of purchasing and using raw materials.

22.9 A (6,200 × 0.8) – 380

22.10 D $(2 \times 40 \times 12{,}000) \div 400^2$

22.11 A Number of materials requisitions

22.12 B 20,290/560

22.13 A (3.0 × 10.60) + (1.3 × 36.20)

22.14 Both statements are false.

22.15 C Job costing would be most appropriate for a car repairer.

22.16 $22.96

179,070/7,800

22.17 D Both descriptions describe a by-product.

22.18 A A composite cost unit may be used in service costing.

22.19 C 24 – (180,000/20,000)

22.20 A 15/60

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Appendix: Mathematical tables

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Present value table

Present value of 1 ie $(1+r)^{-n}$

where r = discount rate
n = number of periods until payment

Periods	Discount rates (r)									
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239

Periods										
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065

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Annuity table

Present value of annuity of 1, ie $\frac{1-(1+r)^{-n}}{r}$

where r = discount rate
n = number of periods.

Periods (n)	Discount rates (r) 1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606

Periods (n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675

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Mock Exam 1
(Specimen Exam)

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Foundations in Accountancy

MA2

Managing Costs and Finances

Mock Examination 1

Specimen Exam

Questions	
Time allowed	2 hours
ALL 50 questions are compulsory and must be attempted	

DO NOT OPEN THIS EXAM UNTIL YOU ARE READY TO START UNDER EXAMINATION CONDITIONS

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ALL 50 questions are compulsory and MUST be attempted

Each question is worth 2 marks.

1 **Which costing method would be MOST suitable for an accountancy firm?**

- ○ Contract costing
- ○ Job costing
- ○ Batch costing
- ○ Process costing

(2 marks)

---

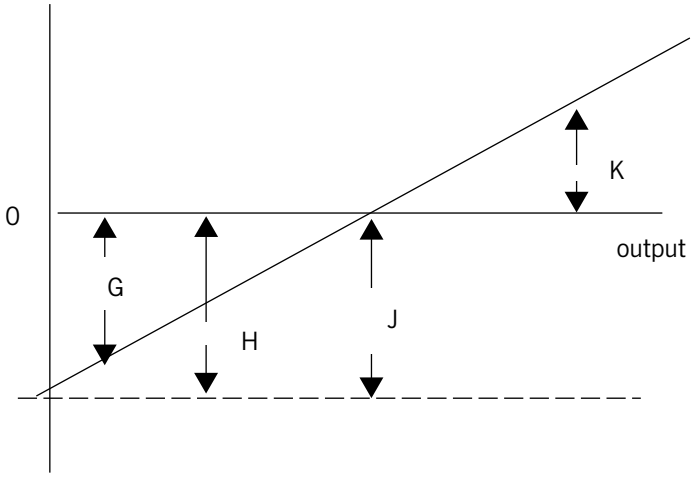
2 **Is each of the following statements, about establishing the trend of a time series, true or false?**

	True	*False*
Where the time series is approximately linear, the line of best fit can be estimated on a scatter graph.	○	○
Where the time series is not approximately linear, moving averages can be calculated.	○	○

(2 marks)

---

3 Four vertical lines have been labelled G, H, J and K at different levels of activity on the following profit-volume chart:



Which line represents the total contribution at that level of activity?

- ○ Line G
- ○ Line H
- ○ Line J
- ○ Line K

(2 marks)

---

4 **Which TWO items in the following list relate to the term 'by-product' in process costing?**

- ☐ Has low sales value relative to joint products
- ☐ Accounted for by crediting the net realisable value to the work-in-progress account
- ☐ Equivalent units of output are less than the actual good units of output
- ☐ Charged with a share of joint production costs

(2 marks)

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5 The following data are available for product X.

	Period Budget	*Period Actual*
Sales units	5,000	5,200
	$	$
Sales revenue	50,000	57,200
Manufacturing cost	30,000	31,200

What is the sales price variance?

- ○ $5,200 adverse
- ○ $5,000 favourable
- ○ $5,200 favourable
- ○ $7,200 favourable

(2 marks)

---

6 A company orders a particular raw material in order quantities of 250 units. No safety inventory is held, the inventory holding cost is $3 per unit per annum and the annual demand is 2,500 units.

What is the total annual inventory holding cost of the material?

- ○ $375
- ○ $750
- ○ $3,750
- ○ $7,500

(2 marks)

---

7 A company has two production cost centres, Cutting and Finishing.

The overheads and operating hours for the two cost centres are:

Cutting:	$210,000	60,000 machine hours	4,000 labour hours
Finishing:	$200,000	5,000 machine hours	14,000 labour hours

From the information given what should be the basis for overhead absorption?

- ○ Both cost centres should be based on machine hours
- ○ Both cost centres should be based on labour hours
- ○ Based on machine hours for Cutting and labour hours for Finishing
- ○ Based on labour hours for Cutting and machine hours for Finishing

(2 marks)

---

8 A company uses the production units method to depreciate the machinery in its factory. A machine that cost $166,200 has an estimated residual value of $5,000 at the end of its six-year useful operating life.

Output from the machine is estimated as 124,000 units over the six years. 15% of the total output will be manufactured in each of Years 1, 2 and 3 with 20% of the total in Years 4 and 5.

What is the total depreciation of the machine in Year 4?

$ ☐

(2 marks)

---

9 12,000 kg of materials, costing $86,090, were input to a manufacturing process in a period during which conversion costs totalled $39,320. Losses in the period were 960 kg with no saleable value. The normal loss is 10% of input.

What was the total manufacturing cost per kg of expected output (to two decimal places of $)?

- ○ $9.68
- ○ $10.45
- ○ $11.36
- ○ $11.61

(2 marks)

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10 **Which of the following may be included in the cost accounts but would be excluded from the financial accounts?**

- ○ Depreciation of equipment
- ○ Distribution expenses
- ○ Factory manager's salary
- ○ Notional rent

(2 marks)

---

11 A company is evaluating a project that requires 400 kg of raw material X. The company has 150 kg of X in stock that was purchased six months ago for $55 per kg. The company no longer has any use for X. The inventory of X could be sold for $40 per kg. The current purchase price for X is $53 per kg.

What is the total relevant cost of raw material X for the project?

$ ☐

(2 marks)

---

12 Dipton have apportioned their indirect costs as shown below:

	DEPT A	*DEPT B*	*DEPT C*	*DEPT D*
Rent and rates ($)	10,000	15,000	45,000	12,500
Heat and lighting ($)	15,000	25,000	25,000	20,000
Administration ($)	15,000	35,000	35,000	35,000
Premises insurance ($)	20,000	15,000	15,000	12,500
Total ($)	60,000	90,000	120,000	80,000
Hours worked:				
Direct labour	10,000	15,000	12,500	10,000
Machine	5,000	20,000	15,000	12,500

If direct labour hours are used to absorb the indirect costs, which department will have the highest hourly absorption rate?

- ○ Department A
- ○ Department B
- ○ Department C
- ○ Department D

(2 marks)

---

13 Two statements follow about the purpose of an email system:

(1) The purpose of an email system is to send and receive data a computer can work with.
(2) The purpose of an email system is to send and receive messages quickly and cheaply.

Are the statements true or false?

- ○ Both statements are false
- ○ Both statements are true
- ○ Statement 1 is false and Statement 2 is true
- ○ Statement 1 is true and Statement 2 is false

(2 marks)

---

14 In a transport business, unit costs are calculated per tonne/kilometre. The following data is available:

Average load (tonnes)	*Average distance (kilometres)*	*Number of journeys*
4	40	10
5	60	12
6	65	8

Total costs were $34,295.

What was the cost per tonne/kilometre (to two decimal places of $)?

$ ☐

(2 marks)

---

15 **Which of the following are objectives of cash budgeting?**

(1) To anticipate cash shortages and surpluses
(2) To enable necessary funds to be made available
(3) To monitor trade receivables

- ○ 1 and 2 only
- ○ 1 and 3 only
- ○ 2 and 3 only
- ○ 1, 2 and 3

(2 marks)

16 **Which of the following describes a flexed budget?**

- ○ A budget that is adjusted to the actual level of activity achieved
- ○ A budget that can be varied by any circumstances
- ○ A budget that is adjusted for inflation
- ○ A budget to allow new product development

(2 marks)

17 **Which of the following formulae can be used for calculating labour efficiency ratio?**

- ○ Actual hours ÷ standard hours of actual output
- ○ Budgeted hours ÷ actual hours
- ○ Standard hours of actual output ÷ actual hours
- ○ Standard hours of actual output ÷ budgeted hours

(2 marks)

18 In Year 1 a company incurred the following indirect costs:

Heat and lighting	$35,000
Rent and rates	$45,000

These costs are apportioned to Department A and Department B on the basis of floor area occupied. A occupies 10,000 square metres and B occupies 5,000 square metres. In Year 2 the indirect costs will rise by 20% and the size of the building will be increased by one third. Department C will occupy the new area.

In Year 2 how much of the indirect costs will be apportioned to Department B?

- ○ $20,000
- ○ $24,000
- ○ $32,000
- ○ $48,000

(2 marks)

19 **Which of the following items are treated differently in cash accounting compared with accruals accounting?**

(1) Depreciation
(2) Sales income
(3) Expenditure on materials

- ○ 1 only
- ○ 1 and 2 only
- ○ 2 and 3 only
- ○ 1, 2 and 3

(2 marks)

20 A hotel had total costs of $1,500,000 for a period. There were a total of 120,000 occupied bed-nights in the period.

What is the cost per service unit (to two decimal places)?

$ ☐

(2 marks)

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21 **Which TWO of the following tasks related to cash handling need to be separated (ie the same employee cannot perform both tasks), in order to prevent fraud from being committed and concealed?**

- ☐ Access to liquid assets
- ☐ Filing of procedures
- ☐ Investment of surplus funds
- ☐ Recording of transactions **(2 marks)**

---

22 A company manufactures a single product which is sold for $70.00 per unit. Unit costs are:

	$/Unit
Variable production	29.50
Fixed production	21.00
Variable selling	4.80
Fixed selling	9.00

20,000 units of the product were manufactured in a period during which 19,700 units were sold.

Using marginal costing, what was the total contribution made in the period?

$ ☐ **(2 marks)**

---

23 A company planned to produce 4,000 units of Product X during a particular year and budgeted its fixed production overheads for the year at $20,000. During the year it actually produced 4,200 units of Product X and it incurred fixed production overheads of $21,840. A predetermined fixed production overhead absorption rate per unit is applied.

Which of the following statements is true?

- ○ Fixed overheads were under-absorbed by $840
- ○ Fixed overheads were over-absorbed by $840
- ○ Fixed overheads were over-absorbed by $1,000
- ○ Fixed overheads were under-absorbed by $1,840 **(2 marks)**

---

24 A firm has discovered that the cost of a raw material will increase.

If nothing else changes what is the effect of this on margin of safety and breakeven point?

- ○ The margin of safety will decrease and the breakeven point will increase
- ○ The margin of safety will increase and the breakeven point will increase
- ○ The margin of safety will decrease and the breakeven point will decrease
- ○ The margin of safety will increase and the breakeven point will decrease **(2 marks)**

---

25 **Which of the following unbudgeted events could lead to a favourable cash flow variance?**

- ○ Extended credit given to customers
- ○ Reduced depreciation charges
- ○ Extended credit given by suppliers
- ○ Taking advantage of early settlement discounts offered by suppliers **(2 marks)**

---

26 The cost accountant in a factory has set out the following summary data for the last two periods:

	Period 1	*Period 2*
Total production costs ($)	47,328	51,652
Output (units)	2,400	2,900

What is the BEST estimate of the variable production costs per unit of output (to two decimal places)?

- ○ $17.81
- ○ $19.72
- ○ $18.68
- ○ $ 8.65 **(2 marks)**

---

27 The occupancy rates for a hotel over a two-year period are shown below:

	Year 1				*Year 2*			
	Qtr 1	*Qtr 2*	*Qtr 3*	*Qtr 4*	*Qtr 1*	*Qtr 2*	*Qtr 3*	*Qtr 4*
Percentage of bedrooms occupied	60%	75%	90%	60%	65%	85%	95%	55%

Which of the following statements is correct?

- ○ The occupancy rate for Year 1 is higher than Year 2 for quarters 3 and 4.
- ○ The occupancy rate for Year 2 is higher than Year 1 for quarters 1, 2 and 3.
- ○ The occupancy rate for Year 2 is higher than Year 1 for all four quarters.
- ○ The occupancy rate for Year 1 is higher than Year 2 for quarters 1, 2 and 3. **(2 marks)**

---

28 The following information relates to Product X for Month 1:

Opening inventory	Nil
Production	900 units
Sales	800 units

If marginal costing rather than absorption costing is used what is the effect on profit and inventory valuation?

- ○ Profit higher Inventory valuation higher
- ○ Profit higher Inventory valuation lower
- ○ Profit lower Inventory valuation higher
- ○ Profit lower Inventory valuation lower **(2 marks)**

---

29 All sales of a company are on credit. Budgets for a period include:

Sales	$724,000
Opening trade receivables	$206,900
Closing trade receivables	$241,600

$4,360 of the opening trade receivables are budgeted to be written off as bad debts during the period.

What are the budgeted cash receipts from sales in the period?

- ○ $684,940
- ○ $689,300
- ○ $754,340
- ○ $758,700 **(2 marks)**

---

30 The following cost details relate to a single product manufactured by Business X:

	Per Unit
Direct materials (5 kg)	$30
Direct labour (11 hours)	$77
Production overheads	$45

During the next period direct labour will be restricted to 340,000 hours and only 140,000 kg of material will be available. Demand is expected to be 30,000 units.

What will be the limiting factor for the next period?

- ○ Material only
- ○ Labour only
- ○ Material and labour
- ○ Neither material nor labour **(2 marks)**

---

31 A company has calculated that its production volume ratio is 103.5% and that its efficiency ratio is 90.0%.

What is the capacity utilisation ratio (to the nearest whole number)?

[ ] % **(2 marks)**

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32 **When is the breakeven point achieved?**

- ○ When the level of contribution is equal to total costs
- ○ When the total variable costs are equal to total contributions
- ○ When the total variable costs are equal to total fixed costs
- ○ When the total contribution is equal to total fixed costs **(2 marks)**

33 **Which TWO of the following statements relating to relevant cost concepts in decision making are correct?**

- ☐ Materials can never have an opportunity cost whereas labour can
- ☐ The annual depreciation charge is not a relevant cost
- ☐ Fixed costs would have a relevant cost element if a decision causes a change in their total expenditure
- ☐ Materials already held in inventory never contribute to relevant cost **(2 marks)**

34 The following data relates to a raw material:

Date		*Units*	*Unit Price* $	*Value* $
1 Jan	Balance b/f	100	5.00	500.00
3 Mar	Issue	40		
4 Jun	Receipt	50	5.50	275.00
6 Jun	Receipt	50	6.00	300.00
9 Sept	Issue	70		

If the LIFO method of pricing is used, what is the value of the issue on 9 September?

- ○ $350
- ○ $395
- ○ $410
- ○ $420 **(2 marks)**

35 A firm with a cost of capital of 12% per annum is considering investing $20,000 now in order to receive 10 annual sums of $4,000 (commencing in one year's time). The annuity factor for 12% over 10 years is 5.65.

What is the net present value of the investment?

- ○ $2,600 positive
- ○ $2,600 negative
- ○ $20,000 positive
- ○ $22,600 positive **(2 marks)**

36 A company operates a piecework payment scheme. Workers receive $0.60 for each unit produced. However the company guarantees that each worker will receive at least $45 per day.

Shown below is the number of units produced by worker A during a recent week:

Day	*Monday*	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
Units produced	90	70	75	60	90

What are worker A's earnings for the week?

$ ☐ **(2 marks)**

37 An investment made now would yield $15,972 in three years if compound interest is earned at an annual rate of 10%.

What is the amount of the investment now?

- ○ $12,000
- ○ $13,200
- ○ $10,909.10
- ○ $11,643.59 **(2 marks)**

---

38 A company manufactures and sells four products. Sales demand cannot be met owing to a shortage of skilled labour. Details of the four products are:

	Product A	*Product B*	*Product C*	*Product D*
Sales demand (units)	1,500	2,000	1,800	1,900
Contribution ($/unit)	2.80	2.60	1.90	2.40
Contribution/sales (%)	30	40	50	45
Skilled labour (hours/unit)	1.4	1.2	0.9	1.0

In what order should the products be made in order to maximise profit?

- ○ Product A, Product B, Product D, Product C
- ○ Product B, Product D, Product C, Product A
- ○ Product C, Product D, Product B, Product A
- ○ Product D, Product B, Product C, Product A **(2 marks)**

---

39 **What is an investment centre?**

- ○ Part of a business involved in financial services
- ○ Part of a business that utilises equipment and machinery
- ○ Part of a business where management are responsible both for revenues and for operating costs
- ○ Part of a business where management are responsible for decisions regarding the purchase of non-current assets **(2 marks)**

---

40 **When communicating written information, which of the following determine(s) the choice of method used?**

(1) Comparative cost
(2) Degree of confidentiality
(3) Speed of delivery

- ○ 1 only
- ○ 3 only
- ○ 1 and 2 only
- ○ 1, 2 and 3 **(2 marks)**

---

41 **Would each of the following appear as an item in a cash budget?**

	Yes	*No*	
Depreciation of a non-current asset	○	○	
Payment for the purchase of a non-current asset	○	○	**(2 marks)**

---

42 **In an interlocking accounting system what would be the entry in the cost accounts to record the charging of indirect production labour costs?**

	Debit	*Credit*	
○	Cost ledger control	Production overhead	
○	Production overhead	Wages	
○	Wages	Financial ledger control	
○	Work-in-progress	Wages	**(2 marks)**

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43 A company manufactures a single product. Unit costs are:

	$/Unit
Variable production	14.75
Fixed production	8.10
Variable selling	2.40
Fixed selling	5.35

400,000 units of the product were manufactured in a period, during which 394,000 units were sold. There was no inventory of the product at the beginning of the period.

Using marginal costing, what is the total value of the finished goods inventory at the end of the period?

$ ☐ **(2 marks)**

---

44 Ordering lead times and weekly usage of a raw material are:

	Lead time	*Weekly usage*
Minimum	2 weeks	400 kg
Maximum	3 weeks	500 kg
Average	2½ weeks	450 kg

The economic order quantity of the material is 1,800 kg and the reorder level is 1,500 kg.

What is the minimum inventory control level?

- ○ 300 kg
- ○ 375 kg
- ○ 675 kg
- ○ 700 kg

(2 marks)

---

45 **Which of the following is NOT a feature of certificates of deposit?**

- ○ Fixed term
- ○ Issued by a bank
- ○ Non-negotiable
- ○ Specified interest rate

(2 marks)

---

46 **Which of the following describes the margin of safety?**

- ○ The total sales units up to break-even sales volume
- ○ The difference in units between the expected sales volume and the break-even sales volume
- ○ The difference between sales value and variable costs
- ○ The difference between total costs and the fixed costs at break-even sales volume

(2 marks)

---

47 **Do each of the following statements explain how a cash budget can be used as a mechanism for control?**

	Yes	*No*
Actual cash flows can be compared with budgeted cash flows to reveal variations from what was expected	○	○
Cash budgets can be revised on a regular basis for forecasting purposes	○	○

(2 marks)

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48 In the manufacture of Chemical X there is a normal loss of 10% of the material input into the process. 340 litres of Chemical X were manufactured in a period during which there was an abnormal loss of 5% of the material input into the process.

How many litres of material were input into the process during the period?

☐ litres **(2 marks)**

---

49 The following information has been extracted from the records of a firm:

	Production department		
	Budget	*Result*	*Variance*
Direct materials	$1,231	$1,648	$427 Adv
Direct labour	$2,156	$1,972	$184 Fav
Direct expenses	$125	$142	$17 Adv

Consider the following statements:

(1) Overtime premium of direct labour caused direct expenses to increase.

(2) Lower skilled direct labour was used causing increased usage of direct materials.

(3) Higher quality of direct labour and materials were used enabling fewer direct labour hours to be required.

(4) Higher skilled direct labour was used enabling less direct material to be used.

Which of the statements are consistent with the variances shown?

- ○ 2 and 3 only
- ○ 3 and 4 only
- ○ 1, 2 and 4 only
- ○ 1, 2, 3 and 4 **(2 marks)**

---

50 **Which of the following are usually treasury functions?**

(1) Credit control
(2) Currency management
(3) Debt collection
(4) Investment of surplus funds

- ○ 1 and 3
- ○ 1 and 4
- ○ 2 and 3
- ○ 2 and 4 **(2 marks)**

(Total = 100 marks)

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Answers to Mock Exam 1 (Specimen Exam)

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1 Job costing

2 Both statements are true.

3 Line J

4 The correct answers are:

- Has low sales value relative to joint products
- Accounted for by crediting the net realisable value to the work-in-progress account

A by-product is an incidental product from a process and has an insignificant value compared to the main product. The net realisable value of a by-product can be deducted from the cost of production of the main product.

5 $5,200 favourable

Sales price/unit = $50,000 ÷ 5,000 = $10

5,200 units were sold which would generate $52,000.

The actual revenue was $57,200 so the sales price variance is $57,200 – $52,000 = $5,200 favourable.

6 $375

Total annual inventory holding cost is average inventory × holding cost. As no safety inventory is held, average inventory is (0 + 250)/2 = 125. Annual inventory holding cost is therefore 125 × 3 = $375.

7 Based on machine hours for Cutting and labour hours for Finishing

Overhead absorption should be based on machine hours for Cutting and labour hours for Finishing.

8 $32,240

The amount to be depreciated is the cost less the residual value, so $166,200 – $5,000 = $161,200. 20% output will be manufactured in Year 4 so the depreciation charge for Year 4 will be $161,200 × 20% = $32,240.

9 $11.61

Expected output is 90% of 12,000 as normal loss is 10% of output, so 10,800 kg. Total manufacturing cost is materials cost + conversion costs = $125,410. So total manufacturing cost/kg = 125,410/10,800 = $11.61

10 Notional rent

Notional rent would not be included in the financial accounts because it is a nominal charge for rent on premises that are owned – no rent has actually been paid or accrued for since the premises are owned.

11 $19,250

The cost to purchase product X is a sunk cost and therefore irrelevant. However it could be sold for $55/kg, so $6,000 – this is a relevant cost. As there are already 150 kg of X in stock, the company needs another 250 kg of X for the project, which will cost $13,250 (250 × 53). The total relevant cost of the project is therefore $19,250.

12 Department C

The hourly absorption rate can be calculated for each department: department A: 6 (60,000/10,000); department B: 6 (90,000/15,000); department C 9.6 (120,000/12,500); department D: 8 (80,000/10,000).

13 Both statements are true.

14 $4.12

The cost per tonne/km can be calculated as follows:

34,295/{(4 × 40 × 10) + (5 × 60 × 12) + (6 × 65 × 8)} = $4.12.

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15 1 and 2 only

Cash budgeting helps businesses to anticipate cash shortages and surpluses and enables necessary funds to be made available. Cash budgeting does not monitor trade receivables.

16 A budget that is adjusted to the actual level of activity achieved

A flexed budget is a budget that is adjusted to the actual level of activity achieved.

17 Standard hours of actual output/actual hours.

18 $24,000

The total indirect costs in Year 2 will be $96,000 (80,000 × 1.2). As the floor area will increase by 1/3, the total floor area will be 20,000 square metres (15,000 × 1.3333). Department B still occupies 5,000 square metres of the building so in Year 2, the amount of indirect costs apportioned to department B will be 96,000 × (15,000/20,000) = $24,000.

19 1, 2 and 3

All the items are treated differently in cash accounting compared with accruals accounting, since accruals accounting takes account of depreciation of non-current assets, as well as trade receivables and trade payables which impact on sales and purchases costs.

20 $12.50

The cost per service unit is the total cost of $1,500,000 divided by the total occupied bed-nights of 120,000 = $12.50.

21 The correct answers are:

- Access to liquid assets
- Recording of transactions

Access to liquid assets and recording of transactions are tasks that need to be separated because if an employee can do both of these, it is very easy for that employee to misappropriate cash by not recording it on the company's systems.

22 $703,290

In marginal costing, contribution is selling price less variable costs so 70 – 29.50 – 4.80 = $35.70. As 19,700 units were sold, the total contribution will be 19,700 × 35.70 = $703,290.

23 Fixed overheads were under-absorbed by $840

The overhead absorption rate is 20,000/4,000 = $5/unit. As 4,200 units were produced, the absorbed overheads are $21,000. The company actually incurred fixed production overheads of $21,840. Fixed overheads were therefore under-absorbed by $840.

24 The margin of safety will decrease and the breakeven point will increase.

25 Extended credit given by suppliers

Extended credit given by suppliers could lead to a favourable cash flow variance.

26 $ 8.65

The best estimate of variable production costs is (51,652 – 47,328)/(2,900 – 2,400) = $8.65.

27 The occupancy rate for Year 2 is higher than Year 1 for quarters 1, 2 and 3.

28 Profit would be lower and inventory valuation would be lower.

29 $684,940

The budgeted cash receipts can be calculated by taking opening trade receivables of $206,900 and the sales in the period of $724,000 and deducting the $4,360 budgeted to be written off and the closing trade receivables of $241,600. This gives budgeted cash receipts of $684,940.

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30 Material only

Material is the limiting factor.

Direct materials required: 5 kg × 30,000 units = 150,000 kg (140,000 kg available).

Direct labour required: 11 hours × 30,000 units = 330,000 hours (340,000 hours available).

31 115

The capacity utilisation ratio can be calculated as 115 [(103.5/90) × 100].

32 When the total contribution is equal to total fixed costs

The breakeven point is when total contribution equals fixed costs. At this point, no profit or loss is made.

33 The correct answers are:

- The annual depreciation charge is not a relevant cost
- Fixed costs would have a relevant cost element if a decision causes a change in their total expenditure

Relevant costs are cash flows, so depreciation cannot be a relevant cost in decision making. Fixed costs are generally irrelevant to a decision, unless they are directly attributable fixed costs, in which case they will be relevant to a decision.

34 $410

Under the LIFO method, we take the last inventory in store first. In this case, (50 × $6) (being receipts on 6 June) + (20 × $5.50) (being part of the receipt on 4 June) = $410.

35 $2,600 positive

The NPV can be calculated as the initial investment of $20,000 less the present value of the annuity (4,000 × 5.65). The NPV of the investment is therefore $2,600 positive.

36 $243

Earnings for Monday are 90 × 0.60 = $54. Earnings on Tuesday are 70 × 0.60 = $42, but worker A is guaranteed at least $45 each day so will get $45 on Tuesday. This also occurs on Wednesday and Thursday as worker A's actual earnings on these days are less than $45. On Friday, earnings are 90 × 0.60 = $54. Total earnings for the week are therefore 54 + 45 + 45 + 45 + 54 = $243.

37 $12,000

The value of the investment now is 15,972/(1.1 × 1.1 × 1.1) = $12,000.

38 Product D, Product B, Product C, Product A

	Product A	*Product B*	*Product C*	*Product D*
Contribution per unit	$2.80	$2.60	$1.90	$2.40
Labour hours required per unit	1.4	1.2	0.9	1.0
Contribution per hour of labour	$2.00	$2.17	$2.11	$2.40
Ranking	4	2	3	1

39 Part of a business where management are responsible for decisions regarding the purchase of non-current assets

An investment centre is a profit centre with additional responsibilities for capital investment and possibly for financing, and whose performance is measured by its return on investment.

40 1, 2 and 3

All three factors determine the communication method used.

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41 Depreciation of a non-current asset No

Payment for the purchase of a non-current asset Yes

Only the payment of a non-current asset will appear because this is the only item for which cash has actually been spent. Depreciation is a non-cash charge made to the statement of profit or loss (income statement) to match the cost of a non-current asset to its useful life.

42 *Debit* Production overhead *Credit* Wages

43 $88,500

6,000 units are left unsold. The value of these using marginal costing will be 6,000 × 14.75 = $88,500.

44 375 kg

The minimum inventory control level is the re-order level less (average lead time × average usage) so 1,500 – (2.5 × 450) = 375 kg.

45 Non-negotiable

Certificates of deposit are negotiable.

46 The difference in units between the expected sales volume and the break-even sales volume

The margin of safety is the difference in units between the expected sales volume and the breakeven sales volume.

47 Actual cash flows can be compared with budgeted cash flows to reveal variations from what was expected: Yes

Cash budgets can be revised on a regular basis for forecasting purposes: No

Using effective cash budgeting, actual cash flows can be compared with budgeted cash flows to reveal variations from what was expected.

48 400 litres

There is a normal loss of 10% so if 340 litres were produced, 340/0.9 must have been input. However, there was an abnormal loss of 5% so of the total material input, only 85% was chemical X. Therefore, 340/0.85 = 400 litres must have been input into the process.

49 2 and 3 only

The following statements are consistent with the variances shown.

Lower skilled direct labour was used causing increased usage of direct materials.

Higher quality of direct labour and materials were used enabling fewer direct labour hours to be required.

50 2 and 4

Currency management and the investment of surplus funds are usually treasury functions.

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Mock Exam 2

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Foundations in Accountancy
MA2
Managing Costs and Finances

Mock Examination 2

Questions	
Time allowed	2 hours
ALL 50 questions are compulsory and must be attempted	

DO NOT OPEN THIS EXAM UNTIL YOU ARE READY TO START UNDER EXAMINATION CONDITIONS

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ALL 50 questions are compulsory and MUST be attempted

Each question in this section is worth 2 marks.

1 **Which of the following statements concerning management information is/are correct?**

1 A management information user should have all the information he/she needs to do his/her job properly.

2 A management information report must be relevant for a variety of purposes.

3 A management information report should contain a lot of detail to ensure complete accuracy.

A 1 only
B 1 and 2
C 2 and 3
D 3 only **(2 marks)**

---

2 **Which of the following are justifications for the widespread use of computers in the provision of management information?**

1 Speed of processing
2 Accuracy of processing
3 Volume and complexity of processing requirements

A 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3 **(2 marks)**

---

3 A company, which uses marginal costing, normally manufactures 1,000 units of a product in a period. The product is sold for $50 per unit. Costs for the 1,000 units are:

Direct materials	$16,300
Direct labour	$9,800
Fixed overheads	$21,600

How much profit will be expected if 1,100 units of the product are manufactured and sold in a period?

$ ☐ **(2 marks)**

---

4 A particular cost has been classified as 'semi-variable'.

How will the average cost per unit of activity be affected by a 20% reduction in the level of activity?

A Decrease by less than 20%
B Decrease by more than 20%
C Increase by less than 25%
D Increase by more than 25% **(2 marks)**

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5 A wholesaler uses the first-in first-out (FIFO) method of pricing inventory issues at each month end. The following details, relating to Product Z, are provided for a month:

Opening balance 860 units at a total cost of $1,892
Purchases 1,000 units at a total cost of $2,250
Sales 910 units

What is the cost of sales of Product Z in the month?

A $2,002.00
B $2,004.50
C $2,045.00
D $2,047.50 **(2 marks)**

---

6 Machine hours are used to absorb overheads in a production cost centre. Overheads allocated and apportioned to the cost centre are:

	$
Allocated	13,122
Apportioned	7,920
Reapportioned from service cost centers	2,988

216,000 units of product are manufactured at a rate of 120 units per machine hour.

What is the overhead absorption rate per machine hour (to two decimal places)?

$ ☐ **(2 marks)**

---

7 **Which of the following are relevant in the calculation of the maximum inventory control level?**

1 Maximum lead time
2 Minimum usage
3 Reorder level
4 Reorder quantity

A 1 and 2
B 3 and 4 only
C 1, 3 and 4
D 2, 3 and 4 **(2 marks)**

---

8 An incentive scheme is in operation for each direct worker in a factory. The basic rate of pay is $8 per hour for an eight-hour day with a bonus if hours worked are less than the standard hours for the output achieved. The bonus is 50% of the time saved against standard, paid at the basic rate. A single product is manufactured and the standard time is 10 minutes per unit.

What is the bonus for a worker who manufactures 60 units in an eight-hour day?

A $0
B $4
C $8
D $16 **(2 marks)**

---

9 The following data relates to production activity in a cost centre for a period:

	Budget	*Actual*
Output (units)	9,600	9,400
Labour (hours)	2,400	2,320

What was the efficiency ratio in the period (to one decimal place)?

☐% **(2 marks)**

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10 **Which TWO of the following are expenses that may be part of the cost accounts, but not of the financial accounts?**

- ☐ Cash discounts available to customers
- ☐ Interest charged to products based on average inventory
- ☐ Notional rent for the use, by different cost centres, of company-owned buildings
- ☐ Trade discounts received from suppliers **(2 marks)**

11 In a period where opening inventory was 5,000 units and closing inventory 8,000 units, a firm had a profit of $130,000 using absorption costing. **If the fixed overhead absorption rate was $4 per unit, what would profit be under a system of marginal costing?**

$ ☐ **(2 marks)**

12 **In which of the following manufacturing environments would job costing be appropriate?**

1 Production is carried out in accordance with the special requirements of each customer
2 Products are mass produced for inventory
3 Joint products are manufactured

A 1 only
B 1 and 2
C 3 only
D 2 and 3 **(2 marks)**

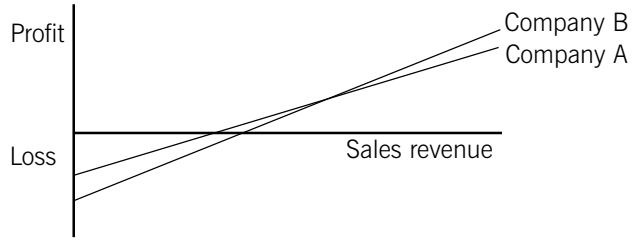
13 **When is service costing used?**

A When indirect costs are a small proportion of total costs
B When overhead absorption is straightforward
C When the absence of a physical product makes it impossible to determine unit costs
D When the output is intangible **(2 marks)**

14 **What will be the effect on the margin of safety if unit variable costs and total fixed costs both increase, assuming no change in selling price or sales volume?**

A Decrease
B Increase
C Stay the same
D Impossible to determine without more information **(2 marks)**

15 **Which statement is true with reference to the following profit/volume (P/V) chart?**



A Company A has lower break-even sales revenue than Company B.
B Company A has a higher contribution to sales ratio than Company B.
C Company A has higher fixed costs than Company B.
D Company A has higher profit than Company B. **(2 marks)**

16 **Which TWO of the following are relevant in capital investment decision-making using discounted cash flow methods of appraisal?**

- ☐ Annual depreciation
- ☐ Cost of capital
- ☐ Sunk costs
- ☐ Timing of future cash flows

(2 marks)

17 **What is the value after three years, to the nearest $, of $100 invested now at a compound rate of interest of 6% per annum?**

A $18
B $19
C $118
D $119

(2 marks)

18 Discounted cash flow analysis is being applied to a project with the following results:

Rate of interest % per annum	*Net present value*
	$
13	9,362
19	(2,015)

Using the above results, what is the best approximation of the internal rate of return of the project?

A 13.8%
B 14.1%
C 17.9%
D 20.6%

(2 marks)

19 A capital investment project requires expenditure of $90,000 in Year 0, followed by cash inflows of $30,000 at the end of each of the four years of the project's life. The project will have a terminal value of $60,000.

What is the payback period of the investment project (to the nearest whole year)?

☐ year(s)

(2 marks)

20 An overhead absorption rate of $12.00 per direct labour hour was established based on a budget of 2,100 hours. Actual direct labour hours worked were 2,180 and actual overhead expenditure was $25,470.

What was the over/under-absorption of overhead?

A $270 under-absorbed
B $690 over-absorbed
C $960 over-absorbed
D $960 under-absorbed

(2 marks)

21 **Are the following statements relating to management information true or false?**

	True	*False*
Management information should have some value otherwise it would not be worth the cost of collecting and communicating it.	○	○
Management information only needs to be accurate enough for its purpose.	○	○

(2 marks)

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22 **Which of the following is a cost centre manager?**

A A person responsible for costs only
B A person responsible for revenue only
C A person responsible for costs and revenue only
D A person responsible for costs, revenue and investment **(2 marks)**

23 12,000 kg of a material were input to a process in a period. The normal loss is 10% of input. There is no opening or closing work-in-progress. Output in the period was 10,920 kg.

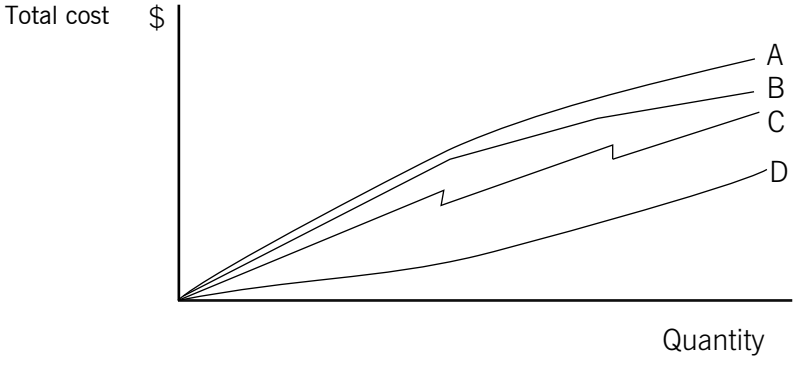
What was the abnormal gain/loss in the period?

A Abnormal gain of 120 kg
B Abnormal loss of 120 kg
C Abnormal gain of 1,080 kg
D Abnormal loss of 1,080 kg **(2 marks)**

24 **What is a cost ledger control account?**

A An account in the cost ledger to record financial accounting items
B An account in the financial ledger to record cost accounting items
C An account that summarises outstanding payables balances
D An account that summarises outstanding receivables balances **(2 marks)**

25 **Which line on the following graph represents the cost of a raw material where bulk discounts apply at intervals on all purchases?**



A Line A
B Line B
C Line C
D Line D **(2 marks)**

26 15,000 units of a product are currently manufactured in a factory per period. Variable costs are $4.50 per unit and fixed costs are $63,000 per period. If the capacity of the factory is increased to enable the manufacture of 20,000 units of product per period total costs would be:

Variable	$88,000
Fixed	$68,000

What would be the reduction in the total cost per unit if the capacity of the factory is increased as described?

A $0.80
B $0.90
C $1.05
D $1.15 **(2 marks)**

27 Stores ledger accounts and/or bin cards may include details of:

1 Receipts and issues
2 Inventory quantity
3 Unit price
4 Inventory value

Which of the items would appear on the stores ledger account but not on the bin card?

A 3 only
B 4 only
C 1 and 2
D 3 and 4 **(2 marks)**

---

28 Products A and B are manufactured jointly. Production costs in the joint process totaled $102,000 in a period and output was:

Product A 12,000 units (sold at $6.00 per unit)
Product B 22,000 units (sold at $4.00 per unit)

Joint costs are apportioned on the basis of realisable value.

What share of the joint costs in the period would be apportioned to Product B?

A $40,800
B $45,900
C $56,100
D $66,000 **(2 marks)**

---

29 **What term is used to represent the benefit sacrificed when one course of action is chosen in preference to an alternative?**

A Avoidable cost
B Direct cost
C Incremental cost
D Opportunity cost **(2 marks)**

---

30 A company manufactures and sells four products. Sales demand cannot be met owing to a shortage of skilled labour. Details of the four products are:

	Product A	*Product B*	*Product C*	*Product D*
Sales demand (units)	1,500	2,000	1,800	1,900
Contribution ($/unit)	2.80	2.60	1.90	2.40
Contribution/sales (%)	30	40	50	45
Skilled labour (hours/unit)	1.4	1.2	0.9	1.0

In what order should the products be made in order to maximise profit?

A Product A, Product B, Product D, Product C
B Product B, Product D, Product C, Product A
C Product C, Product D, Product B, Product A
D Product D, Product B, Product C, Product A **(2 marks)**

---

31 **Which of the following sentences is/are INCORRECT?**

(i) Capital receipts and capital payments take place in the normal course of business

(ii) Revenue receipts and payments take place when a business sells its non-current assets or settles its non-current liabilities

(iii) Exceptional receipts and payments occur at the end of every six months

A None of the above
B (i) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

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32 **Which of the following terms correctly describes the delaying of payments to suppliers whilst speeding up payment collection from customers?**

A Borrowing and lending
B Lagging and shorting
C Pulling and pushing
D Leading and lagging **(2 marks)**

---

33 The following are examples of labour costs incurred in production cost centre X in a factory:

1 Basic wages of machine operatives
2 Cost centre supervisor's wages
3 Wages (including overtime premium) of cleaning staff

Which of the labour costs are direct product costs?

A 1 only
B 2 only
C 1 and 3 only
D 1, 2 and 3 **(2 marks)**

---

34 A new non-current asset, costing $10,000, has a four-year life with an estimated value at the end of its life of 20% of the original investment amount. Two alternative depreciation methods are being considered for the asset:

(i) Reducing balance at 30% per annum
(ii) Machine hour utilisation based on:

Year 1 1,500 hours
Year 2 1,000 hours
Year 3 1,000 hours
Year 4 500 hours

The following statements relate to the above data:

Statement 1: The depreciation charge in Year 1 would be higher using the machine hour method.
Statement 2: The depreciation charge in Year 3 would be lower using the reducing balance method.

Are the statements true or false?

	Statement 1	*Statement 2*
A	True	True
B	True	False
C	False	True
D	False	False

(2 marks)

---

35 A company sold 10,000 units of its single product in a period during which finished goods inventory increased by 2,000 units.

Based on absorption costing, how would the profit in the period and the inventory value at the end of the period compare with those calculated using marginal costing (MC)?

	Profit	*Inventory value*
A	Higher than MC	Higher than MC
B	Higher than MC	Lower than MC
C	Lower than MC	Higher than MC
D	Lower than MC	Lower than MC

(2 marks)

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36 **Which of the following are the key principles that a business should base its cash management policy on?**

(i) Safety
(ii) Exposure
(iii) Profitability
(iv) Liquidity

A (i), (ii), (iii) and (iv)
B (i), (ii) and (iv) only
C (i), (iii) and (iv) only
D (ii), (iii) and (iv) only **(2 marks)**

---

37 A single-product business has the following results for a period:

	$
Sales revenue	268,000 (at $25 per unit)
Less: variable costs	139,360
Contribution	128,640
Less: fixed costs	87,480
Net profit	41,160

What is the breakeven point in units?

A 3,499
B 7,290
C 8,645
D 9,074 **(2 marks)**

---

38 P Co manufactures a single product E. Cost data relating to the product is as follows.

	$ per unit
Selling price	50
Direct material cost	7
Direct labour cost	8
Variable production overhead cost	8
Variable selling overhead cost	2
Fixed overhead cost	10
Profit per unit	15

What is the contribution/sales ratio for product E (to the nearest %)?

A 50%
B 54%
C 30%
D 70% **(2 marks)**

---

39 **Which of the following statements about cash handling procedures are correct?**

(i) Cash handling procedures should prevent fraud or theft.

(ii) Cash handling procedures over payments include prompt banking and proper post-opening arrangements.

(iii) Cash handling procedures relating to receipts include restriction of access to cash and cheques.

A (i) only
B (i) and (iii)
C (ii) and (iii)
D None of the above **(2 marks)**

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40 A business is considering a project requiring an investment of $200,000 now and with estimated cash inflows of $23,000 per annum in perpetuity. The first cash inflow would be received in one year's time. The cost of capital is 10% per annum.

What is the net present value of the investment?

A $2,300
B $3,000
C $20,000
D $30,000 **(2 marks)**

41 **Which of the following statements about certificates of deposit are correct?**

(i) Certificates of deposit are negotiable.
(ii) Title belongs to the seller once the sale is completed.
(iii) Certificates of deposit are issued by the UK government.
(iv) Terms can vary from seven days to five years.

A (i) and (ii) only
B (i), (ii) and (iii)
C (ii) and (iii) only
D (i) and (iv) **(2 marks)**

42 **Which of the following statements about the investment of surplus funds are correct?**

(i) The higher the risk of an investment, the higher the return required by the investor.
(ii) Local authority stocks are considered riskier than convertible loan stocks.
(iii) Government securities are the least risky of marketable UK securities.

A (i) and (ii) only
B (i) and (iii) only
C (ii) and (iii) only
D (i), (ii) and (iii) **(2 marks)**

43 Dunelm (Engineering) Co buys raw materials from suppliers on four weeks' credit and they are delivered immediately. When the raw materials are received, they are held in the warehouse for five weeks before being used in production. The production process takes one week and the completed goods are held for two weeks before finally being sold to credit customers. These customers are allowed a maximum credit period of six weeks but pay after three weeks in order to obtain a discount for prompt settlement.

What is the operating cash cycle of the business?

A 7 weeks
B 10 weeks
C 11 weeks
D 12 weeks **(2 marks)**

44 **Which of the following is NOT an attribute of effective communication?**

A Clarity
B Completeness
C Complexity
D Relevance **(2 marks)**

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45 X Co has the following at the end of March.

Non-current assets	$178,000
Inventory	$2,580
Trade receivables	$5,670
Bank overdraft	$1,300
Trade payables	$4,560

What is X Co's working capital?

A $2,390
B $180,390
C $4,990
D $14,110 **(2 marks)**

---

46 A company has fixed costs per period as follows:

Manufacturing	$56,000
Non-manufacturing	$38,000

Variable costs of the company's single product are $4.20 per unit and the selling price is $7.00 per unit.

What sales revenue (to the nearest $'000) is required in a period to make a profit of $6,000?

A $163,000
B $167,000
C $241,000
D $250,000 **(2 marks)**

---

47 In a 30-day period a restaurant was open for nine hours per day. Costs incurred in the period totalled $65,124. The following additional information is available:

Number of tables available	15
Number of seats per table	4
Customer turnaround	1 hour
Seating occupancy achieved	60%

What was the cost per customer?

A $4.02
B $6.70
C $16.08
D $26.80 **(2 marks)**

---

48 A large hotel contains coffee shops, restaurants and banqueting rooms. These are used by both residents and non-residents of the hotel. The manager of the hotel is responsible for encouraging residents to use the hotel's catering facilities.

Which report will show how effective the manager has been in achieving this objective?

A A report analysing the utilisation of hotel catering facilities per room occupied
B A report showing the amount of money spent in the hotel's catering facilities
C A report showing the number of residents in the hotel at any given time
D A report showing the occupancy of the various catering facilities **(2 marks)**

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49 A product has a budgeted direct material cost of $5 per unit. In a period, production of the product was:

Budget	9,000 units
Actual	8,800 units

$44,380 was incurred on direct materials for the period's production.

What was the direct material variance, comparing actual with the flexed budget?

A $380 Adverse
B $380 Favourable
C $620 Adverse
D $620 Favourable **(2 marks)**

50 A company operates a fleet of vehicles for delivering its goods.

Which of the following cost reports would be most useful for the manager to review to check maintenance expenditure is at a reasonable level for each vehicle?

A Maintenance cost per kilometre driven
B Maintenance cost per insurance value
C Maintenance cost per driver hour
D Maintenance cost per customer order **(2 marks)**

(Total = 100 marks)

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Answers to Mock Exam 2

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1 A A management information user should have all the information he/she needs to do his/her job properly.

2 D Speed of processing, accuracy of processing and volume and complexity of processing requirements are justifications for the widespread use of computers in the provision of management information.

3 $4,690

[(50 – 26.1) × 1,100] – 21,600

4 C Increase by less than 25%.

5 B [(860 × 2.2) + (50 × 2.25)]

6 $13.35

Total overhead = $13,122 + $7,920 + $2,988 = $24,030

Number of machine hours = 216,000/120 = 1,800

Overhead absorption rate per machine hour = $24,030/1,800 = $13.35

7 D Minimum usage, reorder level and reorder quantity are all relevant in the calculation of the maximum inventory control level.

8 C [(10 – 8) × 0.5 × 8]

9 101.3%

(2,350 ÷ 2,320) × 100

10 The correct answers are:

- Interest charged to products based on average inventory
- Notional rent for the use, by different cost centres, of company-owned buildings

Interest charged to products based on average inventory and notional rent for the use, by different cost centres of company-owned buildings are both expenses that may be part of the cost accounts but not part of the financial accounts.

11 $118,000

Marginal cost profit = Absorption cost profit + ((opening inventory – closing inventory) × fixed overhead absorption rate)

= $130,000 + ((5,000 – 8,000) × $4)

= $118,000

12 A Job costing is appropriate in environments where production is carried out in accordance with the special requirements of each customer.

13 D Service costing is used when the output is intangible.

14 A If unit variable costs and total fixed costs both increase, assuming no change in selling price or sales volume, the margin of safety will decrease.

15 A Company A has a lower break-even sales revenue than Company B.

16 The correct answers are:

- Cost of capital
- Timing of future cash flows

Annual depreciation and sunk costs are not relevant costs in discounted cash flow analysis.

17 D (100×1.06^3)

18 C [13 + (6 × 9,362/11,377)]

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19 3 years

(90 ÷ 30)

20 B

		$
Overhead absorbed 2,180 hours @ $12.00	=	26,160
Actual overhead expenditure	=	25,470
Over absorbed	=	690

21 Both statements are true.

22 A A cost centre manager is a person responsible for costs only.

23 A Normal loss = 10% × 12,000 kg
= 1,200 kg

Actual loss = 12,000 kg – 10,920 kg
= 1,080 kg

Abnormal gain = Actual loss – normal loss
= 1,080 – 1,200
= 120 kg

24 A A cost ledger control account is an account in the cost ledger to record financial accounting items.

25 C Line C

26 B (63,000/15,000 – 68,000/20,000) + (4.5 – 88,000/20,000)

27 D The stores ledger and/or bin cards may include details of unit price and inventory value.

28 C Realisable value of product A = 12,000 × $6.00 = $72,000

Realisable value of product B = 22,000 × $4.00 = $88,000

Total realisable value = $160,000

Joint costs apportioned to product B = $202,000 × $\frac{88,000}{160,000}$ = $56,100

29 D Opportunity cost

30 D

	Product A $2.80/1.4	*Product B* $2.60/1.2	*Product C* $1.90/0.9	*Product D* $2.40/1.0
Contribution per skilled labour hour	$2.00	$2.17	$2.11	$2.40
Order	4th	2nd	3rd	1st

The correct answer is therefore D.

31 D Capital receipts and payments take place when a business sells its non-current assets or settles its non-current liabilities.

Exceptional receipts and payments do not occur at regular intervals. Revenue receipts and payments take place in the normal course of business.

32 D Leading and lagging

33 A Basic wages of machine operatives

34 C Statement 1 is false. Statement 2 is true. The depreciation charge in year 1 would be the same for both methods.

35 A Under absorption costing, both profit and inventory value would be higher.

36 C Safety, profitability and liquidity

37 B 87,480/(128,640 ÷ 268,000/25)

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38 A The contribution/sales ratio (C/S ratio) is another term used to describe the profit/volume ratio (P/V ratio).

$$\text{C/S ratio} = \frac{\text{contribution per unit}}{\text{selling price per unit}}$$

$$= \frac{\$(50-7-8-8-2)}{\$50} \times 100\%$$

$$= 50\%$$

39 A Cash handling procedures should prevent fraud or theft. Cash handling procedures over *receipts* include prompt banking and proper post-opening arrangements. Cash handling procedures over *payments* include restriction of access to cash and cheques.

40 D (23,000 ÷ 0.1) – 200,000

41 D Title belongs to the seller until the sale is completed when it passes to the bearer. Certificates of deposit are issued by banks and building societies, not the UK government.

42 B Convertible loan stocks are considered riskier than local authority stocks.

43 A 7 weeks

Buys raw material from suppliers on credit	
Held in warehouse	5 weeks
Production process	1 week
Completed goods are held	2 weeks
Maximum credit period for customers	6 weeks
To obtain a discount for prompt settlement paid after	3 weeks

The operating cash cycle is: 5 + 1 + 2 + 3 – 4 = 7 weeks

44 C Effective communication should be as simple as possible – complete ideas should be explained clearly.

45 A The working capital is the current assets (inventory + trade receivables) less the current liabilities (bank overdraft + trade payables) = $2,390.

46 D

	$
Fixed costs ($56,000 + $38,000)	94,000
Required profit	6,000
Total contribution	100,000

Contribution per unit ($7.00 – $4.20) = $2.80

Therefore number of units = 100,000/2.80 = 35,714

Sales revenue = 35,714 units × $7 = $250,000 (to the nearest $'000)

47 B [$65,124 ÷ (30 × 9 × 15 × 4 × 0.6 customers)]

48 A

49 A

	$
8,800 units should have cost	44,000
But did cost	44,380
	380 Adverse

50 A Maintenance cost per kilometre driven would be the best to choose, since maintenance costs are expected to increase with distance travelled.

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REVIEW FORM

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